Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 — Unit 3 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

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· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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Unit_03 · Unit 3

1 UE26MA141A · Unit_03: Unit 3

This unit master organizes source material for Engineering Mathematics - I, Unit_03, officially titled Unit 3. It includes 13 video transcripts and 18 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26MA141A, Unit_03 (Unit 3). Bucket folder present: yes. Videos: 13; slide decks: 18.

EV0000001

2 Formation of PDE

This lesson is indexed from the source video titled Formation of PDE. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000001. Formation of PDE · source slide page 1.

EV0000175

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Contents



1 Formation of PDE by elimination of arbitrary functions

2 Solutions of Partial Differential Equations

3 Solution of First-Order Equations

Figure G000002. Formation of PDE · source slide page 2.

EV0000178

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Formation of PDE by elimination of arbitrary functions

1. Partial differential equations can be obtained by eliminating arbitrary functions from a given equation
2. A first-order partial differential equation can be derived from an equation involving a single arbitrary function, while a second-order equation may arise from an expression containing two arbitrary functions
3. However, it is not always guaranteed that an nth-order partial differential equation can be formed from an equation involving n arbitrary functions
4. In some cases, higher-order partial derivatives may be required to eliminate all n arbitrary functions, leading to a partial differential equation of order higher than n. Moreover, such a resulting higher-order equation may not be unique

Figure G000003. Formation of PDE · source slide page 3.

EV0000181

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Example: Formation of a PDE by eliminating an arbitrary function

Eliminate the arbitrary function from $z = f(x + y)$ to obtain a first-order partial differential equation.

Answer: Let $u = x + y$. Then $z = f(u)$.

Differentiating with respect to x:

$$\frac{\partial z}{\partial x} = f'(u) \cdot \frac{\partial u}{\partial x} = f'(x + y) \cdot 1 = f'(x + y)$$

Differentiating with respect to y:

$$\frac{\partial z}{\partial y} = f'(u) \cdot \frac{\partial u}{\partial y} = f'(x + y) \cdot 1 = f'(x + y)$$

From above,

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$$

So, the required PDE is:

$$\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = 0$$

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Figure G000004. Formation of PDE · source slide page 4.

EV0000184

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Example: 2

Obtain a second-order partial differential equation by eliminating the arbitrary functions from

$$u = f(x + ct) + g(x - ct).$$

Solution Differentiating the given equation partially with respect to x and t, we obtain

$$\frac{\partial u}{\partial x} = f'(x + ct) + g'(x - ct), \quad \frac{\partial u}{\partial t} = cf'(x + ct) - cg'(x - ct),$$

$$\frac{\partial^2 u}{\partial x^2} = f''(x + ct) + g''(x - ct), \quad \frac{\partial^2 u}{\partial t^2} = c^2 f''(x + ct) + c^2 g''(x - ct).$$

Hence, we obtain the equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Figure G000005. Formation of PDE · source slide page 5.

EV0000187

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Example: 3

Given $z = f(x^2 + y)$, where f is an arbitrary function, form the partial differential equation by eliminating the arbitrary function

Solution:

1. Differentiate with respect to x:

$$\frac{\partial z}{\partial x} = f'(x^2 + y) \cdot 2x$$

2. Differentiate with respect to y:

$$\frac{\partial z}{\partial y} = f'(x^2 + y) \cdot 1 = f'(x^2 + y)$$

3. Eliminate f' by dividing the first equation by the second:

$$\frac{\frac{\partial z}{\partial x}}{\frac{\partial z}{\partial y}} = \frac{2x f'(x^2 + y)}{f'(x^2 + y)} = 2x$$

Required PDE:

$$\frac{\partial z}{\partial x} - 2x\frac{\partial z}{\partial y} = 0$$

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Figure G000006. Formation of PDE · source slide page 6.

EV0000190

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Example: 4

Given $z = f(x^2 + y^2) + x$, where f is an arbitrary function, form the partial differential equation by eliminating the arbitrary function

Solution:

1. Differentiate with respect to x: $\frac{\partial z}{\partial x} = f'(x^2 + y^2) \cdot 2x + 1$
2. Differentiate with respect to y: $\frac{\partial z}{\partial y} = f'(x^2 + y^2) \cdot 2y$
3. Eliminate f' by solving the second equation for $f'(x^2 + y^2)$:

$$f'(x^2 + y^2) = \frac{1}{2y}\frac{\partial z}{\partial y} \quad (\text{for } y \neq 0)$$

Substitute into the first equation: $\frac{\partial z}{\partial x} = 2x\left(\frac{1}{2y}\frac{\partial z}{\partial y}\right) + 1 = \frac{x}{y}\frac{\partial z}{\partial y} + 1$

Required PDE:

$$\frac{\partial z}{\partial x} - \frac{x}{y}\frac{\partial z}{\partial y} = 1$$

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Figure G000007. Formation of PDE · source slide page 7.

EV0000193

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solutions of Partial Differential Equations

- A solution of a partial differential equation is a relation between the dependent and independent variables that satisfies the partial differential equation
- It is known that a partial differential equation is formed by eliminating arbitrary constants and arbitrary functions. These relations are solutions of the partial differential equation formed
- There are two types of solutions for partial differential equations:
  1. Solution containing arbitrary constants
  2. Solution containing arbitrary functions

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Figure G000008. Formation of PDE · source slide page 8.

EV0000196

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Complete Solution & General Solution

- A solution of a partial differential equation which contains as many arbitrary constants as the number of independent variables is called the **complete solution**
- A solution of a partial differential equation, which contains as many arbitrary functions as the order of the equation, is called the **general solution**

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Figure G000009. Formation of PDE · source slide page 9.

EV0000199

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Example

Suppose we have the relation:

$$z = f(x + y)$$

where f is an arbitrary (differentiable) function.
Now, $\frac{\partial z}{\partial x} = f'(x + y)$; $\frac{\partial z}{\partial y} = f'(x + y)$. Therefore,

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$$

The partial differential equation: $\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = 0$

Conclusion:

The equation $\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = 0$ is a partial differential equation formed by eliminating the arbitrary function f from the original relation. The original relation $z = f(x+y)$ is a solution to this partial differential equation

Figure G000010. Formation of PDE · source slide page 10.

EV0000202

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solution of First-Order Equations

- An equation containing x, y, z, p, q defines a first order partial differential equation, that is

$$F(x, y, z, p, q) = 0 \tag{1}$$

- **General solution:** A relation of the form $\phi(u, v) = 0$, where ϕ is an arbitrary function of $u = u(x, y, z)$, $v = v(x, y, z)$ and satisfies Eq. (1) is called a *general solution*
- **Particular solution:** The solution obtained by determining the arbitrary function in the general solution by using some specified condition is called a *particular solution*

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Figure G000011. Formation of PDE · source slide page 11.

EV0000205

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000174 · EV0000175 · EV0000176
2. Source slide page 2 (extracted text): Contents 1 Formation of PDE by elimination of arbitrary functions 2 Solutions of Partial Differential Equations 3 Solution of First-Order Equations 2 / 11 EV0000177 · EV0000178 · EV0000179
3. Source slide page 3 (extracted text): Formation of PDE by elimination of arbitrary functions 1 Partial differential equations can be obtained by eliminating arbitrary functions from a given equation 2 A first-order partial differential equation can be derived from an equation involving a single arbitrary function, while a second- order equation may arise from an expression containing two arbi- trary functions 3 However, it is not always guaranteed that an nth-order partial differ- ential equation can be formed from an equation involving n arbitrary functions 4 In some cases, higher-order partial derivatives may be required to eliminate all n arbitrary functions, leading to a partial differential equation of order higher than n. Moreover, such a resulting higher- order equation may not be unique 3 / 11 EV0000180 · EV0000181 · EV0000182

4. Source slide page 4 (extracted text): Example: Formation of a PDE by eliminating an arbitrary function Eliminate the arbitrary function from z = f(x + y) to obtain a first-order partial differential equation. Answer: Let u = x + y. Then z = f(u). Differentiating with respect to x: ∂z ∂u = f′(u) · = f′(x + y) · 1 = f′(x + y) ∂x ∂x Differentiating with respect to y: ∂z ∂u = f′(u) · = f′(x + y) · 1 = f′(x + y) ∂y ∂y From above, ∂z ∂z = ∂x ∂y So, the required PDE is: ∂z −∂z = 0 ∂x ∂y 4 / 11 EV0000183 · EV0000184 · EV0000185

5. Source slide page 5 (extracted text): Example: 2 Obtain a second-order partial differential equation by eliminating the arbitrary functions from u = f(x + ct) + g(x −ct). Solution Differentiating the given equation partially with respect to x and t, we obtain ∂u ∂u = f′(x + ct) + g′(x −ct), = cf′(x + ct) −cg′(x −ct), ∂x ∂t ∂2u ∂2u = f′′(x + ct) + g′′(x −ct), = c2f′′(x + ct) + c2g′′(x −ct). ∂x2 ∂t2 Hence, we obtain the equation ∂2u ∂2u = c2 ∂t2 ∂x2 5 / 11 EV0000186 · EV0000187 · EV0000188

6. Source slide page 6 (extracted text): Example: 3 Given z = f(x2 + y), where f is an arbitrary function, form the partial differential equation by eliminating the arbitrary function Solution: 1 Differentiate with respect to x: ∂z = f′(x2 + y) · 2x ∂x 2 Differentiate with respect to y: ∂z = f′(x2 + y) · 1 = f′(x2 + y) ∂y 3 Eliminate f′ by dividing the first equation by the second: ∂z 2xf′(x2 + y) ∂x = = 2x ∂z f′(x2 + y) ∂y Required PDE: ∂z −2x∂z = 0 ∂x ∂y 6 / 11 EV0000189 · EV0000190 · EV0000191

7. Source slide page 7 (extracted text): Example: 4 Given z = f(x2 + y2) + x, where f is an arbitrary function, form the partial differential equation by eliminating the arbitrary function Solution: 1 Differentiate with respect to x: ∂z = f′(x2 + y2) · 2x + 1 ∂x 2 Differentiate with respect to y: ∂z = f′(x2 + y2) · 2y ∂y 3 Eliminate f′ by solving the second equation for f′(x2 + y2): 1 ∂z f′(x2 + y2) = (for ∂y = 0) 2y ∂y Substitute into the first equation: ∂z = 2x 1 ∂z + 1 = x ∂z + 1 ∂x 2y ∂y y ∂y Required PDE: ∂z −x ∂z = 1 ∂x y ∂y 7 / 11 EV0000192 · EV0000193 · EV0000194

8. Source slide page 8 (extracted text): Solutions of Partial Differential Equations A solution of a partial differential equation is a relation between the dependent and independent variables that satisfies the partial differential equation It is known that a partial differential equation is formed by eliminating arbitrary constants and arbitrary functions. These relations are solutions of the partial differential equation formed There are two types of solutions for partial differential equations: 1. Solution containing arbitrary constants 2. Solution containing arbitrary functions 8 / 11 EV0000195 · EV0000196 · EV0000197

9. Source slide page 9 (extracted text): Complete Solution & General Solution A solution of a partial differential equation which contains as many arbitrary constants as the number of independent variables is called the complete solution A solution of a partial differential equation, which contains as many arbitrary functions as the order of the equation, is called the general solution 9 / 11 EV0000198 · EV0000199 · EV0000200

10. Source slide page 10 (extracted text): Example Suppose we have the relation: z = f(x + y) where f is an arbitrary (differentiable) function. Now, ∂z = f′(x + y); ∂z = f′(x + y). Therefore, ∂x ∂y ∂z ∂z = ∂x ∂y The partial differential equation: ∂z −∂z = 0 ∂x ∂y Conclusion: The equation ∂z −∂z = 0 is a partial differential equation formed by ∂x ∂y eliminating the arbitrary function f from the original relation. The orig- inal relation z = f(x+y) is a solution to this partial differential equation 10 / 11 EV0000201 · EV0000202 · EV0000203

11. Source slide page 11 (extracted text): Solution of First-Order Equations An equation containing x, y, z, p, q defines a first order partial differential equation, that is F(x, y, z, p, q) = 0 (1) General solution: A relation of the form $\phi(u, v) = 0$, where ϕ is an arbitrary function of u = u(x, y, z), v = v(x, y, z) and satisfies Eq. (1) is called a general solution Particular solution: The solution obtained by determining the arbitrary function in the general solution by using some specified condition is called a particular solution 11 / 11 EV0000204 · EV0000205 · EV0000206

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:04.548] Hello everyone. Welcome back to the course on Engineering Mathematics One. [00:00:13.028] In this session, we will discuss formation of partial differential equation by elimination of arbitrary functions, solutions of partial differential equations, and solution of first order equations. [00:00:36.164] So let's start with formation of a partial differential equation by elimination of arbitrary functions. [00:00:46.852] Remember, there are two ways to form a PDE. That is by elimination of arbitrary constants and by elimination of arbitrary functions. [00:01:09.540] If a given equation contains a single arbitrary function, we can always form a first order PDE. In a similar manner, a second order equation may arise from an expression containing [00:01:29.508] Two arbitrary functions. [00:01:34.084] Remember, it is not always guaranteed that an nth order PDE can be formed from an equation involving n arbitrary functions. In some cases, higher order. [00:01:53.764] Partial derivatives may be required to eliminate all n arbitrary functions. That leads to a PDE of harder higher than n. In these cases, the obtain PDE may not be unique. [00:02:19.588] Let's take up some of the examples. [00:02:23.812] For a partial differential equation from an arbitrary function z equals f of x plus y. Observe that in this question we have f to be an arbitrary function. [00:02:44.164] So let us take x plus y to be u. Then the given function becomes z equals f of u. Because there is only one arbitrary function. [00:03:00.196] We proceed by finding first order partial derivatives. So let us first compute dou z by doux. That's going to be f dash of u times dou by dou x. Since u equals x plus y, dou u by dou x becomes 1. [00:03:18.084] So dou z by dou x equals f dash of x plus y. Similarly, dou z by dou y equals f dash of x plus y. So from these two expressions after equating, we get dou z by dou x equals [00:03:37.252] dou z by dou y so the required p d will be dou z by doux minus dou z by dou y equals zero so in a given question we have a single arbitrary function and we got a first order p d e [00:04:01.348] Let's look at the second question. In the given problem, there are two arbitrary functions, namely f and g. So usually we find second order derivatives. So differentiating the given expression partially with respect to x and t. [00:04:21.220] Because U is a function of X and T. [00:04:27.524] So w by dou x becomes f dash of x plus ct plus g dash of x minus c t and w by dou t equals f dash of x plus c t times c plus [00:04:45.636] g dash of x minus ct times minus c because the coefficient of minus c t with respect to t is minus c. [00:04:58.692] Since there are two arbitrary functions, we need second derivative also. Let us compute dou square u by dou x square. We get f dash of x plus ct plus g double dash of x minus ct.

EV0000002 · EV0000003 · EV0000004 · EV0000005

Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.676] Similarly, dou square u by dou t square equals c square of f double dash x plus c t plus c square of g double dash x minus c t. In dou square u by dou t square, if you take c square common, we are left out with [00:05:35.556] F double dash X plus C T plus G double dash X minus C T which is same as dou square u by dou x square hence the second order PDE will be dou square u by dou t square equals [00:05:51.876] C

square times dou square u by dou x square. [00:05:59.748] Let's look at the third problem. Given z equals f of x square plus y, where f is an arbitrary function. So we need to form a PDE. [00:06:16.068] Observe carefully in this question we have a single arbitrary function, so we can expect first order PDE and maybe more. Differentiate partially with respect to x. So we get douz by doux equals f dash of x square plus y times 2x. [00:06:34.916] Similarly douz by dou y equals f dash of x square plus y. Observe dou z by doux and double dou z by dou y. So how do you eliminate f dash of x square plus y square from these two? [00:06:51.364] So we can go for division. So eliminate f dash by dividing the first equation by the second. So that leads to douz by doux divided by dou z by dou y equals 2x. So finally the first order PDE after eliminating f from the given function will be. [00:07:09.668] dou z by dou x minus minus two x times dou z by dou y equals zero similar to third problem let us look at one more problem z equals f of x square plus y square plus x when f is an arbitrary function [00:07:28.868] From the PDU for this particular function. Since z is a function of x and y, we can find out douz by doux and double dou z by dou y. [00:07:40.260] That's same as f dash of x square plus y square times 2x plus 1 and dou z by dou y equals f dash of x square plus y square times 2y. So how do we eliminate f dash from these two? [00:07:57.828] We cannot solve like previous problem. [00:08:02.148] So use expression number two and solve for f dash of x square plus y square, which is same as 1 over 2y times dosed by 2y. So y obviously non-zero. Substitute f dash of x square plus y square in the first expression. [00:08:18.852] That leads to dou z by dou x equals 2x of 1 by 2y dou z by dou y plus 1. So which is same as after cancellation of 2 we get x by y dou z by dou y plus 1. Hence the required PDE will be [00:08:36.836] dou z by dou x minus x by y times dou z by dou y equals one so remember there are two ways to form a pd first one by elimination of arbitrary constants [00:08:56.004] And second by elimination of arbitrary functions. After understanding this, let's move on to solutions of partial differential equations. Remember, a solution of a PDE is nothing but [00:09:15.684] A relation between dependent variable, independent variable that satisfies the given PDE. [00:09:26.244] It is known that a PD can be formed in two ways, one by eliminating arbitrary constants, and second one by eliminating arbitrary functions. And these relations are solutions of partial differential equation which is formed. [00:09:48.356] There are two types of solutions for partial differential equation. Solution containing arbitrary functions and solution containing arbitrary concepts. So this idea leads to complete solution and general solution of a PDE.

EV0000006 · EV0000007 · EV0000008 · EV0000009

Transcript block 3 (source interval 000600-000899 seconds): [00:10:10.596] Let's look at what is a complete solution and what is general solution. A solution of a PDE is said to be complete solution if the number of arbitrary consents is same as the number of independent variables present in a PDE. [00:10:31.204] Similarly, a solution of a PDE is said to be a general solution whenever the number of arbitrary functions is the same as the order of the equation. [00:10:45.828] So the idea of general solution can be seen again in Lagrange's linear equation. [00:10:56.836] In order to understand solution of a PDE, let's look at an example. Let Z equals f of x plus y. We need to form a PDE by eliminating this arbitrary function f. [00:11:13.444] As we know from previous idea, let us find out douz by doux and dou z by dou y. It is very easy to see that we can eliminate f dash of x space y by equating dou z by doux and dou z by dou y. [00:11:27.332] So the required or the obtained PDE for this case is douz by dou x minus douz by dou y equals zero. So how do we talk about solution of a PDE through this example? [00:11:41.732] As you can see the given question is z equals f of x plus y and the obtained answer is dou z by dou x minus dou z by dou equals zero.

So what is the connection between these two? [00:11:55.172] The equation dou z by dou x minus dou z by dou y equals zero is a partial differential equation formed by eliminating arbitrary function from the original function. [00:12:09.796] So this original function that is z d equals f of x plus y x plus y is a solution of the given pd or the obtained pd that is dou z by doux minus douz by dou y equals zero. So this is how we can understand a solution of a PD. [00:12:32.452] Let's look at general solution as well as particular solution. [00:12:39.236] So an equation containing XYZPQ defines a first order PDE which we have seen already in the session number one. A relation of the form phi of UV equals zero where phi is an arbitrary function of u and v. [00:12:57.988] and satisfies equation one is called a general solution. In a similar manner, the solution obtained by determining the arbitrary function in the general solution by using some specified condition is called a particular solution. [00:13:17.124] We will look at the idea of general solution once again in Lagrange's linear equation. So in this session we learned how to form a PDE by elimination of arbitrary functions. [00:13:33.092] We understood the idea of solution of PDEs. We also learned solution of first order equations. [00:13:42.340] Thank you.

EV0000010 · EV0000011 · EV0000012 · EV0000013

Full transcript: 1586 normalized words, 52 timestamped segments, 831.0 seconds duration. Low-confidence segments flagged: 0.

EV0000002 · EV0000003 · EV0000004 · EV0000005

3 Introduction

This lesson is indexed from the source video titled Introduction. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000014 · EV0000015

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000012. Introduction to Partial differentiation · source slide page 1.

EV0000208

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Contents



1 Introduction to Partial Differential Equations

Figure G000013. Introduction to Partial differentiation · source slide page 2.

EV0000211

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Definition: Differential Equation

A **Differential Equation** is an equation that involves:

- an **independent variable**,
- a **dependent variable**, and
- **derivatives of the dependent variable with respect to the independent variable**.

Example (ODE):

$$\frac{dy}{dx} + y = e^x$$

- x: independent variable
- y: dependent variable
- $\frac{dy}{dx}$: derivative of y with respect to x

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Figure G000014. Introduction to Partial differentiation · source slide page 3.

EV0000214

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

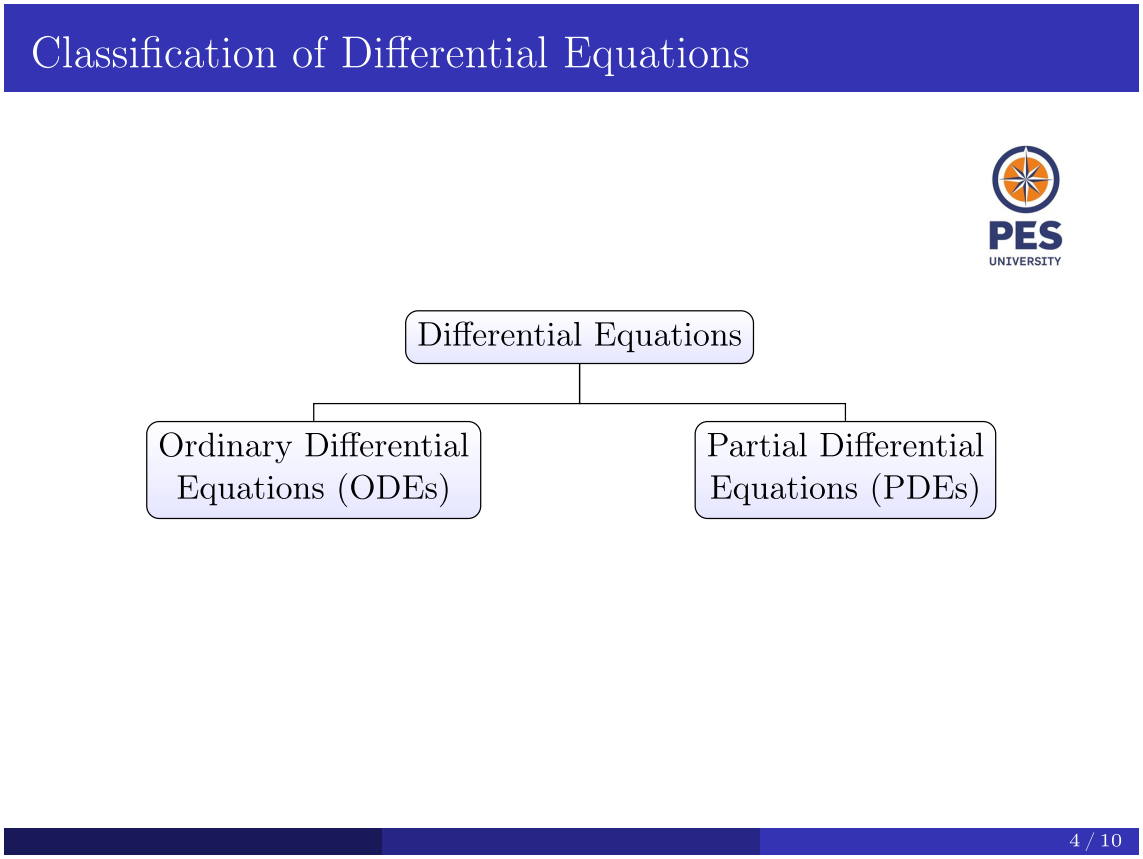


Figure G000015. Introduction to Partial differentiation · source slide page 4.

EV0000217

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ODEs vs PDEs

Ordinary Differential Equations (ODEs)

- Having Derivatives with respect to only **one independent variable**
- The unknown function depends on a **single variable**

Example:

$$\frac{dy}{dx} + y = e^x$$

Partial Differential Equations (PDEs)

- Involves **partial derivatives** with respect to **two or more independent variables**
- The unknown function depends on **multiple variables**

Example:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

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Figure G000016. Introduction to Partial differentiation · source slide page 5.

EV0000220

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differential Equations

- A partial differential equation (PDE) is an equation that involves partial derivatives of a function with several independent variables
- In contrast to ordinary differential equations (ODEs), which have one independent variable, PDEs involve two or more

Examples:

One-dimensional heat conduction equation: $u_t = u_{xx}$

Laplace equation: $u_{xx} + u_{yy} = 0$

One-dimensional wave equation: $u_{tt} = u_{xx}$

Figure G000017. Introduction to Partial differentiation · source slide page 6.

EV0000223

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Why Use PDEs?

Many physical and social problems have more than two independent variables. Because of this, partial differential equations (PDEs) are the best tool to study these problems. PDEs help us understand many real-world phenomena, including:

- Waves, like those in vibrating strings
- Heat flow
- Fluid movement
- Electromagnetic fields

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Figure G000018. Introduction to Partial differentiation · source slide page 7.

EV0000226

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Order and Degree of Partial Differential Equations

- **Order:** The order of a partial differential equation is the order of the highest order partial derivative occurring in the equation
- **Degree:** The degree of a partial differential equation is the degree of the highest order partial derivative occurring in the equation after the equation has been made free of radicals and fractions with respect to the partial derivatives

  Example: $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z \quad (\text{Order} = 1, \text{Degree} = 1)$

  $a^2\left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}\right) = \frac{\partial^2 z}{\partial t^2} \quad (\text{Order} = 2, \text{Degree} = 1)$

  $\left(\frac{\partial z}{\partial x}\right)^3 + \frac{\partial^2 z}{\partial y^2} = \cos(x+y) \quad (\text{Order} = 2, \text{Degree} = 1)$

  $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0 \quad (\text{Order} = 2, \text{Degree} = 1)$

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Figure G000019. Introduction to Partial differentiation · source slide page 8.

EV0000229

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Standard Notations

Let z be a dependent variable and x, y be independent variables, that is, $z = z(x, y)$. Then,

- $p = \partial z/\partial x = z_x$
- $q = \partial z/\partial y = z_y$
- $r = \partial^2 z/\partial x^2 = z_{xx}$
- $s = \partial^2 z/\partial x \partial y = z_{xy}$
- $t = \partial^2 z/\partial y^2 = z_{yy}$

Figure G000020. Introduction to Partial differentiation · source slide page 9.

EV0000232

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First and Second order PDE's

- An equation containing x, y, z, p, q defines a first order partial differential equation, that is

$$f(x, y, z, p, q) = 0 \tag{1}$$

  This equation is linear, if it is linear in p, q
- An equation containing x, y, z, p, q, r, s, t defines a second order partial differential equation, that is

$$g(x, y, z, p, q, r, s, t) = 0 \tag{2}$$

  This equation is linear, if it is linear in p, q, r, s, t

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Figure G000021. Introduction to Partial differentiation · source slide page 10.

EV0000235

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000207 · EV0000208 · EV0000209
2. Source slide page 2 (extracted text): Contents 1 Introduction to Partial Differential Equations 2 / 10 EV0000210 · EV0000211 · EV0000212
3. Source slide page 3 (extracted text): Definition: Differential Equation A Differential Equation is an equation that involves: an independent variable, a dependent variable, and derivatives of the dependent variable with respect to the independent variable. Example (ODE): dy + y = ex dx x: independent variable y: dependent variable dx:dy derivative of y with respect to x 3 / 10 EV0000213 · EV0000214 · EV0000215
4. Source slide page 4 (extracted text): Classification of Differential Equations Differential Equations Ordinary Differential Equations (ODEs) Partial Differential Equations (PDEs) 4 / 10 EV0000216 · EV0000217 · EV0000218

5. Source slide page 5 (extracted text): ODEs vs PDEs Ordinary Differential Equations (ODEs) Having Derivatives with respect to only one independent variable The unknown function depends on a single variable Example: dy + y = ex dx Partial Differential Equations (PDEs) Involves partial derivatives with respect to two or more inde- pendent variables The unknown function depends on multiple variables Example: ∂u = α∂2u ∂t ∂x2 5 / 10 EV0000219 · EV0000220 · EV0000221
6. Source slide page 6 (extracted text): Partial Differential Equations A partial differential equation (PDE) is an equation that involves partial derivatives of a function with several independent variables In contrast to ordinary differential equations (ODEs), which have one independent variable, PDEs involve two or more Examples: One-dimensional heat conduction equation: ut = uxx Laplace equation: uxx + uyy = 0 One-dimensional wave equation: utt = uxx 6 / 10 EV0000222 · EV0000223 · EV0000224
7. Source slide page 7 (extracted text): Why Use PDEs? Many physical and social problems have more than two independent variables. Because of this, partial differential equations (PDEs) are the best tool to study these problems. PDEs help us un- derstand many real-world phenomena, including: Waves, like those in vibrating strings Heat flow Fluid movement Electromagnetic fields 7 / 10 EV0000225 · EV0000226 · EV0000227
8. Source slide page 8 (extracted text): Order and Degree of Partial Differential Equations Order: The order of a partial differential equation is the order of the highest order partial derivative occurring in the equation Degree: The degree of a partial differential equation is the degree of the highest order partial derivative occurring in the equation af- ter the equation has been made free of radicals and fractions with respect to the partial derivatives Example: x ∂z + y ∂z = z (Order = 1, Degree = 1) ∂x ∂y a2 ∂2z + ∂2z = ∂2z (Order = 2, Degree = 1) ∂x2 ∂y2 ∂t2 ∂z 3 ∂2z + = cos(x + y) (Order = 2, Degree = 1) ∂x ∂y2 ∂2z ∂2z + = 0 (Order = 2, Degree = 1) ∂x2 ∂y2 8 / 10 EV0000228 · EV0000229 · EV0000230
9. Source slide page 9 (extracted text): Standard Notations Let z be a dependent variable and x, y be independent variables, that is, z = z(x, y). Then, p = ∂z/∂x = zx q = ∂z/∂y = zy r = ∂2z/∂x2 = zxx s = ∂2z/∂x∂y = zxy t = ∂2z/∂y2 = zyy 9 / 10 EV0000231 · EV0000232 · EV0000233
10. Source slide page 10 (extracted text): First and Second order PDE's An equation containing x, y, z, p, q defines a first order partial differential equation, that is f(x, y, z, p, q) = 0 (1) This equation is linear, if it is linear in p, q An equation containing x, y, z, p, q, r, s, t defines a second order partial differential equation, that is g(x, y, z, p, q, r, s, t) = 0 (2) This equation is linear, if it is linear in p, q, r, s, t 10 / 10 EV0000234 · EV0000235 · EV0000236

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:04.132] Hello everyone. Welcome back to the course on Engineering Mathematics 1. In this session, we'll understand partial differential equations. [00:00:18.500] Introduction to partial differential equations. [00:00:25.796] Before we understand what is a partial differential equation, let's look at what is a differential equation. [00:00:40.036] An equation involving independent variable, dependent variable and derivatives of the dependent variable with respect to independent variable is called a differential equation. As an example, [00:01:00.484] dy by dx plus y equals e power x. In this case x is the independent variable, y is dependent variable and the derivative of y [00:01:16.772] with respect to x is nothing but dy by dx. [00:01:26.788] Let's understand the classification of differential

equations. [00:01:35.140] Differential equations are classified into two based on the number of independent variables present in it. [00:01:45.156] The first one ordinary differential equations and partial differential equations. [00:01:59.044] Let's look at the difference between ordinary differential equations and partial differential equations. [00:02:10.340] In the case of ordinary differential equations, we see that the derivatives are with respect to only one independent variable. Also, the unknown function depends on a single variable. As you can see in this example, so y is dependent variable. [00:02:30.276] And X is independent variable. In a similar way, in partial differential equations we see that the partial derivatives are with respect to two or more independent variables. Also [00:02:47.844] The unknown function depends on multiple variables. In this example, dou u by dou t equals alpha dou square u by dou x square. In this case u is dependent variable and x and t are independent variables. [00:03:09.604] So mathematically we can define a partial differential equity in this manner. [00:03:18.628] A partial differential equation is an equation that involves partial derivatives of a function with several independent variables. As in the case of ordinary differential equations, we have [00:03:36.932] only one independent variable. In the case of partial differential equations, we have two or more independent variables. [00:03:49.124] Some of the examples of partial differential equations are one dimensional heat conduction equation, Laplask equation, and one dimensional wave equation. [00:04:08.164] Let's look at why do we use partial differential equations. As we have seen earlier, one of the applications of ordinary differential equations is modeling. In a similar manner, [00:04:24.548] Partial differential equations help us in understanding many real world phenomena like waves, those in vibrating strings, heat flow, fluid movement, electromagnetic fields, etc. [00:04:48.932] Let's understand the order and degree of a partial differential equation. As in the case of ordinary differential equations, the order of a partial differential equation is the order of the

EV0000014 · EV0000015 · EV0000016 · EV0000017

Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.308] Highest order partial derivative present in it. Similarly, the degree of a partial differential equation is the degree of the highest order partial derivative present in it, provided equation has been made free of radicals and fractions with respect to. [00:05:26.084] The partial derivatives. As you can see, some of the examples to answer the order and degree of a partial differential equation. In the first example, you can see order is also one, degree is also one. In the second example, order and degree are two, one, and one respectively. [00:05:46.852] In the third example, order and degree are two and one respectively. In the fourth example, order is two and degree is one. [00:05:58.980] So this is uh a standard notation which we use throughout the chapter. Let's z be a dependent variable and x and y be independent variables. Then pqr s t are denoted by dou z by doux, dou z by dou y [00:06:17.860] dou square z divided by dou x square, dou square z divided by dou x dou y and dou square z divided by dou y square. So these are the standard notations which we use throughout the chapter. [00:06:38.340] Let's understand what is a first order PDE and the second order PDE. So based on the standard notations, an equation containing X, Y, Z, P, and Q defines a first order partial differential equation. [00:06:58.404] Remember, this first order partial differential equation is said to be linear if it is linear in P and Q. It means we cannot have the product of P and Q, we cannot have P squared, we cannot have Q square, and so on. Similarly, [00:07:15.748] An equing XYZ PQRST defines a second order partial differential equation. That is G of XYZ PQRST equals zero. So this is said to be a linear. If it is linear in PQRST. [00:07:35.940] So in this session, we understood some of the basics of partial differential equations. Thank you.

EV0000018 · EV0000019 · EV0000020 · EV0000021

Full transcript: 750 normalized words, 29 timestamped segments, 472.896 seconds duration. Low-confidence segments flagged: 0.

EV0000014 · EV0000015 · EV0000016 · EV0000017

4 Problems on formation of PDE by eliminating arbitrary functions

This lesson is indexed from the source video titled Problems on formation of PDE by eliminating arbitrary functions. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000022 · EV0000023

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000022. Problems on formation of PDE by eliminating arbitrary functions · source slide page 1.

EV0000238

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Contents



1 Problems on formation of PDE by eliminating arbitrary functions

Figure G000023. Problems on formation of PDE by eliminating arbitrary functions · source slide page 2.

EV0000241

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem 1

$z = (x + y)\,\phi(x^2 - y^2)$

Solution: Differentiating

$$z_x = 1 \cdot \phi + (x + y)2x \cdot \phi' \tag{1}$$

$$z_y = 1 \cdot \phi + (x + y)(-2y)\phi' \tag{2}$$

From (2),

$$\phi' = \frac{\phi - z_y}{2y(x + y)} \tag{3}$$

Substituting (3) in (1):

$$z_x = \phi + 2x(x + y)\left[\frac{\phi - z_y}{2y(x + y)}\right]$$

$$p = \phi + \frac{x}{y}(\phi - q)$$

$$p = \left(\frac{x + y}{y}\right)\phi - \frac{x}{y}q$$

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Figure G000024. Problems on formation of PDE by eliminating arbitrary functions · source slide page 3.

EV0000244

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Problem 1 (contd.)

From the given equation, $\phi = \frac{z}{x+y}$. Substituting ϕ,

$$p = \frac{x+y}{y} \cdot \frac{z}{x+y} - \frac{x}{y}q = \frac{z}{y} - \frac{x}{y}q$$

or $\quad yp + xq = z$

Figure G000025. Problems on formation of PDE by eliminating arbitrary functions · source slide page 4.

EV0000247

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Problem 2

$z = x^n f\left(\frac{y}{x}\right)$

Solution: By differentiation,

$$z_x = nx^{n-1} f + x^n \left(-\frac{y}{x^2}\right) f'$$

$$z_y = x^n \cdot \frac{1}{x} f' \qquad \text{or} \quad f' = \frac{z_y}{x^{n-1}}$$

Eliminating f',

$$z_x = nx^{n-1} f - x^{n-2} y \cdot \frac{z_y}{x^{n-1}}$$

$$xp = nx^n f - yq$$

$$\text{or} \quad xp = nz - yq$$

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Figure G000026. Problems on formation of PDE by eliminating arbitrary functions · source slide page 5.

EV0000250

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Problem 3

$$xyz = f(x + y + z)$$

Solution: Differentiating w.r.t. x and y

$$yz + xyz_x = 1 \cdot f' + f' \cdot z_x \tag{1}$$
$$xz + xyz_y = 1 \cdot f' + f' \cdot z_y \tag{2}$$

From (2),

$$f' = \frac{xz + xyz_y}{1 + z_y} = \frac{xz + xyq}{1 + q} \tag{3}$$

Figure G000027. Problems on formation of PDE by eliminating arbitrary functions · source slide page 6.

EV0000253

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Problem 3 (contd.)

Put (3) in (1)

$$yz + xyp = (1+p)f' = (1+p)\left(\frac{xz + xyq}{1+q}\right)$$

$$(1+q)(yz + xyp) = (1+p)(xz + xyq)$$

or $\quad x(y-z)p + y(z-x)q = z(x-y)$

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Figure G000028. Problems on formation of PDE by eliminating arbitrary functions · source slide page 7.

EV0000256

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem 4

$z = f(x)\, g(y)$

Solution: Differentiating w.r.t. x and y, we get

$$z_x = f'(x)\, g(y)$$
$$z_y = f(x)\, g'(y)$$

$$z_x \cdot z_y = f'(x)\, g(y) \cdot f(x)\, g'(y) = f(x)g(y)f'(x)g'(y) = z\, f'(x)g'(y)$$

But

$$z_{xy} = f'(x)g'(y)$$

so

$$z_x \cdot z_y = z\, z_{xy}$$

or

$$pq = zs$$

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Figure G000029. Problems on formation of PDE by eliminating arbitrary functions · source slide page 8.

EV0000259

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000237 · EV0000238 · EV0000239
2. Source slide page 2 (extracted text): Contents 1 Problems on formation of PDE by eliminating arbitrary functions 2 / 8 EV0000240 · EV0000241 · EV0000242
3. Source slide page 3 (extracted text): Problem 1 z = (x + y) ϕ(x2 −y2) Solution: Differentiating zx = 1 · ϕ + (x + y)2x · ϕ′ (1) zy = 1 · ϕ + (x + y)(−2y)ϕ′ (2) From (2), ϕ −zy ϕ′ = (3) 2y(x + y) Substituting (3) in (1): ϕ −zy zx = ϕ + 2x(x + y) 2y(x + y) x p = ϕ + (ϕ −q) y x + y p = ϕ −x q y y 3 / 8 EV0000243 · EV0000244 · EV0000245
4. Source slide page 4 (extracted text): Problem 1 (contd.) From the given equation, ϕ = x+y.z Substituting ϕ, x + y z z p = · −x q = −x q y x + y y y y or yp + xq = z 4 / 8 EV0000246 · EV0000247 · EV0000248
5. Source slide page 5 (extracted text): Problem 2 z = xnf y x Solution: By differentiation, zx = nxn−1f + xn −y f′ x2 1 zy zy = xn · xf′ or f′ = xn−1 Eliminating f′, zy zx = nxn−1f −xn−2y · xn−1 xp = nxnf −yq or xp = nz −yq 5 / 8 EV0000249 · EV0000250 · EV0000251

6. Source slide page 6 (extracted text): Problem 3 xyz = f(x + y + z) Solution: Differentiating w.r.t. x and y yz + xyzx = 1 · f′ + f′ · zx (1) xz + xyzy = 1 · f′ + f′ · zy (2) From (2), xz + xyzy xz + xyq f′ = = (3) 1 + zy 1 + q 6 / 8 EV0000252 · EV0000253 · EV0000254
7. Source slide page 7 (extracted text): Problem 3 (contd.) Put (3) in (1) xz + xyq yz + xyp = (1 + p)f′ = (1 + p) 1 + q (1 + q)(yz + xyp) = (1 + p)(xz + xyq) or x(y −z)p + y(z −x)q = z(x −y) 7 / 8 EV0000255 · EV0000256 · EV0000257
8. Source slide page 8 (extracted text): Problem 4 z = f(x) g(y) Solution: Differentiating w.r.t. x and y, we get zx = f′(x) g(y) zy = f(x) g′(y) zx · zy = f′(x) g(y) · f(x) g′(y) = f(x)g(y)f′(x)g′(y) = z f′(x)g′(y) But zxy = f′(x)g′(y) so zx · zy = z zxy or pq = zs 8 / 8 EV0000258 · EV0000259 · EV0000260

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:04.772] Hello everyone. Welcome back to the course on engineering mathematics one. In this session, we will solve problems on formation of partial differential equation by eliminating arbitrary functions. [00:00:24.196] In one of the classes we understood what is the procedure to form a PDE by eliminating arbitrary functions. So let's solve problems on this concept. [00:00:37.572] Let us note on the first problem z equals x plus y phi of x square minus y square. So this expression contains a single arbitrary function, namely phi. [00:00:58.500] So as per the uh procedure, let us differentiate z partially with respect to x and y. [00:01:09.668] So we need to use uh the product rule when we differentiate z partially with respect to x. So let us take y as the first function and the derivative of x plus y with respect to x is 1 plus x plus y is a second function. [00:01:28.804] And phi becomes phi dash. Phi dash means phi dash of x square minus y square and x square become 2x. Let us call this as expression 1. In a similar manner, uh differentiate z partially with respect to y. So we get phi as the first function. [00:01:47.492] X plus Y becomes 1 with respect to Y and pi dash of X square minus Y square times minus 2y and this X plus Y is a another function. So x2 is the derivative of Z partially with respect to Y. [00:02:08.132] So let's look at one and two. How to eliminate file. [00:02:15.364] So from expression number two, we can solve for phi dash. That is so phi dash becomes phi minus z y divided by two y of x plus y. It means you can take this term to right hand side left hand side, it become plus. [00:02:34.564] x plus y times u y into phi dash. Then you can shift this z sub x y to right hand side so it becomes phi minus z y. So when you solve for phi dash we'll get phi minus douz by dou y divided by two y of x plus y [00:02:52.996] So what can be done for that? Let us substitute three in uh expression number one that is uh dou z by doux. So we get z suffix x equals five, that is one into five plus [00:03:09.892] Write x plus y and two x as it is. In place of this phi dash, we substitute phi minus douze by dou y over two y of x plus y. [00:03:24.420] So what is this equal to? By the standard notation, we understood that those z by doux is same as p and this y as it is. So you can cancel this x plus y, x plus y, we can cancel uh this two and this two. [00:03:43.588] So we are left out with the x by y of phi minus q because dou z by dou y equals q as per the standard notation. So when we solve this, uh we get uh phi plus x by y [00:04:03.396] phi minus x y by y times q which you can see here so these two are nothing but x plus y over y of phi [00:04:18.820] So still phi is seen in the problem. So how do we eliminate this? So for that, we need to replace phi as z divided by x plus y. If you look at the problem. [00:04:33.668] So let us solve for phi. Say phi means phi of x square minus y square. This is z over x plus y. So when you substitute that into previous expression, we'll get t equals x plus y divided by y times z by x plus y. So this value minus [00:04:52.132] X by Y times Q. So we can cancel X plus Y and Xpress Y. So

we get Z by Y minus X by Y times Q. So when you solve this, we get this as the PD. You can clearly see that that's this is a first order partial differential equation.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.612] That's the answer. I hope you understood the concept. [00:05:20.932] Let us look at the second problem. Z equals x power n times f of y by x. Even this function contains a single arbitrary function, namely f. So as per the procedure. [00:05:38.628] We differentiate z partially with respect to x. So we get dou z by doux equals so we need to apply product rule. So f is the first function. [00:05:52.356] And the derivative of x power n with respect to x is nothing but n times x into x power n minus 1 plus x power n as the uh first function and the derivative of f of y by x becomes f dash. [00:06:08.644] of y by x then uh y by x becomes minus y by x square with respect to x. So what is this equal to? Okay in a similar manner let us compute to dou z by dou y so for douz by dou y no need to apply product rule because x becomes a constant [00:06:26.244] And f becomes f dash and y by x becomes 1 by x. Y is 1 and x is constant. So from this expression we can solve for f dash, that is uh dou z by dou y divided by if we use laugh indices, this is x power n minus 1. [00:06:45.764] Let us substitute this into this expression. Let us call one. So z subic x. So nx n minus one time f minus. Again, this is going to be x power n minus two times y. [00:07:00.676] In place of f dash, we put uh dou z by dou y over x power n minus one. So in the next step we can see this is nothing but x power n uh divided by x. [00:07:14.500] And this is going to be if you use law indices for x power n minus 2 and x power n minus 1, you get x power minus 1. So that leads to 1 by x. So if you cross multiply and replace douz by doux as p, finally we get uh x times p. [00:07:32.996] nx n f minus y into q because those that better y is q as per the standard notation so if you observe this particular step still f is same so how do we eliminate this f [00:07:51.588] So if you look at the given problem, so this x power n 10 f is nothing but even set. So that is why this is the answer for the given problem, and even this is a first order partial differential equation. [00:08:15.076] Let us take another problem xyz equals f of x plus y plus z. So differentiate given problem partially with respect to x, and you know z is a function of x and y. So when we apply product rule, so we need to [00:08:35.108] So y z times one plus xy and z becomes dou z by doux on the right hand side. So this is uh f becomes f dash. [00:08:53.412] of one because x plus y play z here plus f dash of dou z by doux because of this in a similar manner uh given problem with respect to y we get the x times z [00:09:08.196] plus x y dou z by dou y equals f dash plus f dash into dou z by dou y [00:09:16.932] So let us use equation number two and solve for f dash. So f dash becomes x plus x y dou z by dou y divided by one plus z y. Because uh here we take it is uh f dash of one plus dou z by dou y. [00:09:36.388] It is nothing but this. So finally this is same as Xz XY Q got dou z by dou y is q and one plus q. [00:09:47.940] So at last uh we substitute uh expression three in expression number one. So that gives uh yz x y p equals one plus p into f dash. So that's what you can see from expression number one. When you take f dash common get one plus p.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.764] So one plus b as it is in base of f dash as per uh expression number three we put x plus x y q over one plus q. And uh this can be simplified to this particular step. Uh you can just uh shift. [00:10:27.140] one plus q to left hand side y z plus x y p equals one plus p time x z plus x y q so when we simplify this so that leads to this particular step so that is x of y minus z p plus [00:10:44.836] Y of z minus x q equals z of x minus y. So you decide what is the order of this linear differential equation because of presence

of P and Q. So order is again one. [00:11:02.148] So let's look at another problem Z equals f of x times g of y. Now observe carefully this expression contains two arbitrary functions namely f and g. So we may need to proceed for second order partial derivatives. [00:11:22.820] So as usual, let us differentiate uh the given problem partially with respect to x and y. So we get douz by 2x, g of y is constant, and f of x becomes f dash of x. [00:11:36.068] With respect to y, we get douz by dou y equals f of x times g dash f i because f of x is constant. So if we observe these two uh in the current state, we cannot eliminate arbitrary functions, namely f and g. So what do we do? We proceed for [00:11:55.556] Second order partial derivatives. Observe carefully from either of these two. You can say either ZXY or it could be ZYX. No, this is nothing but mixed partial derivative. It is nothing but [00:12:14.244] F dash of x times g dash of y. Even the second is also the same. So z y with respect is even this is uh f dash of x times g dash of y. Okay. So that is nothing but this. [00:12:31.364] dou square z divided by doux dot dou y. So f dash of x time g dash of y. So again how to eliminate f and g [00:12:44.548] Let us take the product of uh dou z by doux or dou z by dou y so that leads to this particular expression and this is same as f of x dot g of y dot f dash of x then g dash of y. [00:13:01.764] So this is nothing but Z. The given problem is f of x dot g of y equals f dash of x dot g dash of y. So from this expression [00:13:14.020] So f dash of x dot g dash of y is nothing but this mixed partial derivative those squares are divided by doux dot dou y. So this expression leads to this particular step. As you can see, as per the standard notation. [00:13:33.892] So this is P and this is Q. This is nothing but S. So S becomes those squares that are divided by 2x dot douy. So in this session we understood how to solve the problems on formation of a partial differential equation. [00:13:51.396] by eliminating arbitrary functions. Thank you.

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Full transcript: 1884 normalized words, 50 timestamped segments, 839.0 seconds duration. Low-confidence segments flagged: 0.

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5 Session-3

This lesson is indexed from the source video titled Session-3. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000034 · EV0000035

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.524] Hello everyone. Welcome back to the course on engineering mathematics one. [00:00:10.980] In this session, we will discuss Lagrange's linear equation and working rule to solve Lagrange's linear equation. [00:00:27.460] So what is a Lagrangian's linear equation? The linear first order partial differential equation of the form capital P of XYZ times small p plus q of xyz q equals r of xyz. [00:00:47.300] So this small b and q are dou z by dou x and dou z by dou y respectively. And this expression is called Lagrangius equation in two independent variables, namely x and y. [00:01:05.188] Remember the general solution of Lagrangian's linear equation is given by pi of u comma v equals zero where phi is an

arbitrary function. [00:01:21.572] And U and V are the two linearly independent solutions of the acceler equation which is given by Tx divided by capital P equals dy by capital Q equals dz by R. [00:01:41.284] After understanding Lagrange's linear equation and its general solution, let's look at the working rule to solve Lagrange's linear equation. There are three steps in the procedure. [00:01:58.532] Let us consider the Lagrange linear equation capital B times small P plus Q times small Q equals R. [00:02:10.692] From the given problem, we need to identify capital B, capital U and capital R. Then we write axillary equation in this form. Remember there are three ways to solve this axillary equation. [00:02:28.836] One is by method of grouping, second by method of multipliers, and the third is the differential concept. So once we find out the two linearly independent solutions, u and v by using this actual equation. [00:02:48.292] We write the general solution in the form φ of u comma v equals 0. Let us discuss method of proving, method of multipliers, a differential concept one by one. [00:03:04.164] So remember we will solve the axillary equations by the method of grouping or method of multipliers or both to get two independent solutions, namely u equals a and u equals b where ab are arbitrary constants. [00:03:27.972] So what is method of grouping in Lagrange's method? So how do we apply method of grouping for a problem? [00:03:37.988] So once we finish writing XL equation, if we want to apply method of grouping, we will write two ratios at a time like this. [00:03:50.916] It could be dx by capital P equals Dy by Q or Dy by capital Q equals dz by R, Dx by B equal DZ by R. So it is first and second ratio, or this is second and third ratio, or this is first and last ratio. [00:04:11.716] So from these three, if we are able to separate variables, then we say that integration is possible. So if integration is possible, after separating the variables, we say that method of grouping works. [00:04:31.652] So let's take an example how to apply method of grouping. [00:04:39.172] Assume the accelerary equations are given by dx over small x, dy over y and dzy z So it means for a given problem capital P is X, capital Q is Y. [00:04:57.476] And R is that. Let us solve this problem by method grouping. It is easy to see that the variables have already been separated. So from the first and second, we can write

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:15.588] dx by x equals dy by y so after integration we get log x equals log y plus log c1 so after using log m plus log v so this leads to x by y equals c2 or it would c one also [00:05:35.428] Similarly, if we take a second and third ratio, we get y by z equal to c2. If we take first and third, we take x by z equal to c3. So for this, there are three possibilities, but we can take any two for the problem. [00:05:54.308] So if you take first and second uh second and third ratios, the general solution for this problem is given by f of x by y comma x by z equals zero, or it could be phi of x by y and x by z equals zero. [00:06:14.052] So if you look at the procedure. [00:06:18.564] So once you are able to separate variables, we can integrate each group to obtain two independent solutions. So once we find out two independent solutions, we write the general solution as phi of u comma v0. So this is how we can apply method of grouping for a problem. [00:06:42.244] So let's look at uh how do we apply method of multipliers for a problem in Lagrange's equation. So as usual, when we are not able to apply method of grouping, we switch over to method of multipliers. [00:07:00.868] So what do we do in method of multiple? So as usual, this is accelerate equation. [00:07:10.244] So in method of multipliers, the task is to choose three things. We call them as lm. They are either constants or functions of XYZ. So on what basis we choose LMN. For that we need to look at each ratio. [00:07:29.284] by the property of ratio proportion we can write this axle equation to be LDX plus m dy plus ndz divided

by lp plus mq plus r so this is capital p so what has been done here [00:07:47.076] So once we choose LMN by inspection, we multiply DX by L or multiply dy by m and dz by n as well as multiply lm by this pq respectively and we can write each ratio in this form. [00:08:08.164] Now the condition for selection of LMN is this denominator should be zero. It means make sure that after choose LMN, this denominator becomes zero. [00:08:24.356] Therefore, this can be written as this numerator LDX plus mdy plus ndz equals zero and we see that it is easy to integrate. So once we can integrate this, the solution of this differential equation gives u equals c1 where u is a function of xyz as we have seen earlier. [00:08:45.092] So for a problem to get another solution, if you are not able to apply grouping, we proceed with multipliers. So similarly we choose another set of multipliers namely L1, M1, N1. [00:08:59.620] And find a second solution B equals C2. So once we find U and B, we write the general solution in this form. So this is a way to apply method of multipliers. So remember the condition for the selection of element is going to be [00:09:19.236] L times capital B plus M times capital U plus N times capital R should be zero. [00:09:26.948] In order to understand this, let's take an example. Suppose we have dx over y minus z. It means for a given problem capital P is Y minus Z and capital Q is Z minus X and R is going to be X minus Y. [00:09:47.268] Observe carefully and choose a multipliers. So that when you use the property of racial proportion, you should get zero. Obviously, you can observe that you can cancel y and minus y, you can cancel minus z plus z and minus six plus six.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.508] Only if you multiply each of these three by one one one. It means to get one independent solution the multipliers are one one one. Then each ratio becomes one into dx and one into dy and one into dz divided by one times y minus z. [00:10:25.444] as 1 times z minus x and 1 times x minus y that gives 0. So we can equate dx plus dy plus 0 and integration of this becomes x plus y plus z equals c 1. [00:10:41.700] So for another set of solution, we can try with one more set of multipliers. So this is how we apply method of multiplier for a problem. [00:10:56.388] As I mentioned, there is one other method to find the solution in Lagrange's method. It's a rare case, of course. It is called differential concept in Lagrange's equation. [00:11:11.012] So as usual we first write the acceler equations that is dx by capital P, Dy by capital Q and Dz by R. Sometimes instead of grouping or using multipliers directly, we use differential concept to form [00:11:28.708] a linear combination of differential that is easier to integrate. [00:11:36.580] Remember in this case also we have to choose multipliers. The multipliers may be chosen such that the numerator that is LDX plus MDY plus N T Z is an exact differential of the denominator LP plus MQ plus NR. [00:11:54.980] Once we achieve this, we will equate this with the fraction of acceler equation that is dx by p equals dy by q and dz by r. So what is the meaning of numerator as the exact differential of denominator? [00:12:13.828] For example, in the denominator you may be having x square plus y square plus z square. Suppose if you are able to get two times x into dx plus y into dy. [00:12:31.844] plus z into dz in this case we say that the numerator is exact differential of denominator and this will be equated with a fraction of dx by capital B equals dy by capital U or Dz by R anyone so to get the answer. [00:12:52.676] So in this session uh we understood what is Lagrangius equation. What are the three different types to solve Lagrangian's equation? In the next session, we will solve problems on Lagrangian's equation. Thank you.

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Full transcript: 1559 normalized words, 47 timestamped segments, 790.528 seconds duration. Low-confidence segments flagged: 0.
EV0000034 · EV0000035 · EV0000036 · EV0000037

6 Lagranges Linear equation_continued

This lesson is indexed from the source video titled Lagranges Linear equation_continued. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000046 · EV0000047

Graphs and diagrams

ENGINEERING MATHEMATICS - I
Unit - 3: Partial Differential Equations

Department of Science and Humanities

PES UNIVERSITY

Figure G000047. Lagrange s Linear Equation Continued · source slide page 1.

EV0000313

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Contents



1 Problems on Lagrange's Linear Equation

Figure G000048. Lagrange s Linear Equation Continued · source slide page 2.

EV0000316

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial y} = ly - mx$

The auxiliary equations are $\frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx}$
Using multipliers x, y, z, we get
Each fraction

$$= \frac{x\,dx + y\,dy + z\,dz}{x(mz-ny) + y(nx-lz) + z(ly-mx)} = \frac{x\,dx + y\,dy + z\,dz}{0}$$

$$\implies x\,dx + y\,dy + z\,dz = 0$$

On integration, it gives

$$x^2 + y^2 + z^2 = c_1 \qquad (1)$$

Again using multipliers l, m, n, we get
Each fraction

$$= \frac{l\,dx + m\,dy + n\,dz}{l(mz-ny) + m(nx-lz) + n(ly-mx)} = \frac{l\,dx + m\,dy + n\,dz}{0}$$

$$\implies l\,dx + m\,dy + n\,dz = 0$$

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Figure G000049. Lagrange s Linear Equation Continued · source slide page 3.

EV0000319

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial y} = ly - mx$ (contd.)

On integration, it gives

$$lx + my + nz = c_2 \qquad (2)$$

Hence, from (1) and (2), the required general solution is

$$\phi(x^2 + y^2 + z^2, lx + my + nz) = 0$$

Figure G000050. Lagrange s Linear Equation Continued · source slide page 4.

EV0000322

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $x(z^2-y^2)\frac{\partial z}{\partial x}+y(x^2-z^2)\frac{\partial z}{\partial y}=z(y^2-x^2)$

Solution. The auxiliary equations are

$$\frac{dx}{x(z^2-y^2)}=\frac{dy}{y(x^2-z^2)}=\frac{dz}{z(y^2-x^2)} \qquad (1)$$

Using multipliers x, y, z, each fraction of (1) is equal to

$$\frac{x\,dx+y\,dy+z\,dz}{x^2(z^2-y^2)+y^2(x^2-z^2)+z^2(y^2-x^2)}=\frac{x\,dx+y\,dy+z\,dz}{0}$$

$$\implies x\,dx+y\,dy+z\,dz=0$$

On integration

$$x^2+y^2+z^2=C_1 \qquad (2)$$

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Figure G000051. Lagrange s Linear Equation Continued · source slide page 5.

EV0000325

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $x(z^2-y^2)\frac{\partial z}{\partial x}+y(x^2-z^2)\frac{\partial z}{\partial y}=z(y^2-x^2)$ (contd.)

Again, (1) can be written as

$$\frac{dx}{x(z^2-y^2)}=\frac{dy}{y(x^2-z^2)}=\frac{dz}{z(y^2-x^2)}$$

$$=\frac{\frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z}}{(z^2-y^2)+(x^2-z^2)+(y^2-x^2)}=\frac{\frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z}}{0}$$

$$\implies \frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z}=0$$

$$\implies \log x+\log y+\log z=\log C_2$$

$$\implies \log xyz=\log C_2 \implies xyz=C_2 \qquad (3)$$

From (2) and (3), the general solution is

$$\phi(x^2+y^2+z^2,xyz)=0$$

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Figure G000052. Lagrange s Linear Equation Continued · source slide page 6.

EV0000328

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $(y^2 + z^2)p - xyq + zx = 0$

Solution:
Auxiliary equations are

$$\frac{dx}{y^2 + z^2} = \frac{dy}{-xy} = \frac{dz}{-zx}$$

From the 2nd and 3rd fractions,

$$\frac{dy}{y} = \frac{dz}{z} \quad \text{or} \quad \frac{y}{z} = c_1$$

Choosing multipliers as x, y, z:

$$x\,dx + y\,dy + z\,dz = x(y^2 + z^2) + y(-xy) + z(-zx) = 0$$

Figure G000053. Lagrange s Linear Equation Continued · source slide page 7.

EV0000331

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $(y^2+z^2)p - xyq + zx = 0$ (contd.)

Integrating,

$$x^2 + y^2 + z^2 = c_2$$

The general solution is

$$\phi\left(\frac{y}{z},\, x^2+y^2+z^2\right) = 0$$

Figure G000054. Lagrange s Linear Equation Continued · source slide page 8.

EV0000334

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$

Here the auxiliary equations are

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$$

Taking the last two members, we have

$$\frac{dy}{y} = \frac{dz}{z}$$

which on integration gives

$$\log y = \log z + \log a$$

or

$$\log \frac{y}{z} = \log a \quad \text{or} \quad \frac{y}{z} = a \qquad \cdots (1)$$

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Figure G000055. Lagrange s Linear Equation Continued · source slide page 9.

EV0000337

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$ (contd.)

Using x, y, z as multipliers, we get
Each fraction

$$= \frac{x\,dx + y\,dy + z\,dz}{x(x^2 + y^2 + z^2)} = \frac{dz}{2xz}$$

or

$$\frac{2x\,dx + 2y\,dy + 2z\,dz}{x^2 + y^2 + z^2} = \frac{dz}{z}$$

which on integration gives

$$\log(x^2 + y^2 + z^2) = \log z + \log b$$

or

$$\log\left(\frac{x^2 + y^2 + z^2}{z}\right) = \log b \quad \text{or} \quad \frac{x^2 + y^2 + z^2}{z} = b \qquad \cdots(2)$$

From (1) and (2), the general solution is

$$\phi\left(\frac{x^2 + y^2 + z^2}{z}, \frac{y}{z}\right) = 0$$

Figure G000056. Lagrange s Linear Equation Continued · source slide page 10.

EV0000340

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000312 · EV0000313 · EV0000314
2. Source slide page 2 (extracted text): Contents 1 Problems on Lagrange's Linear Equation 2 / 10 EV0000315 · EV0000316 · EV0000317
3. Source slide page 3 (extracted text): Solve (mz −ny) ∂z + (nx −lz)∂z = ly −mx ∂x ∂y The auxiliary equations are dx = dy = dz mz−ny nx−lz ly−mx Using multipliers x, y, z, we get Each fraction x dx + y dy + z dz x dx + y dy + z dz = = x(mz −ny) + y(nx −lz) + z(ly −mx) 0 =⇒x dx + y dy + z dz = 0 On integration, it gives x2 + y2 + z2 = c1 (1) Again using multipliers l, m, n, we get Each fraction l dx + m dy + n dz l dx + m dy + n dz = = l(mz −ny) + m(nx −lz) + n(ly −mx) 0 =⇒l dx + m dy + n dz = 0 3 / 10 EV0000318 · EV0000319 · EV0000320
4. Source slide page 4 (extracted text): Solve (mz −ny) ∂z + (nx −lz)∂z = ly −mx (contd.) ∂x ∂y On integration, it gives lx + my + nz = c2 (2) Hence, from (1) and (2), the required general solution is ϕ(x2 + y2 + z2, lx + my + nz) = 0 4 / 10 EV0000321 · EV0000322 · EV0000323

5. Source slide page 5 (extracted text): Solve x(z2 −y2) ∂z + y(x2 −z2)∂z = z(y2 −x2) ∂x ∂y Solution. The auxiliary equations are dx dy dz = = (1) x(z2 −y2) y(x2 −z2) z(y2 −x2) Using multipliers x, y, z, each fraction of (1) is equal to x dx + y dy + z dz x dx + y dy + z dz = x2(z2 −y2) + y2(x2 −z2) + z2(y2 −x2) 0 =⇒x dx + y dy + z dz = 0 On integration x2 + y2 + z2 = C1 (2) 5 / 10 EV0000324 · EV0000325 · EV0000326

6. Source slide page 6 (extracted text): Solve x(z2 −y2) ∂z + y(x2 −z2)∂z = z(y2 −x2) (contd.) ∂x ∂y Again, (1) can be written as dx dy dz = = x(z2 −y2) y(x2 −z2) z(y2 −x2) dx + dy + dz dx + dy + dz x y z x y z = = (z2 −y2) + (x2 −z2) + (y2 −x2) 0 =⇒dx + dy + dz = 0 x y z =⇒log x + log y + log z = log C2 =⇒log xyz = log C2 =⇒xyz = C2 (3) From (2) and (3), the general solution is ϕ(x2 + y2 + z2, xyz) = 0 6 / 10 EV0000327 · EV0000328 · EV0000329

7. Source slide page 7 (extracted text): Solve (y2 + z2)p −xyq + zx = 0 Solution: Auxiliary equations are dx dy dz = = y2 + z2 −xy −zx From the 2nd and 3rd fractions, dy dz y = or = c1 y z z Choosing multipliers as x, y, z: x dx + y dy + z dz = x(y2 + z2) + y(−xy) + z(−zx) = 0 7 / 10 EV0000330 · EV0000331 · EV0000332

8. Source slide page 8 (extracted text): Solve (y2 + z2)p −xyq + zx = 0 (contd.) Integrating, x2 + y2 + z2 = c2 The general solution is y ϕ , x2 + y2 + z2 = 0 z 8 / 10 EV0000333 · EV0000334 · EV0000335

9. Source slide page 9 (extracted text): Solve (x2 −y2 −z2)p + 2xyq = 2xz Here the auxiliary equations are dx dy dz = = x2 −y2 −z2 2xy 2xz Taking the last two members, we have dy dz = y z which on integration gives log y = log z + log a or y y log = log a or = a · · · (1) z z 9 / 10 EV0000336 · EV0000337 · EV0000338

10. Source slide page 10 (extracted text): Solve (x2 −y2 −z2)p + 2xyq = 2xz (contd.) Using x, y, z as multipliers, we get Each fraction x dx + y dy + z dz dz = = x(x2 + y2 + z2) 2xz or 2x dx + 2y dy + 2z dz dz = x2 + y2 + z2 z which on integration gives log(x2 + y2 + z2) = log z + log b or x2 + y2 + z2 x2 + y2 + z2 log = log b or = b · · · (2) z z From (1) and (2), the general solution is x2 + y2 + z2 y ϕ , = 0 z z 10 / 10 EV0000339 · EV0000340 · EV0000341

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:04.836] Hello everyone. Welcome back to the course on Engineering Mathematics 1. In this session, we will solve problems on Lagrange's linear equation. [00:00:20.676] So from theory we understood that the Lagrangius linear equation is given by capital B of XYZ times small p that is dou z by doux plus q of xyz [00:00:42.532] Q equals R of XYZ. [00:00:49.604] So in order to solve Lagrangius equation we will use actual equation which is nothing but dx divided by capital B. So dy divided by q and az divided by r [00:01:08.388] So we can use either grouping or multipliers or both or in some special cases we use exact differential concept also. [00:01:20.132] On noting these idea, let us solve problems on Lagrangian's linear equation. Let us look at uh yz square divided by x times small p plus x q equals y square. [00:01:38.532] So this is Lagrangius linear equation, so where this is uh capital P, this is uh capital Q, and this is R. After the selection of PQR, we write the acceler equations. [00:01:54.532] That is dx divided by y square z over x, dy by x and dz by y square. In order to make simplification easy, let us simplify the first term. [00:02:09.188] to be x times dx divided by y square z equals dy by x z and dz by y square. [00:02:20.388] So we have to think uh which method is appropriate after writing such a equation. Should we go for grouping or should we go for multipliers or can go for both. [00:02:34.116] When we observe these uh actual equations, we

can solve this problem by method of grouping because we can separate the variables and uh we can perform integration. [00:02:45.892] So let's take let us take uh first two fractions uh that is uh x dx uh divided by y square z equals dy and xz. So after cancellation of uh this z and this z and cross and multiplication or after separating the variables. [00:03:04.996] we'll get x square dx and uh y square dy. So variables have been separated. So we can perform integration. So integration becomes uh x power 3 by 3 equals y power 3 by 3 plus c [00:03:20.580] And this is same as x power 3 minus y power 3 equals 3c, which is same as some constant a. So the first solution for this problem is going to be x power 3 minus y power 3 equals a. So this looks like u of xyz equals a. [00:03:39.844] Let us uh search one more solution. Again, when we'll observe this uh exer equation, we can go for grouping again. So if you take a look at this, uh if you take first and the last ratios. [00:03:54.628] We see that x dx divided by y square z equals dy dz divided by y square. After cancellation of uh y square and this y square, that leads to x into dx equals z time dz. Again variables have been separated. [00:04:15.204] So on integration uh we get x square by 2, then uh z square by 2 plus some constant c and this is same as x square minus z square equals 2 c and this is same as x square minus uh z square equals b and this is our another solution we have x y z equals b. [00:04:35.364] As you know from theory, uh once we find out uh these two linearly independent solutions, uh we write the general solution to be phi of u, comma v 0 so which is same as phi of x power 3 minus y power 3 and x square minus z square equals 0. [00:04:55.460] So in this case uh we took uh the first and second fractions and first and last fractions. If it is possible to take second and third, it is also allowed. So that could be another answer for this problem, if it is possible. So here phi is an arbitrary.

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.516] Let us take a look at the second problem. It is uh yzp minus xzq equals xy. So as we know, this is uh capital P, this is capital Q, and uh this is XY. So acceler equations are dx over yz. [00:05:36.452] Dy divided by minus x and dz by xy. Again we can easily decide we can go for method of grouping in order to find both the solutions. [00:05:53.700] So from the first and second fractions, uh we get dx by y equals uh dy by minus x. So once we separate variables that leads to x dx plus y dy zero. [00:06:08.164] After integration that becomes uh x square by 2 plus uh y square by 2 equal to some constant c1. That is same as x square plus y square equal to 2 c1. So that is same as this. [00:06:27.460] So 2k becomes C1. So this is the first solution for the problem. So to find the second uh solution again we can go for uh uh the first and third fractions uh that gives [00:06:41.636] Dx by Y Z and Dz by XY After cancellation of this y and this y variables are separated. So that gives DX by Z and Dz by X. So [00:07:00.036] This is same as [00:07:05.924] x into dx and z into dz. So x square by two equals z square by two plus some k2. So this is nothing but x square [00:07:23.140] minus z square equals uh 2k2 so which is same as another constant c2. So finally the general solution for this problem is uh phi of x square plus y square and uh z square minus z square equals 0. [00:07:42.948] So this is how we can apply method of grouping for a problem to solve Lagrangius linear equation. Let us uh take out another problem Z of Z square plus XY times p x minus q, everything is a product. [00:08:00.452] So unless we simplify, we will not able to say whether this is a grandious linear equation or not. So once we simplify we see that the equation reduces to uh Xz of Z square plus XY, then uh minus Yz of I mean this is uh capital P [00:08:20.228] This is going to be capital Q and this is going to be R. So while looking at the actual equation, if you go for first and two fractions after cancellation of uh Z square plus XY and this Z square plus XY we get

[00:08:38.084] uh dx by x equals dy by minus y. And you can cancel this z also. So on integration we get uh log x equals uh minus log y. [00:08:52.996] So for convenient purpose we can say constant to be log c1. So this is log x plus log y equals log c1. So this gives uh log of xy equals log c1. [00:09:10.564] So that is MS uh XY equals C1. [00:09:17.156] So now how do you find out second uh solution that is b equals to b let us take out the first and third fractions so we get the x power 3 dx and uh z power 3 plus xy z tim z. [00:09:35.268] So this is uh slightly different from uh last uh previous problems. If you take a look at right hand side, unless we solve for this xy, we cannot say this is purely a function of z. In order to do that, we can uh use the previous answer. [00:09:55.108] that is xy equal to c1 so when you replace x x y by c1 so we get this there so x power 3 dx equals z power 3 plus c1 z time dz now you can see the left hand side we have a x and dx on the right hand side we have z and dz so now you can integrate

EV0000050 · EV0000051 · EV0000052 · EV0000053

Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.276] So on integration we will get x power 4 by 4. This is uh z power 4 by 4, c1 is constant, z becomes z square by 2 plus another constant, namely c2. So this will be simplified to this x power 4 minus z power 4 plus 2 c1 equals c2. [00:10:35.044] So again we have to concentrate on C1. So again go back to C1, C1 is XY. So the final answer for this problem is going to be phi of XY and X power 4 minus Z power 4 minus 2 XYZ square equals 0. [00:10:52.708] So these are the problems on Lagrangius linear equation and uh to find out the solution we use uh the method of grouping. Thank you.

EV0000054 · EV0000055 · EV0000056 · EV0000057

Full transcript: 1418 normalized words, 40 timestamped segments, 667.989 seconds duration. Low-confidence segments flagged: 0.

EV0000046 · EV0000047 · EV0000048 · EV0000049

7 Lagranges Linear equation_continued

This lesson is indexed from the source video titled Lagranges Linear equation_continued. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000058 · EV0000059

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.460] Hello everyone. Welcome back to the course on Engineering Mathematics One. [00:00:11.236] In the last in the last class, we solved problems on Lagrangius linear equation using method of grouping. So in this session, we'll solve problems on Lagrangius linear equation by using the method of multipliers as well as [00:00:30.404] Differential concept. [00:00:39.556] Let us solve the problem Mz minus N y times douz by doux plus nx minus Lz times douz by dou y equals L y minus mx. As we can see this is uh capital P [00:00:58.564] And this is small p and this is q and this is small q this is r after choosing p q r the axillary equations are dx over uh capital p, dy over capital q and dz by r [00:01:21.476] It is easy to see that we can't apply the method of grouping for this case because we are not able to

separate the variables. [00:01:31.460] In order to proceed with the method of multipliers, we need to choose uh element. They are either constants or functions of xyz. So to find the first solution, we choose a multipliers as x, y, and z. [00:01:48.932] Then each fraction of this expression 1 becomes x into dx plus y into dy plus z into dz. It means we multiply this dx by x, dy by y and dz by z. Even in the denominator, we multiply this capital P by X. [00:02:09.124] capital Q by Y and R by Z. So we can clearly see that all six terms in the denominator get cancel out. So that leads to X into DOX X into DX plus Y into Dy plus Z times Dz equals 0. [00:02:29.092] And it is easily integrable. So on integration we get x square by two plus y square by two plus z square by two equals some constant k1. [00:02:46.756] So this is same as x square plus y square plus z square equals 2k1. So we assume 2k1 to be some c1. Therefore, the first solution for this case is x square plus y square plus z square equals c1. [00:03:06.884] So this is like U of XYZ equals C1. So let us see how to find out another solution B. In a similar manner, we choose another set of multipliers, namely LMN. [00:03:24.036] In this case, again we get the denominator as 0. So this each fraction of 1 reduces to LDX plus Mdy plus N dz equals 0. LMN are concept. So on integration we get [00:03:41.124] Lx plus MY plus Lz equals C2. So this is the second solution. This is nothing but V of XYZ equals C2. So finally we are done with both the solutions, both independent solutions, namely U of XYZ equals C1. [00:04:00.164] and V of XYZ equals C2 and hence the general solution for the given problem is p of u comma v equals 0. So this is how we can apply method of multipliers for in order to solve Lagrangius linear equation. [00:04:18.564] Let's look at another problem. X of Z square minus Y square. So this is capital B. This is small p. [00:04:31.876] Y of x square minus z square, which is capital Q, this is small q, and this is r. So the axillary equations are dx over capital B, dy by q and dz by r. [00:04:48.132] Again, it is easy to see that we can't apply method of grouping, so we go for method of multipliers. By choosing multipliers as xyz, then each fraction of one becomes x into dx plus y into dy plus z into dz.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.660] Obviously in the denominator denominator all six term gets cancel out. So we get zero. So this implies x into dx plus y dy plus z zero on integration we get. [00:05:22.628] U of x y z equals c one to be x square plus y square plus z square equals c one. [00:05:31.524] So what is another set of multipliers? [00:05:38.180] It is easier to see from the given problem if uh x square if x is not there, you can cancel uh z square, or if y is not there and z is not there, you can cancel uh z square and minus z square and uh minus y square plus y square. [00:05:58.500] and uh plus x square and minus x square. So logically we can choose another set of multipliers to find the second solution that is 1 by x, 1 by y, and 1 by z. This is another set of multipliers for finding the second [00:06:18.628] which is nothing but v of x y z equals c2. So when we do that in the numerator we get uh dx by x plus uh dy by y plus dz by z equals zero so when we do integration uh it gets log log x plus log y plus log z [00:06:38.244] For convenient purpose, uh instead of C2 we write log because all terms on the left hand side containing log. So log a plus log b plus log c becomes log z becomes log x y z plus log c2. [00:06:56.004] then xy z becomes c2. So this is the second solution for the given problem and hence the general solution is given by phi of x square plus y square plus z square and xyz equals zero. [00:07:13.028] So in these two problems we applied only multipliers to find both the solutions. [00:07:20.868] Let's take up another problem y square plus z square of p minus x y q plus zx equals 0. So before choosing capital P, capital Q, capital R, we need to shift this last term to right hand side so it becomes minus ZX. [00:07:40.036] So which is nothing but uh R value. So

based on this uh the axillar equations are dx by y square plus z square and dy by minus xy and dz by minus zx. [00:07:55.076] It is very easy to see that to find the first solution we can go for second and third ratio. So that gives dy by y and uh t z by dz after cancellation of uh minus x and this minus x. So on integration we get uh y by z equals c1 because we get uh log y. [00:08:17.060] equals log z plus log c one. So you can say log y minus log z equals log c one. Log a minus log b is uh log a by b so based on that we get y by z equals c one. [00:08:37.220] So no more grouping. So we are going for method of multipliers. So in this problem we apply both method of grouping as well as method of multipliers. [00:08:53.380] So choosing multipliers as xyz, so as usual denominator becomes zero, so we can say x dx plus y dy plus z equals zero on integration as per previous problems we get uh x square plus y square plus square equal to c2. [00:09:12.420] And hence the general solution is given by phi of y by z x square plus y square plus z square equals zero. [00:09:27.652] Let us take another problem x square minus y square plus I'm sorry minus z square of p plus 2xy q equals 2 zx. [00:09:40.932] So the accelerary equations are EX divided by x square minus y square minus z square equals dy by 2xy and z by 2x. Obviously by two by using the second and third fractions or ratios.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:00.516] We get uh dy by y and dz by z so on integration. So that gives y by z equals a. So what do we do for one more solution? [00:10:16.996] Should we go for grouping again or we can go for multipliers or we should go for differential concept. [00:10:26.052] So no more grouping. Even multipliers doesn't work. So let us go for differential concept. As you know from theory, this is similar to method of multipliers. So the difference is uh once we choose the multipliers. [00:10:46.756] So we make sure that so this L D X plus M D Y and N D Z divided by [00:11:03.492] LP plus MQ plus Nr. This is equated with a fraction of this is [00:11:14.884] dx by p or dy by q or dz by r in order to get the integral. It means once we achieve this, we're going to equate this either with the first one or second one or third one. Depends on the problem. [00:11:33.188] So let us see what happens to this problem when we use this concept. That is differential concept. So let us choose multipliers as xyz, then uh each fraction becomes uh x dx plus y dy plus z dz. [00:11:52.356] So it means uh how do we do simplification? So you should multiply this is something like x into dx plus y into dy. [00:12:05.284] plus z divided by so multiply this by x so that gives x power three minus uh xy square minus x z square multiply second by y so we'll get a two x y square [00:12:25.124] The last one by z we get 2x square. So this is nothing but uh so same numerator in the denominator we get uh x power 3 and minus 1 and plus 2. So this is uh xy square and similarly x z square. [00:12:45.444] Again same numerator. In the denominator you can take x common. So that gives uh x square plus uh y square plus z square. So this will be equated with a fraction of this x and equation. It could be first one, second one, or third one. [00:13:05.796] So unfortunately for this problem we cannot equate this with the first fraction because it doesn't have x. So we can equate this either with the second one or third one because it has got a single x here. [00:13:19.396] So assume that uh we are equating this with uh the last one. So after cancellation of this x and this x are multiplying or taking two on the left hand side. [00:13:38.052] So that gives a 2x dx plus 2y dy and 2z dz divided by x square plus y square plus z square equals z by z. We should observe carefully. So whatever we have on the left hand side, so we observe that the numerator. [00:13:57.348] is the exact differential of denominator. So what's the meaning of that? So this right hand side can be written as P of X plus Y square plus Z square. [00:14:12.580] In the denominator, as usual, we have uh x square plus y square plus z

square. On the right hand side we have uh dz by z. Observe the left hand side carefully, it is like the numerator is the exact differential of denominator. [00:14:29.828] So we do this uh to get an integral. So the integration of this uh gives log of x square plus y square plus z square and right hand side we get log z plus uh log b for convenient purpose because other two terms are log. [00:14:49.060] Therefore, if you use log a plus log b we get the solution as x square plus y square plus z square divided by z equals b. So this is the second solution for the given problem. So uh in this problem to get the first solution we use the as usual grouping method.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.996] To get the second solution we use the differential concept method. So I hope you got a clarity on solving problems on Lagrangius linear equation. Thank you.

EV0000070 · EV0000071 · EV0000072 · EV0000073

Full transcript: 1908 normalized words, 55 timestamped segments, 925.568 seconds duration. Low-confidence segments flagged: 0.

EV0000058 · EV0000059 · EV0000060 · EV0000061

8 Method of separation of variables

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Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000057. Method of separation of variables · source slide page 1.

EV0000343

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Contents



1 Method of Separation of Variables

Figure G000058. Method of separation of variables · source slide page 2.

EV0000346

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Method of Separation of Variables

- It is based on the idea that a solution can be written as a product of functions, each depending on only one independent variable
- Separation of variables provides closed-form solutions for many fundamental partial differential equations—the one-dimensional heat equation, the wave equation, and Laplace's equation—by reducing each PDE to a set of ordinary differential equations

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Figure G000059. Method of separation of variables · source slide page 3.

EV0000349

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Working Rule

1. For a P.D.E. in the function u of two independent variables x and y, assume that the required solution is separable, i.e.,
$$u(x, y) = X(x)Y(y), \tag{1}$$
where $X(x)$ is a function of x alone and $Y(y)$ is a function of y alone
2. Substitution of u from (1) and its derivatives reduces the P.D.E. to the form
$$f(X, X', X'', \ldots) = g(Y, Y', Y'', \ldots) \tag{2}$$
which is separable in X and Y
3. Since the L.H.S. of (2) is a function of x alone and the R.H.S. of (2) is a function of y alone, then (2) must be equal to a common constant, say k

Figure G000060. Method of separation of variables · source slide page 4.

EV0000352

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Working Rule (contd.)



1. Thus (2) reduces to

$$f(X, X', X'', \ldots) = k \tag{3}$$

$$g(Y, Y', Y'', \ldots) = k \tag{4}$$

2. The determination of a solution to P.D.E. reduces to the determination of solutions to two O.D.E.s (with appropriate conditions).

Figure G000061. Method of separation of variables · source slide page 5.

EV0000355

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Example

Use the separation of variables technique to solve

$$3u_x + 2u_y = 0 \quad \text{with} \quad u(x,0) = 4e^{-x}.$$

Solution: Assume $u(x,y) = X(x)Y(y)$.
Then the P.D.E. becomes $3X'Y + 2XY' = 0$ or

$$\frac{X'}{X} = -\frac{2}{3}\frac{Y'}{Y} = k \quad \text{(constant)}$$

Solving $X' - kX = 0$, we get $X(x) = c_1 e^{kx}$.
Similarly, $Y' + \frac{3}{2}kY = 0$, so $Y(y) = c_2 e^{-\frac{3}{2}ky}$.
So $u(x,y) = X(x)Y(y) = c_1 e^{kx} \cdot c_2 e^{-\frac{3}{2}ky} = ce^{kx}e^{-\frac{3}{2}ky} = ce^{k(x-\frac{3}{2}y)}$.
Given that $4e^{-x} = u(x,0) = X(x)Y(0) = ce^{kx}$,
Thus $c = 4$, $k = -1$.
Hence, the required solution is

$$u(x,y) = 4e^{-\frac{1}{2}(2x-3y)}$$

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Figure G000062. Method of separation of variables · source slide page 6.

EV0000358

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Example

Use separation of variables to solve

$$5\,u_x \;-\; 4\,u_y \;=\; 0, \qquad u(x,0) = 3e^{2x}.$$

Solution. Assume $u(x,y) = X(x)\,Y(y)$. Then,
$u_x = X'(x)Y(y), \quad u_y = X(x)Y'(y)$, so $5X'Y \;-\; 4X\,Y' = 0$
$\Longrightarrow \; \frac{X'}{X} = \frac{4}{5}\,\frac{Y'}{Y} = k.$
Solve ODEs.

$$X'/X = k \;\Rightarrow\; X(x) = A\,e^{kx},$$
$$\frac{Y'}{Y} = \frac{5}{4}k \;\Rightarrow\; Y(y) = B\,e^{\frac{5}{4}k\,y}.$$

Combine.

$$u(x,y) = AB\,e^{kx}\,e^{\frac{5}{4}k\,y} = C\,e^{k\,(x+\frac{5}{4}y)}.$$

Initial condition.

$$u(x,0) = C\,e^{kx} = 3e^{2x} \;\Longrightarrow\; C = 3,\; k = 2.$$

Final answer. $u(x,y) = 3\,e^{2x+\frac{5}{2}y}$.

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Figure G000063. Method of separation of variables · source slide page 7.

EV0000361

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000342 · EV0000343 · EV0000344
2. Source slide page 2 (extracted text): Contents 1 Method of Separation of Variables 2 / 7 EV0000345 · EV0000346 · EV0000347
3. Source slide page 3 (extracted text): Method of Separation of Variables It is based on the idea that a solution can be written as a product of functions, each depending on only one independent variable Separation of variables provides closed-form solutions for many fun- damental partial differential equations—the one-dimensional heat equation, the wave equation, and Laplace's equation—by reducing each PDE to a set of ordinary differential equations 3 / 7 EV0000348 · EV0000349 · EV0000350

4. Source slide page 4 (extracted text): Working Rule 1 For a P.D.E. in the function u of two independent variables x and y, assume that the required solution is separable, i.e., u(x, y) = X(x)Y (y), (1) where X(x) is a function of x alone and Y (y) is a function of y alone 2 Substitution of u from (1) and its derivatives reduces the P.D.E. to the form f(X, X′, X″, . . .) = g(Y, Y ′, Y ″, . . .) (2) which is separable in X and Y 3 Since the L.H.S. of (2) is a function of x alone and the R.H.S. of (2) is a function of y alone, then (2) must be equal to a common constant, say k 4 / 7 EV0000351 · EV0000352 · EV0000353

5. Source slide page 5 (extracted text): Working Rule (contd.) 1 Thus (2) reduces to f(X, X′, X″, . . .) = k (3) g(Y, Y ′, Y ″, . . .) = k (4) 2 The determination of a solution to P.D.E. reduces to the determi- nation of solutions to two O.D.E.s (with appropriate conditions). 5 / 7 EV0000354 · EV0000355 · EV0000356

6. Source slide page 6 (extracted text): Example Use the separation of variables technique to solve 3ux + 2uy = 0 with u(x, 0) = 4e−x. Solution: Assume u(x, y) = X(x)Y (y). Then the P.D.E. becomes 3X′Y + 2XY ′ = 0 or X′ = −2 Y ′ = k (constant) X 3 Y Solving X′ −kX = 0, we get X(x) = c1ekx. 2 ky. Similarly, Y ′ + 32kY = 0, so Y (y) = c2e−3 2 ky = cekxe−32 ky = cek(x−32 y).So u(x, y) = X(x)Y (y) = c1ekx · c2e−3 Given that 4e−x = u(x, 0) = X(x)Y (0) = cekx, Thus c = 4, k = −1. Hence, the required solution is u(x, y) = 4e−12 (2x−3y) 6 / 7 EV0000357 · EV0000358 · EV0000359

7. Source slide page 7 (extracted text): Example Use separation of variables to solve 5 ux −4 uy = 0, u(x, 0) = 3e2x. Solution. Assume u(x, y) = X(x) Y (y). Then, ux = X′(x)Y (y), uy = X(x)Y ′(y), so 5X′Y −4X Y ′ = 0 X′ 4 Y ′=⇒ = = k. X 5 Y Solve ODEs. X′/X = k ⇒X(x) = A ekx, Y ′ 5 5 k = ⇒Y (y) = B e 4 y. Y 4k Combine. 5 5 u(x, y) = AB ekx e 4 k y = C ek (x+ 4 y). Initial condition. u(x, 0) = C ekx = 3e2x =⇒C = 3, k = 2. 5 Final answer. u(x, y) = 3 e2x+ 2 y. 7 / 7 EV0000360 · EV0000361 · EV0000362

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.620] Hello everyone. Welcome back to the course on Engineering Mathematics One. [00:00:11.108] In last two classes, we learned how to solve problems on Lagrange's linear equation by using method of multipliers or method of grouping or both. [00:00:27.652] In some special cases, uh we also applied differential method. [00:00:36.100] So in this session, we will learn what is a method of separation of variables. How do we apply this for a problem? And what is its advantage. [00:00:55.332] This method is also called product method because we assume the solution of the given PDE in the form of a product is depending on only one independent variable. As an application, [00:01:14.532] Separation of variables provides closed form solutions for many fundamental partial differential equations, namely one dimensional heat equation, the wave equation, and the Laplace equation. [00:01:32.900] By reducing each PDE to a set of ordinary differential equations. [00:01:41.508] In a future class, we will see how to apply this method of separation of variables to find out the various possible solutions of one dimensional heat equation. Before that, we'll understand. [00:02:00.132] How to apply this method for a problem. [00:02:07.460] Let us assume that [00:02:11.620] U equals f of x y as usual u is dependent and x and y are independent variables. If the given PD contains x and y to be independent variables, we need to assume [00:02:28.388] the solution of the given PDE in the form of a product that is u equals x of x times y of y where x is a function of x only and this is a function of y only. [00:02:48.452] After we assume the solution of the given problem in this manner, we need to substitute this in a given PDA. Then the partial derivatives present in the equation

should be. [00:03:05.796] reduced to ordinary derivatives. So how do you do that? Suppose we have dou u by dou x in the problem. So once we replace u to be product of capital X capital Y. [00:03:25.604] Assuming u is a function of x and y. So this is even as we can take y c outside and dou x by doux can be written as dx by dx. So this is how we can convert. [00:03:43.492] ordinary partial derivative into ordinary derivative in a similar manner if we have dou u by dou y [00:03:58.788] Because y is x y, this can be written as in this case we take x outside and written as d capital y divided by d small y. So this is how we convert the partial derivative. [00:04:17.028] In terms of ordinary derivative. [00:04:21.732] Or the partial derivatives are reduced to ordinary derivative in this manner. [00:04:28.420] After we do this again we substitute these in the given problem, then we have to make sure that the LHS is a function of X only and RHS is a function of Y only or vice versa. In other words [00:04:49.092] The expression containing ordinary derivatives should be rearranged in such a way that left hand side is a function of x only and rhs is a function of y only. That's what you can see in expression number two.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.564] After we have this scenario that is f of x equals g of y, so mathematically they both have some common constant k. Therefore, we equate both sides to some common constant k. [00:05:24.260] Let us say that f of x equals k as well as g of y equals k. Remember, this is nothing but ordinary derivatives. [00:05:40.644] We need to solve these two by known methods. So it could be variable separable or it could be linear equation or homogeneous or exact. We solve these two for capital X and capital Y. [00:05:58.340] So these two are the solutions of the given ordinary differential equation. It means the one what you got from step number two. So finally, the solution of the given PDE becomes U equals capital X into capital Y. [00:06:14.628] So the solution what we found from the previous step should be substituted back to the assumption. So that gives the solution for the given PDE. So this is how we apply method of grouping, sorry, uh method of separation of variables for a problem. [00:06:33.124] In order to understand this uh procedure, let's look at one or two examples. [00:06:42.660] Use the method of separation of variables to solve this partial differential equation, which is nothing but three times dou u by doux plus two times. [00:07:01.732] dou u by dou y equals zero so conditions are u of x comma zero equals four of e power minus x so it is not necessary to have conditions like this we can apply this method for problems without this condition also [00:07:23.076] So let's see uh how do we use this condition in later stage? So when we observe a problem, uh obviously u is a function of x and y, so obviously we have to assume the solution in the form of a product. [00:07:39.492] That is u equal to x of x times y of y. Then we have to substitute this in a given PDE where the partial derivative should be converted into ordinary derivative. So it means uh as we know from the procedure. [00:07:58.884] This uh three dou by dou x uh becomes three capital y e capital X divided by D small X plus second term will be two times X into P y by [00:08:18.532] P y equals zero. So in order to make LHS is a function of X only and RHS is a function of Y only, we divide through what by XY. This is what we assume. So when we divide through what by XY, [00:08:36.868] So we get uh 3 divided by x after cancellation of y. Then dx by x. We just ship this right hand side. So minus uh 2 by y. Then we get uh dy by y. [00:08:53.380] So it can also be written as uh something like 1 by x. So dx by dx then uh minus uh 2 by 3 send 1 by y. So d capital y divided by dy. [00:09:12.292] So we can clearly see that left hand side is a function of X and RHS is a function of Y. [00:09:21.956] So we need to equate both to some common constant k. It means uh we write uh one by x by t x equals k. Another partition

minus uh two by three. [00:09:40.164] One by capital Y by DY equals k. So we need to solve these two using known methods for capital X as well as capital Y. [00:09:56.900] So obviously it is easy to see that uh this can be written as a dx by x equals uh k time dx. So on integration uh we get uh log x equals uh kx some

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:15.332] K1 constant for convenient purpose. Then X becomes uh C1 E power KX where C1 becomes e power K1. So this is the first solution. [00:10:32.996] so which is same as uh what you see here similarly when we solve this uh we get so this is uh dy by y so minus two by three becomes so minus three by two then k times dy [00:10:52.868] After integration it is log y equals minus 3 by 2 k y plus some constant k2. Therefore y equals c2 times e power minus 3 by 2 times y. [00:11:12.868] So this is nothing so KY. So this is nothing but what you see here. So these are the two solutions of the ordinary differential equations which we found here. So finally the solution for the given PDE will be. [00:11:32.516] U equals XY. So U equal to XY becomes take the product of these two. So C1 and C1. So you can call this as some A. E power uh A times E power B minus B, E power A minus B. You can take uh K. [00:11:52.068] outside so that is same as uh x minus three by two of y so in the problem if you don't have any conditions like this [00:12:07.876] Uh we can say we can stop here. That is u of x y to be the this is a solution for the given problem. So whenever we have a condition we need to proceed. Obviously, we need to find out uh uh a value as well as this k value. So let's see how do you find that. [00:12:25.732] So when we apply the condition, uh if you observe here 4e power minus 6 equal to u of x comma 0, it means wherever we have y, we put 0. So when we do that, so it actually leads to [00:12:45.444] four e power minus x equals so in this uh term take y to be zero so a e power k is only x [00:13:01.220] So obviously you compare A becomes four and K becomes minus one. So that's what we see here. Seven is substitute both in the given problem. I mean if you substitute both A and C K value. [00:13:16.708] We get this. So finally this is a solution for the given problem by using method of separation of variables. I think now you understood the meaning of this statement in the pro in the data. [00:13:33.732] It means uh by reducing each PDE to a set of ordinary differential equations as a each depending on only one independent variable. So it means uh u of x dot x dot y. [00:13:51.716] And this is how we can convert partial derivative into ordinary derivative. And this will answer for the given problem. [00:14:02.052] So to get a clarity on this method, let us solve one more problem, uh, which is nothing but five times dou by dou x minus four dou by dou y equals zero. And the condition is u of x. [00:14:18.852] comma zero equals three times e power two x. So as usual uh when we use the procedure so this is going to be [00:14:35.044] y capital Y capital X divided by D small X minus four then capital X capital Y divided by D small Y equals zero [00:14:54.852] So to eliminate y in the first and x in the second, we should divide through by the assumption, which is xy. So when we do that, we get uh phi by x, bx by bx equals uh

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:14.244] four by y by. So this can also be written as one by x dy by sorry d x by x equals four by phi [00:15:34.564] 1 by y py and this d y. So remember this is not compulsory to shift this phi to the right hand side. It is not compulsory. So we can uh do the problem uh you can solve this problem without taking phi on the right hand side. [00:15:55.908] So when we equate uh this left hand side to k and this right hand side to k, so what do we get? So that leads to dx by x equals uh k times dx. [00:16:12.612] So

which is same as uh log x equals uh kx plus some constant k1. Therefore x is uh some a1 k. So this is the first solution. [00:16:30.916] or this is a solution of this ordinary differential equation that is 1 by x dx by x. In a similar manner how do we solve second one? So when we solve we get a dy by y equals 4 by 5 becomes uh 5 by 4 [00:16:50.724] then k d y so on integration log y equals uh phi k by four plus some k2 [00:17:05.796] Software solving for Y. So we get uh some C two E power Pi by four then KY where C two is uh [00:17:24.356] that e power k to constant. So finally the solution for the given pd becomes uh as you can see e power k of x plus pi by four times y. So when we use a condition [00:17:41.476] We are equating this part to three times e power two x equals c as it is. Of course in our case, okay, you can call some c e power kx. When y is zero, so this part disappears. [00:18:00.676] So on comparing we get uh c equal to 3 and uh k equal to 2. So finally the solution for the given problem is 3 times e power 2x plus pi by 2 of y. So how did we get this after finding uh c and k. [00:18:19.652] uh it becomes u equals uh three e power this is actually two of [00:18:33.188] two of x plus pi by four times y. So this is uh three not one by three. So this is uh e power and uh [00:18:50.052] one by two. So this is two, this is uh four x plus five y. So this is one by two. So therefore three power this is two x plus five by two y. [00:19:09.060] So this is how we can uh solve to get the final answer. So in this session uh we understood what is the method of separation of variables, what happens, and we also learned how to solve problems using this method. [00:19:29.220] So in the next class, we apply this method to find out the various possible solutions of one dimensional heat equation. Thank you.

EV0000086 · EV0000087 · EV0000088 · EV0000089

Full transcript: 2199 normalized words, 71 timestamped segments, 1184.384 seconds duration. Low-confidence segments flagged: 0.

EV0000074 · EV0000075 · EV0000076 · EV0000077

9 Modeling of One dimensional Heat equation

This lesson is indexed from the source video titled Modeling of One dimensional Heat equation. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000090 · EV0000091

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000071. Modeling of one dimensional heat equation · source slide page 1.

EV0000385

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Contents



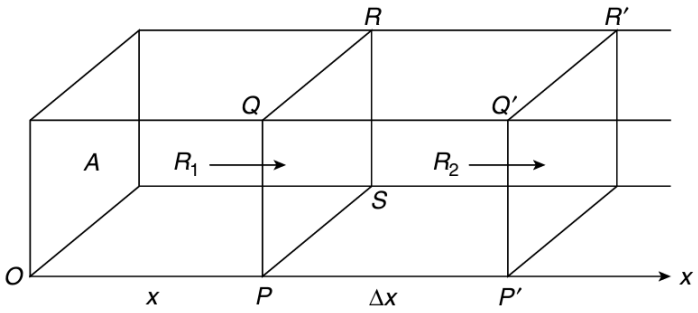
1 Applications of PDE – Modelling of the one-dimensional heat equation

Figure G000072. Modeling of one dimensional heat equation · source slide page 2.

EV0000388

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Modelling of the one-dimensional heat equation



Consider a long, thin bar (or rod) of constant cross-sectional area A and homogeneous conducting material. Let ρ be the density of the material, c be the specific heat, and k be the thermal conductivity of the material. We assume that the surface of the bar is insulated so that the heat flow is along parallel lines which are perpendicular to the area A.

Figure G000073. Modeling of one dimensional heat equation · source slide page 3.

EV0000391

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Choose one end of the bar as origin and the direction of heat flow as the positive x-axis. Let $u(x,t)$ be the temperature at a distance x from 0. If Δu be the temperature change in the slab of thickness Δx of the bar, and time change Δt, then the quantity of heat in this slab is

$$\begin{aligned}&\text{(specific heat)} \times \text{(mass of the element slab)} \times \text{(change in temperature)}\\ &\qquad = c\left(A\,\rho\,\Delta x\right)\Delta u.\end{aligned}$$

Hence, the rate of change (i.e., increase) of heat in the slab at time t is

$$c\big(A\rho\,\Delta x\big)\lim_{\Delta t\to 0}\frac{\Delta u}{\Delta t} = c\big(A\rho\,\Delta x\big)\frac{\partial u}{\partial t}.$$

Figure G000074. Modeling of one dimensional heat equation · source slide page 4.

EV0000394

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Let R_1 be the rate of inflow of heat at x in the slab and R_2 be the rate of outflow of heat at $x + \Delta x$. Then

$$c(A\rho\,\Delta x)\,\frac{\partial u}{\partial t} = R_1 - R_2, \tag{1}$$

where by Fourier's law

$$R_1 = -kA\left(\frac{\partial u}{\partial x}\right)_x \quad \text{and} \quad R_2 = -kA\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x}.$$

The negative sign reflects the fact that heat flows from regions of higher temperature to regions of lower temperature.

Figure G000075. Modeling of one dimensional heat equation · source slide page 5.

EV0000397

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Since $\dfrac{\partial u}{\partial t}$ is negative and R_1, R_2 are positive, the rate of increase of heat at time t is

$$R_1 - R_2 = kA\left[\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right]. \tag{2}$$

Combining (1) and (2), we get

$$c(A\rho\,\Delta x)\,\frac{\partial u}{\partial t} = kA\left[\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right].$$

Hence

$$\frac{\partial u}{\partial t} = \frac{k}{c\rho}\,\frac{\left(\frac{\partial u}{\partial x}\right)_{x+\Delta x} - \left(\frac{\partial u}{\partial x}\right)_x}{\Delta x}.$$

As $\Delta x \to 0$, the difference quotient becomes a second derivative,

$$\frac{\partial u}{\partial t} = \frac{k}{c\rho}\,\frac{\partial^2 u}{\partial x^2},$$

where $\dfrac{k}{c\rho}$ is the thermal diffusivity of the material.

Figure G000076. Modeling of one dimensional heat equation · source slide page 6.

EV0000400

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Setting

$$\alpha^2 = \frac{k}{c\rho},$$

we obtain the one-dimensional heat equation

$$\boxed{\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}.}$$

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Figure G000077. Modeling of one dimensional heat equation · source slide page 7.

EV0000403

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000384 · EV0000385 · EV0000386
2. Source slide page 2 (extracted text): Contents 1 Applications of PDE – Modelling of the one-dimensional heat equation 2 / 7 EV0000387 · EV0000388 · EV0000389
3. Source slide page 3 (extracted text): Modelling of the one-dimensional heat equation Consider a long, thin bar (or rod) of constant cross-sectional area A and homogeneous conducting material. Let ρ be the density of the material, c be the specific heat, and k be the thermal conductivity of the material. We assume that the surface of the bar is insulated so that the heat flow is along parallel lines which are perpendicular to the area A. 3 / 7 EV0000390 · EV0000391 · EV0000392
4. Source slide page 4 (extracted text): Choose one end of the bar as origin and the direction of heat flow as the positive x-axis. Let u(x, t) be the temperature at a distance x from 0. If Δu be the temperature change in the slab of thickness Δx of the bar, and time change Δt, then the quantity of heat in this slab is (specific heat) × (mass of the element slab) × (change in temperature) = c A ρ Δx Δu. Hence, the rate of change (i.e., increase) of heat in the slab at time t is Δu ∂u c Aρ Δx lim = c Aρ Δx . Δt→0 Δt ∂t 4 / 7 EV0000393 · EV0000394 · EV0000395

5. Source slide page 5 (extracted text): Let R1 be the rate of inflow of heat at x in the slab and R2 be the rate of outflow of heat at x + Δx. Then ∂u c Aρ Δx = R1 −R2, (1) ∂t where by Fourier's law ∂u ∂u R1 = −kA and R2 = −kA . ∂x x ∂x x+Δx The negative sign reflects the fact that heat flows from regions of higher temperature to regions of lower temperature. 5 / 7 EV0000396 · EV0000397 · EV0000398
6. Source slide page 6 (extracted text): ∂u Since is negative and R1, R2 are positive, the rate of increase of ∂t heat at time t is h ∂u ∂u i R1 −R2 = kA ∂x x+Δx − ∂x x . (2) Combining (1) and (2), we get ∂u h ∂u ∂u i c Aρ Δx = kA − . ∂t ∂x x+Δx ∂x x Hence ∂u − ∂u ∂u k ∂x x+Δx ∂x x = . ∂t cρ Δx As Δx →0, the difference quotient becomes a second derivative, ∂u k ∂2u = , ∂t cρ ∂x2 k where is the thermal diffusivity of the material. cρ 6 / 7 EV0000399 · EV0000400 · EV0000401
7. Source slide page 7 (extracted text): Setting k α2 = cρ, we obtain the one-dimensional heat equation ∂u ∂2u = α2 . ∂t ∂x2 7 / 7 EV0000402 · EV0000403 · EV0000404

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.780] Hello everyone. Welcome back to the course on Engineering Mathematics Month. [00:00:11.364] In this session, we will discuss the applications of partial differential equations. So let us understand how to model one dimensional heat equation. [00:00:28.388] So what is the heat equation? Heat equation is a fundamental partial differential equation in mathematics and physics that describes the distribution of heat in a given region over time. [00:00:49.252] Its mathematical formulation is given by dou u by dou t equals alpha square dou square u divided by dou x square. [00:01:07.364] So here u of x comma t is the temperature at a position x and at time t. And this alpha square is thermal diffusivity. It measures the rate at which [00:01:27.236] Heat spreads through a material. So this alpha square is called thermal diffusivity. So let's understand how to model this equation. [00:01:50.244] So this is a modeling of one dimensional heat equation. [00:01:58.276] Observe the diagram [00:02:02.948] Consider a long thin metal rod that has same cross section area that is A along its length. The rod is made of one uniform material that is [00:02:21.028] Every part of the rock is made of exact the same substance so that its properties do not change from place to place. [00:02:33.220] Let ρ be the density of the material, c be the specific heat of the material and k be the thermal conductivity of the material. [00:02:50.148] We have an assumption here. That is we assume that the rod's outer surface is perfectly insulated. It means heat can only travel along the length of the rod. Like this. [00:03:08.228] Let us choose one end of the rod to be origin O and the positive x axis has eight flow. [00:03:26.276] Let uh u of x comma t be the temperature at position x and at time t. So what we do is that uh take a look at uh two points on the rod, it is something like this. [00:03:47.460] So let let us take two points in the dot, uh one at position x and another at position x plus delta x [00:03:58.852] So that uh uh we can call this as a slab or slice. So the width of this slab is delta x. [00:04:10.532] So let us assume the temperature at X is uh U of X comma T, and the temperature at uh X plus delta X is going to be U of X plus delta X comma T. So we need to find out uh [00:04:28.708] The quantity of heat present in the slab. So the change in temperature across this little slab is going to be delta u equals u of x plus delta x comma t minus u of x, comma t. [00:04:49.700] So we need to find out the quantity of heat present in the slab of this rod. So if delta u be the temperature change in the slab of thickness delta x of the bar and change a time chain delta t.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:07.172] Then the quantity of heat present in the slab is going to be the product of specific heat, how much so it can store, and the mass of the element slab, and the change in temperature. Initially, we assume that uh C with the specific heat and uh [00:05:27.108] Change in temperature obviously delta u. So to find out the mass of the element slap, we need to use this formula that is density equals mass by volume. [00:05:42.948] So this implies mass equals density times volume. So density is taken as rho. [00:05:51.268] And this volume is nothing but a times delta x. This is what we have seen in the diagram. So that is why this mass is going to be a times rho time delta x. So obviously [00:06:06.724] rate of change of heat in the slab at time t equals c times a rho delta x uh we should take the limiting as delta t tends to zero so this gives w u by do t so this is nothing but [00:06:26.116] The rate of change of heat in the slab or slice at time t. Okay. [00:06:36.036] Let us discuss R1 and R2. Let R1 be the rate of inflow of heat at X and R2 be the rate of outflow of heat at del X plus delta X. So that the heat stored rate is obviously R1 minus R2. [00:06:56.868] So it means [00:07:00.868] We get R C times A rho delta x dou by dou t equals R1 minus R2. So when we use Fourier's law. So what is Fourier's law? In one dimension, Fourier's law says that the heat flow, heat flux. [00:07:20.996] that is uh heat per unit area per unit time is directly proportional to the negative of the temperature gradient so we get uh minus k times dou u by dou x at x so [00:07:38.052] When we multiply the flux by cross-sectional area that is capital A, we are going to get the total rate of heat flow through the phase. So and this negative sign is because heat flows from a higher temperature to the lower temperature. [00:07:58.628] So with respect to the diagram, uh this is a place where temperatures maximum. When you start moving from left to right, it is easy to see that temperature decreases. [00:08:16.324] So let us compute R1 minus R2, that is a rate of increase of heat at time t. So R1 minus R2 becomes uh KA dou by dou x at x delta x and uh dou u by doux at x. So when you combine one and two. [00:08:36.804] So this can be written as after cancellation of A and A and we can modify this to W by dou t let us shift C and row delta X to the right hand side. So we get this. So finally as delta X tends to 0. [00:08:56.772] So we know from chapter one. So this is going to be dou by dou x of dou u by dou x. So that is why the limit of this part as delta x tends to zero is [00:09:16.356] dou square u divided by dou x square. So finally we arrive at dou u by dou t equals k by c times rho then dou square u by dou x square where this constant it is easy to see that this is positive. [00:09:35.460] And this is called the thermal diffusivity of the material. So let us take this as some alpha square. So finally the one dimensional heat equation is going to be dou u by dou t equals alpha square. [00:09:53.444] those value divided by dou x square so where alpha square equals uh k by 0 is the thermal diffusivity of the material so it talks about heat spreads through a material so on so in this session uh

EV0000094 · EV0000095 · EV0000096 · EV0000097

Transcript block 3 (source interval 000600-000899 seconds): [00:10:13.380] We understood how to derive the expression for dou u by dou t equals dou square alpha square times dou square u by dou x square as a model. Thank you.

EV0000098 · EV0000099 · EV0000100 · EV0000101

Full transcript: 1156 normalized words, 38 timestamped segments, 628.5625 seconds duration. Low-confidence segments flagged: 0.

EV0000090 · EV0000091 · EV0000092 · EV0000093

10 Solution of one dimensional heat equation

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Graphs and diagrams

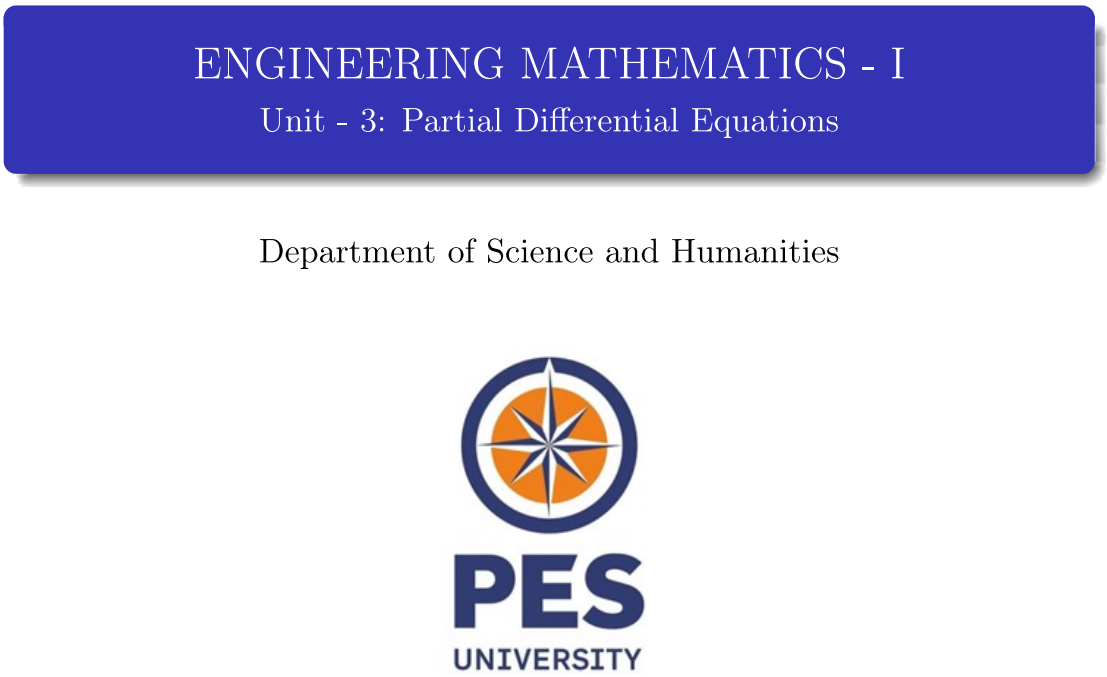


Figure G000078. Solution of one dimensional heat equation · source slide page 1.

EV0000406

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Contents



1 Applications of PDE – Solution of the heat equation by the method of separation of variables

Figure G000079. Solution of one dimensional heat equation · source slide page 2.

EV0000409

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Solution of the heat equation

The heat equation is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \tag{1}$$

Let

$$u = XT \tag{2}$$

where X is a function of x only and T is a function of t only, be a solution of (1).

Then

$$\frac{\partial u}{\partial t} = XT' \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} = X''T$$

Substituting in (1), we have

$$XT' = c^2 X''T$$

Separating the variables, we get

$$\frac{X''}{X} = \frac{1}{c^2}\frac{T'}{T} \tag{3}$$

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Figure G000080. Solution of one dimensional heat equation · source slide page 3.

EV0000412

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solution of the heat equation (contd.)

Now the L.H.S. of (3) is a function of x only and the R.H.S. is a function of t only. Since x and t are independent variables, this equation can hold only when both sides reduce to a constant, say k. The equation (3) leads to the ordinary differential equations

$$\frac{d^2X}{dx^2} - kX = 0 \qquad \text{and} \qquad \frac{dT}{dt} - kc^2T = 0 \tag{4}$$

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Figure G000081. Solution of one dimensional heat equation · source slide page 4.

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Solution of the heat equation (contd.)

Solving equations (4), we get
(i) When k is positive and $= p^2$, say

$$X = c_1 e^{px} + c_2 e^{-px}, \quad T = c_3 e^{c^2 p^2 t}$$

$$u = (c_1 e^{px} + c_2 e^{-px}) \cdot c_3 e^{c^2 p^2 t} \tag{5}$$

(ii) When k is negative and $= -p^2$, say

$$X = c_1 \cos px + c_2 \sin px, \quad T = c_3 e^{-c^2 p^2 t}$$

$$u = (c_1 \cos px + c_2 \sin px) \cdot c_3 e^{-c^2 p^2 t} \tag{6}$$

(iii) When $k = 0$

$$X = c_1 x + c_2, \quad T = c_3$$

$$u = (c_1 x + c_2) c_3 \tag{7}$$

Figure G000082. Solution of one dimensional heat equation · source slide page 5.

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Solution of the heat equation (contd.)

Thus the various possible solutions of the heat equation (1) are:

$$u = (c_1 e^{px} + c_2 e^{-px}) \cdot c_3 e^{c^2 p^2 t} \tag{5}$$

$$u = (c_1 \cos px + c_2 \sin px) \cdot c_3 e^{-c^2 p^2 t} \tag{6}$$

$$u = (c_1 x + c_2) c_3 \tag{7}$$

Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem. Since u decreases as time t increases, the only suitable solution of the heat equation is

$$u = (c_1 \cos px + c_2 \sin px) c_3 e^{-c^2 p^2 t}$$

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Figure G000083. Solution of one dimensional heat equation · source slide page 6.

EV0000421

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000405 · EV0000406 · EV0000407
2. Source slide page 2 (extracted text): Contents 1 Applications of PDE – Solution of the heat equation by the method of separation of variables 2 / 6 EV0000408 · EV0000409 · EV0000410
3. Source slide page 3 (extracted text): Solution of the heat equation The heat equation is ∂u ∂2u = c2 (1) ∂t ∂x2 Let u = XT (2) where X is a function of x only and T is a function of t only, be a solution of (1). Then ∂u ′ ∂2u = XT and = X′′T ∂t ∂x2 Substituting in (1), we have XT ′ = c2X′′T Separating the variables, we get X′′ 1 T ′ = (3) X c2 T 3 / 6 EV0000411 · EV0000412 · EV0000413
4. Source slide page 4 (extracted text): Solution of the heat equation (contd.) Now the L.H.S. of (3) is a function of x only and the R.H.S. is a function of t only. Since x and t are independent variables, this equation can hold only when both sides reduce to a constant, say k. The equation (3) leads to the ordinary differential equations d2X dT −kX = 0 and −kc2T = 0 (4) dx2 dt 4 / 6 EV0000414 · EV0000415 · EV0000416

5. Source slide page 5 (extracted text): Solution of the heat equation (contd.) Solving equations (4), we get (i) When k is positive and = p2, say X = c1epx + c2e−px, T = c3ec2p2t u = (c1epx + c2e−px) · c3ec2p2t (5) (ii) When k is negative and = −p2, say X = c1 cos px + c2 sin px, T = c3e−c2p2t u = (c1 cos px + c2 sin px) · c3e−c2p2t (6) (iii) When k = 0 X = c1x + c2, T = c3 u = (c1x + c2)c3 (7) 5 / 6 EV0000417 · EV0000418 · EV0000419
6. Source slide page 6 (extracted text): Solution of the heat equation (contd.) Thus the various possible solutions of the heat equation (1) are: u = (c1epx + c2e−px) · c3ec2p2t (5) u = (c1 cos px + c2 sin px) · c3e−c2p2t (6) u = (c1x + c2)c3 (7) Of these three solutions, we have to choose that solution which is consis- tent with the physical nature of the problem. Since u decreases as time t increases, the only suitable solution of the heat equation is u = (c1 cos px + c2 sin px)c3e−c2p2t 6 / 6
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SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.332] Hello everyone. Welcome back to the course on engineering mathematics one. [00:00:10.628] In previous classes, we understood what is the method of separation of variables. Why did we use method of separation of variables? We also solved some of the problems using method of separation of variables. [00:00:30.596] In this session also, we will find the various possible solutions of one dimensional heat equation by using method of separation of variables. [00:00:53.316] As in Open Theory, the one the dimensional heat equation is given by dou u by dou t equals c square times dou square u divided by dou x square. So when we observe this [00:01:13.988] So we can write this as u equals f of x and t. It means x and t are independent variables. So as the preliminary step, we will assume the solution of this PDE. [00:01:33.956] In the form of a product product that is u equal to x t where x is a function of x only and t is a function of t only. Then we need to substitute this assumption into the given. [00:01:54.724] So let us uh say the expression number one. So then equation one becomes Let us say dou by dou t of x t equals C square [00:02:14.308] dou square by dou x square of x t. After we substitute u equals x t into the given problem, we need to convert the partial derivatives present in this equation [00:02:32.484] as the ordinary derivatives so this is nothing but the left hand height left hand side can be written as x of dt by dt [00:02:47.716] Then uh C square the right hand side term can be written as uh t of d square x divided by dx square. [00:03:00.356] If you look at the LH system, we need to remove this X. If you look at the right hand side, we need we need to remove this T. So let us divide through what by capital X T, then that leads to 1 by T, T T by D T equals [00:03:20.900] C square by capital X, T square X divided by P x square. [00:03:30.020] This can also be written as 1 by C square t dt by dt equals 1 over x d square x divided by dx square. [00:03:48.260] Observe carefully. We have expressed the left hand side as a function of P and the right hand side as a function of X. So we can equate both sides to some common constant k. So let's do that. [00:04:07.844] So let us take the right hand side first. So this is uh 1 by x t square x divided by e x square equals to some common constant k. And similarly 1 by c square t. [00:04:25.668] D D by D T equals [00:04:35.652] So how do you solve the left hand side term? So this can be written as d square x divided by dx square equals kx. So using the operator this can be written as d square minus k. [00:04:54.852] of x equals zero where capital D stands for D by DX. In a similar manner. So this can be written as DT by dt equals k
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Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.148] C square capital T when you rearrange this is written as T minus K C square of E equals zero. So when you observe this, these are the two ordinary differential equations. [00:05:35.268] So that's what you see in equation number four. Okay so how do you proceed further? [00:05:45.572] Mathematically, K is an arbitrary real constant, but uh different signs of k gives different ordinary differential equations in X and D. So obviously uh it gives a different solutions [00:06:03.364] So now we consider three cases. Case number one, when k is zero, case number two, when k is positive, let us say p square, k is negative, let us say k equal to minus p square. [00:06:21.220] So obviously we get uh three different solutions. So let us look at case one when k 0. So when k 0 this equation 4 becomes p square x divided by a x square equals 0. [00:06:39.556] So this is same as uh B square X zero [00:06:47.812] When you write axelor equation that gives uh m square equals zero. It means uh m equals zero comma zero two times. In a similar manner, this uh second expression becomes [00:07:05.860] This is M as uh D of P equals zero. So when you write accelerary equation, it becomes m equal to zero. So then uh [00:07:21.636] What is the solution of these ordinary differential equations? If you observe X is a dependent and capital T is dependent, so we have to express capital X in terms of small X. As you know from second chapter, when the roots are repeated. [00:07:39.844] So this is how we write the general solution. So x equals uh c1 plus uh c2 x times e power 0x it is not shown. Similarly for 0, we can write uh the solution to be t equals just c3 times e power 0 t. [00:07:59.748] Therefore, what is the general solution for the PD? So as per the theory, we need to substitute this X and Y into the assumption. So the solution for the PD is given by U equals X T. So this is same as uh C1 plus uh C2X. Then [00:08:20.100] See. So this is the solution to the given problem when K is zero. Okay. In the later stage, uh we have to decide whether we accept this or not. We'll discuss this uh after the completion of the problem. Okay [00:08:43.492] So how do we find the solution when k is positive? That is when k equal to k n k is p square. So let me write over here. So when uh we take k to be positive, that is p square. [00:09:00.452] then uh this differential equation becomes d square x divided by e x square minus p square x zero. So this can be written as uh capital D square minus uh p square of [00:09:20.132] X equals 0. So acceler equation becomes M square minus P square 0. So the roots are M equals plus R minus P. So the roots are real and distinct. [00:09:37.540] Similarly from the second expression, we can have dt by t t minus k, which is p square, then c square t 0. This is same as t minus [00:09:58.020] P squared C square of P equals zero. Therefore accelerate equation becomes M equals sorry M minus uh P squared C square zero

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:15.876] So the single root is M equals P square C square. Again, how do we write the general solution when the roots are real or distinct? [00:10:33.732] So for M equal to plus R minus P, the general solution will be okay now for convenient purpose, uh because we have used C1, C2, C3, we will now use C1 dash, C2 dash C3 dash. [00:10:50.372] So for m equal to plus r minus p, the general solution is uh c one double dash e power px plus uh c two dash e power minus p x. [00:11:06.308] Similarly, for T, it is uh E power P square C square T. So these are the two solutions of ordinary differential equations. Therefore, the solution of the PDE is given by [00:11:25.252] U equals xp. So just write in the form of a product. So c1 dash e power p x plus c two dash e power minus px. Then e power [00:11:45.284] P square C square T. So this is the solution for the given problem when K is positive. [00:11:57.508] Again, we need to discuss after the end of the problem. Should we accept this or should we reject this? [00:12:08.388] Okay, let's move on to case number three. When k is negative, say minus b square.

As you can see, when we proceed based on case two, we get this. I mean to say when uh k [00:12:28.292] is negative that is uh k is minus p square [00:12:35.076] then the ordinary differential equations are going to be [00:12:43.364] So actual equations you'll get uh m square plus p square c square zero. So that gives uh m equals plus or minus i p. You see the roots are complex. So alpha zero, beta is p. [00:13:01.668] Similarly for the second expression uh we get m square plus uh p square equals zero. So this will give us uh sorry this is uh m m square so m equal to plus r minus i b. [00:13:24.260] sorry uh the first root will be m plus p square c square zero so this is only m equals uh minus p square c square so this is the root [00:13:37.988] So from the second we get uh m square plus p square zero, so we get m equal to plus or minus i p. So this is a complex root, alpha zero and uh beta is p. So this is a single root. So based on this root uh two roots. [00:13:55.012] We write the general solution to be this as well as this. So based on the previous case, uh we write the c1 double dash, c two double dash, and c three double dash. Okay, so we now understood uh all the three solutions. [00:14:14.724] When K is uh zero, K is positive, and K is negative. And that's what we see in expression number 567. So we got this when K is 0, C1 express C2 play into C3. [00:14:33.668] This is when a case positive C one dash, C two dash and C three dash. Similarly C one double dash, C two double dash and c three double dash. So observe five six seven. [00:14:48.644] These are the various possible solutions of one dimensional heat equation by using method of separation of variables. Now the task is to identify

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.580] Which solution is considered? In the sense Which solution is chosen, which solution is rejected. [00:15:17.764] So how do you do that? So basically in uh heat diffusion problems the temperature u should decrease as a time t [00:15:33.124] Increases. So based on this conclusion, we can choose which solution is accepted and which solution is rejected. If you look at this seventh one, the seventh one is independent of the [00:15:53.924] In other words, since u equals c1x plus uh c2 times c3 is independent of t, this is rejected. It means expression number seven, expression seven is not considered. What about equation number five? [00:16:12.932] Again you can see here as the time t increases this also increases it means as p increases u also increases therefore even the fifth one is also rejected. So look at expression number six. [00:16:31.972] And observe E power minus C square P square t. It is easy to see that e power minus C square P square t decays to zero. [00:16:45.988] As t tends to infinity. So that the temperature remains bounded and vanishes in the long run. So we can say that since u equal to c1 double dash cos p x plus and c2 c2 dash sine px and so on. [00:17:06.020] So this is acceptable solution since in this case [00:17:13.636] U decreases as the time t increases. That's what we see. Therefore, out of these three, the sixth one is considered. Okay, you can read this. [00:17:32.740] Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem. Since u decreases as time decreases, the only suitable solution of the 8th equation is the sixth one. I hope this concept is clear to you. [00:17:53.956] So in this session uh we found out the various possible solutions of one dimensional heat equation by using the method of separation of variables. We also understood out of these three solutions, which one is suitable. [00:18:14.084] Thank you.

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Full transcript: 1955 normalized words, 65 timestamped segments, 1097.5 seconds duration. Low-confidence segments flagged: 0.

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11 Higher order PDE

This lesson is indexed from the source video titled Higher order PDE. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000118 · EV0000119

Graphs and diagrams

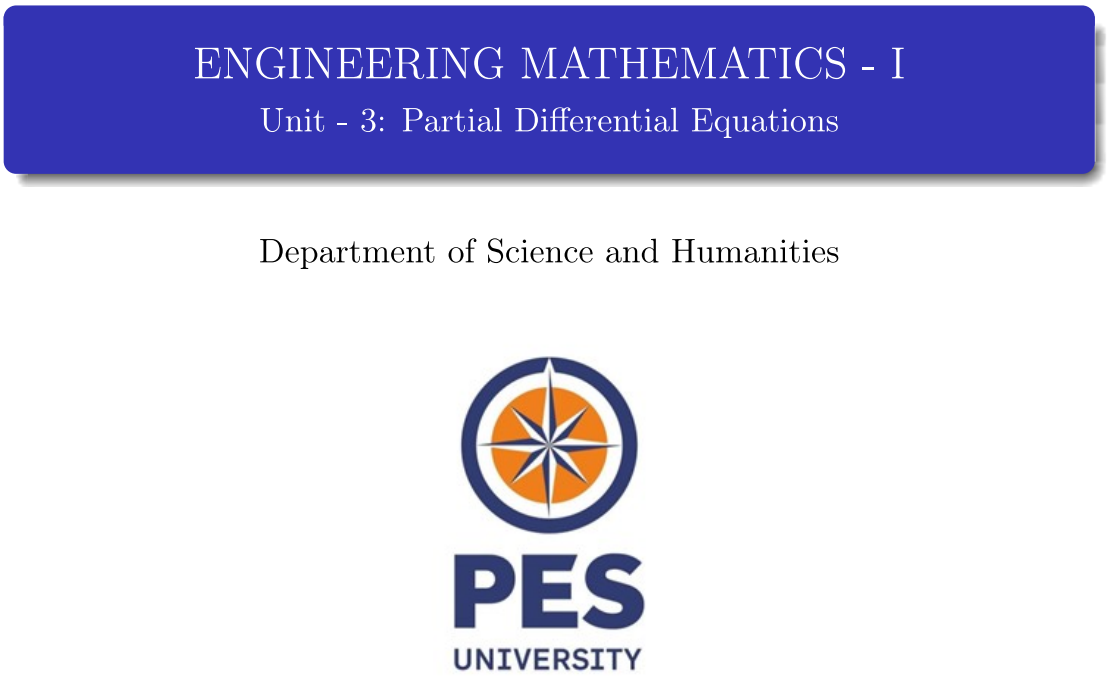


Figure G000084. Higher order PDE · source slide page 1.

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Contents



1 Solution of higher order Linear Partial Differential Equations with constant coefficients

Figure G000085. Higher order PDE · source slide page 2.

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Higher-Order Linear Equations with Constant Coefficients

- We have already learned how to solve higher-order linear ordinary differential equations with constant coefficients. These same methods can also be used to solve higher-order linear partial differential equations with constant coefficients
- For example, consider the following partial differential equation

$$\begin{aligned} A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1}\,\partial y} + A_2 \frac{\partial^n z}{\partial x^{n-2}\,\partial y^2} + \cdots \\ \cdots + A_{n-1} \frac{\partial^n z}{\partial x\,\partial y^{n-1}} + A_n \frac{\partial^n z}{\partial y^n} = f(x,y)\,. \end{aligned} \tag{1}$$

where A_i, $i = 0, 1, \ldots, n$ are constants.

Figure G000086. Higher order PDE · source slide page 3.

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Higher-Order Linear Equations with Constant Coefficients (contd.)

Denote

$$p = \frac{\partial z}{\partial x} = Dz, \quad q = \frac{\partial z}{\partial y} = D'z, \quad D^2z = \frac{\partial^2 z}{\partial x^2}, \quad DD'z = \frac{\partial^2 z}{\partial x \partial y}$$

etc.
Then, Eq. (1) can be written as

$$F(D, D')z = \sum_{r=0}^{n} A_r D^{n-r} (D')^r z = f(x, y). \tag{2}$$

Figure G000087. Higher order PDE · source slide page 4.

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Higher-Order Linear Equations with Constant Coefficients (contd.)



As in the case of ordinary differential equations, the general solution of Eq. (2) can be written as

$$z = (\text{complementary function}) + (\text{particular integral}),$$

where the complementary function is the solution of the homogeneous equation $F(D, D')z = 0$ and the particular integral is a solution satisfying the non-homogeneous equation (2) and does not contain any arbitrary constants.

Figure G000088. Higher order PDE · source slide page 5.

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Complementary Function

Consider the linear, homogeneous equation

$$F(D, D')z = 0. \tag{3}$$

If $F(D, D')$ can be factorised into linear factors of the form

$$(a_i D + b_i D' + c_i),$$

then $F(D, D')$ is said to be *reducible*. Otherwise, it is said to be *irreducible*

Figure G000089. Higher order PDE · source slide page 6.

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Complementary Function (contd.)

$F(D, D')$ may contain some reducible and some irreducible factors. We have the following examples.

$$(D^2 - D'^2)z = (D - D')(D + D')z = 0 \tag{4}$$

$$(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = (D - D')(D - 2D')(D - 3D')z = 0 \tag{5}$$

$$(2D^2 - D')z = 0 \tag{6}$$

$$(D^3 + D^2D' + D'^3)z = 0 \tag{7}$$

$$(D^3 - D^2D' + DD' - D'^2)z = (D - D')(D^2 + D'^2)z = 0 \tag{8}$$

Eqs. (4) and (5) are the reducible forms, Eqs. (6) and (7) are irreducible forms and Eq. (8) has one reducible and one irreducible factor.
Note that we will discuss only the equations which are in "reducible form".

Figure G000090. Higher order PDE · source slide page 7.

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Distinct Roots

Consider the solution of the partial differential equation with one linear factor, that is, the solution of

$$(a_i D + b_i D' + c_i)z = 0, \quad \text{or} \quad a_i p + b_i q = -c_i z. \tag{9}$$

The solution of Eq. (9) is given by

$$\boxed{z = e^{-(c_i x)/a_i}\, \phi_i(b_i x - a_i y)} \tag{10}$$

Figure G000091. Higher order PDE · source slide page 8.

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Distinct Roots (contd.)

Some particular cases:

a) If the linear factor is of the form $(a_iD + b_iD')$, then Eq.(10) simplifies to

$$\boxed{z = \phi_i(b_ix - a_iy)} \tag{11}$$

(b) If the linear factor is of the form $(b_iD' + c_i)$, then we have the differential equation as $(b_iD' + c_i)z = 0$ or $b_iq_i + c_iz = 0$. The solutions of the auxiliary equations are

$$b_ix = k_1 \quad \text{and} \quad z = k_2e^{-(c_iy)/b_i}$$

Hence, the solution of Eq. (9), when $a_i = 0$, is given by

$$k_2 = \phi_i(k_1), \quad \text{or} \quad ze^{(c_iy)/b_i} = \phi_i(b_ix)$$

or

$$\boxed{z = e^{-(c_iy)/b_i}\,\phi_i(b_ix)} \tag{12}$$

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Figure G000092. Higher order PDE · source slide page 9.

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Distinct Roots (contd.)

(c) Similarly, if the linear factor is of the form $(a_iD + c_i)$, then the solution of Eq.(9), when $b_i = 0$, is given by

$$\boxed{z = e^{-\frac{c_i x}{a_i}}\,\phi_i\big(a_i y\big)} \tag{13}$$

The complementary function is given by

$$z = e^{-\frac{c_1 x}{a_1}}\,\phi_1\big(b_1x - a_1y\big) + e^{-\frac{c_2 x}{a_2}}\,\phi_2\big(b_2x - a_2y\big) + \cdots + e^{-\frac{c_n x}{a_n}}\,\phi_n\big(b_nx - a_ny\big)$$

Similarly, the complementary function in the other cases can be written.

Figure G000093. Higher order PDE · source slide page 10.

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Distinct Roots (contd.)

Note: If all terms in $F(D, D')$ have the same total degree, then we can set $m = D/D'$ and look for factors of the equation using m.
Example: Find the solutions of the following PDE's



$$\begin{aligned}
&\text{(i)} && (D^2 - D'^2)z = 0,\\
&\text{(ii)} && (D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = 0,\\
&\text{(iii)} && 3r + 7s + 2t + 7p + 4q + 2z = 0,\\
&\text{(iv)} && 2r - s - t - p + q = 0.
\end{aligned}$$

(i) We have $(D^2 - D'^2)z = (D + D')(D - D')z = 0$.
For the factor $D - D'$, we have $a_1 = 1$, $b_1 = -1$.
For the factor $D + D'$, we have $a_2 = 1$, $b_2 = 1$.
Therefore, using Eq. (11), we obtain the solution as

$$z = \phi_1^*(-x - y) + \phi_2(x - y) = \phi_1(x + y) + \phi_2(x - y).$$

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Figure G000094. Higher order PDE · source slide page 11.

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Distinct Roots (contd.)

(ii) We have

$$(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z$$
$$\Rightarrow (D - D')(D - 2D')(D - 3D')z = 0.$$

For the factor $D - D'$, we have $a_1 = 1$, $b_1 = -1$.
For the factor $D - 2D'$, we have $a_2 = 1$, $b_2 = -2$.
For the factor $D - 3D'$, we have $a_3 = 1$, $b_3 = -3$.
If we set $(D/D') = m$, we get the equation as $m^3 - 6m^2 + 11m - 6 = 0$, whose roots are $m = 1, 2, 3$. Hence, the factors are $D - D'$, $D - 2D'$, $D - 3D'$.
Using Eq. (11), the solution can be written as

$$z = \phi_1^*(-x - y) + \phi_2^*(-2x - y) + \phi_3^*(-3x - y)$$
$$= \phi_1(x + y) + \phi_2(2x + y) + \phi_3(3x + y).$$

Figure G000095. Higher order PDE · source slide page 12.

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Distinct Roots (contd.)

The aforementioned note does not apply to the following two problems.

(iii) We have

$$(3D^2+7DD'+2D'^2+7D+4D'+2)z = (3D+D'+1)(D+2D'+2)z = 0.$$

For the factor $3D + D' + 1$, we have $a_1 = 3$, $b_1 = 1$, $c_1 = 1$.
For the factor $D + 2D' + 2$, we have $a_2 = 1$, $b_2 = 2$, $c_2 = 2$.
Using Eq. (10), the solution can be written as

$$z = e^{-x/3}\phi_1(x - 3y) + e^{-2x}\phi_2(2x - y).$$

Figure G000096. Higher order PDE · source slide page 13.

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Distinct Roots (contd.)

(iv) We have

$$(2D^2 - DD' - D'^2 - D + D')z = (D - D')(2D + D' - 1)z = 0.$$

For the factor $D - D'$, we have $a_1 = 1$, $b_1 = -1$, $c_1 = 0$.
For the factor $2D + D' - 1$, we have $a_2 = 2$, $b_2 = 1$, $c_2 = -1$.
Using Eqs. (10) and (11), the solution can be written as

$$z = \phi_1^*(-x - y) + e^{x/2}\phi_2(x - 2y)$$

$$= \phi_1(x + y) + e^{x/2}\phi_2(x - 2y).$$

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Figure G000097. Higher order PDE · source slide page 14.

EV0000463

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Multiple Factors

Let a factor be of multiplicity 2, that is, we have either of the factors

$$(a_iD + b_iD' + c_i)^2, \quad \text{or} \quad (a_iD + c_i)^2, \quad \text{or} \quad (b_iD' + c_i)^2$$

1. For the equation $(a_iD + b_iD' + c_i)^2 z = 0$, the general solution is given by

$$\boxed{z = e^{-(c_i x)/a_i}\,[x\,\phi_i(b_i x - a_i y) + \psi_i(b_i x - a_i y)]}$$

2. For the equation $(a_iD + c_i)^2 z = 0$, the general solution is given by

$$\boxed{z = e^{-(c_i x)/a_i}\,[x\,\phi_i(a_i y) + \psi_i(a_i y)]}$$

Figure G000098. Higher order PDE · source slide page 15.

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Multiple Factors (contd.)

1. For the equation $(b_iD' + c_i)^2 z = 0$, the general solution is

$$\boxed{z = e^{-(c_i y)/b_i}\left[y\,\phi_i(b_i x) + \psi_i(b_i x)\right]}$$

2. If the factor is of multiplicity m, say $(a_1D + b_1D' + c_1)^m$, then the general solution is

$$\boxed{z = e^{-\frac{c_1 x}{a_1}}\left[\psi_1(b_1x - a_1y) + x\,\psi_2(b_1x - a_1y) + \ldots + x^{m-1}\,\psi_m(b_1x - a_1y)\right]}$$

Similarly, we can write the general solution in the other cases.

Figure G000099. Higher order PDE · source slide page 16.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000423 · EV0000424 · EV0000425
2. Source slide page 2 (extracted text): Contents 1 Solution of higher order Linear Partial Differential Equations with constant coefficients 2 / 16 EV0000426 · EV0000427 · EV0000428
3. Source slide page 3 (extracted text): Higher-Order Linear Equations with Constant Coefficients We have already learned how to solve higher-order linear ordinary differential equations with constant coefficients. These same meth- ods can also be used to solve higher-order linear partial differential equations with constant coefficients For example, consider the following partial differential equation ∂nz ∂nz ∂nz A0 + A1 n−1 + A2 n−2 + · · · ∂xn ∂x ∂y ∂x ∂y2 (1) ∂nz ∂nz · · · + An−1 n−1 + An = f(x, y) . ∂x ∂y ∂yn where Ai, i = 0, 1, . . . , n are constants. 3 / 16 EV0000429 · EV0000430 · EV0000431
4. Source slide page 4 (extracted text): Higher-Order Linear Equations with Constant Coefficients (contd.) Denote ∂z ∂z ∂2z ∂2z p = = Dz, q = = D′z, D2z = , DD′z = ∂x ∂y ∂x2 ∂x∂y etc. Then, Eq. (1) can be written as n F(D, D′)z = X ArDn−r(D′)rz = f(x, y). (2) r=0 4 / 16 EV0000432 · EV0000433 · EV0000434

5. Source slide page 5 (extracted text): Higher-Order Linear Equations with Constant Coefficients (contd.) As in the case of ordinary differential equations, the general solution of Eq. (2) can be written as z = (complementary function) + (particular integral), where the complementary function is the solution of the homogeneous equation F(D, D′)z = 0 and the particular integral is a solution satisfying the non-homogeneous equation (2) and does not contain any arbitrary constants. 5 / 16 EV0000435 · EV0000436 · EV0000437

6. Source slide page 6 (extracted text): Complementary Function Consider the linear, homogeneous equation F(D, D′)z = 0. (3) If F(D, D′) can be factorised into linear factors of the form (aiD + biD′ + ci), then F(D, D′) is said to be reducible. Otherwise, it is said to be irre- ducible 6 / 16 EV0000438 · EV0000439 · EV0000440

7. Source slide page 7 (extracted text): Complementary Function (contd.) F(D, D′) may contain some reducible and some irreducible factors. We have the following examples. (D2 −D′2)z = (D −D′)(D + D′)z = 0 (4) (D3−6D2D′+11DD′2−6D′3)z = (D−D′)(D−2D′)(D−3D′)z = 0 (5) (2D2 −D′)z = 0 (6) (D3 + D2D′ + D′3)z = 0 (7) (D3 −D2D′ + DD′ −D′2)z = (D −D′)(D2 + D′2)z = 0 (8) Eqs. (4) and (5) are the reducible forms, Eqs. (6) and (7) are irreducible forms and Eq. (8) has one reducible and one irreducible factor. Note that we will discuss only the equations which are in "re- ducible form". 7 / 16 EV0000441 · EV0000442 · EV0000443

8. Source slide page 8 (extracted text): Distinct Roots Consider the solution of the partial differential equation with one linear factor, that is, the solution of (aiD + biD′ + ci)z = 0, or aip + biq = −ciz. (9) The solution of Eq. (9) is given by z = e−(cix)/ai ϕi(bix −aiy) (10) 8 / 16 EV0000444 · EV0000445 · EV0000446

9. Source slide page 9 (extracted text): Distinct Roots (contd.) Some particular cases: a) If the linear factor is of the form (aiD + biD′), then Eq.(10) simplifies to z = ϕi(bix −aiy) (11) (b) If the linear factor is of the form (biD′ + ci), then we have the differential equation as (biD′ + ci)z = 0 or biqi + ciz = 0. The solutions of the auxiliary equations are bix = k1 and z = k2e−(ciy)/bi Hence, the solution of Eq. (9), when ai = 0, is given by k2 = ϕi(k1), or ze(ciy)/bi = ϕi(bix) or z = e−(ciy)/bi ϕi(bix) (12) 9 / 16 EV0000447 · EV0000448 · EV0000449

10. Source slide page 10 (extracted text): Distinct Roots (contd.) (c) Similarly, if the linear factor is of the form (aiD + ci), then the solution of Eq.(9), when bi = 0, is given by z = e−cixai ϕi aiy (13) The complementary function is given by z = e−c1xa1 ϕ1 b1x −a1y + e−c2x a2 ϕ2 b2x −a2y + · · · + e−cnx an ϕn bnx −any Similarly, the complementary function in the other cases can be written. 10 / 16 EV0000450 · EV0000451 · EV0000452

11. Source slide page 11 (extracted text): Distinct Roots (contd.) Note: If all terms in F(D, D′) have the same total degree, then we can set m = D/D′ and look for factors of the equation using m. Example: Find the solutions of the following PDE's (i) (D2 −D′2)z = 0, (ii) (D3 −6D2D′ + 11DD′2 −6D′3)z = 0, (iii) 3r + 7s + 2t + 7p + 4q + 2z = 0, (iv) 2r −s −t −p + q = 0. (i) We have (D2 −D′2)z = (D + D′)(D −D′)z = 0. For the factor D −D′, we have a1 = 1, b1 = −1. For the factor D + D′, we have a2 = 1, b2 = 1. Therefore, using Eq. (11), we obtain the solution as z = ϕ∗1(−x −y) + ϕ2(x −y) = ϕ1(x + y) + ϕ2(x −y). 11 / 16 EV0000453 · EV0000454 · EV0000455

12. Source slide page 12 (extracted text): Distinct Roots (contd.) (ii) We have (D3 −6D2D′ + 11DD′2 −6D′3)z ⇒(D −D′)(D −2D′)(D −3D′)z = 0. For the factor D −D′, we have a1 = 1, b1 = −1. For the factor D −2D′, we have a2 = 1, b2 = −2. For the factor D −3D′, we have a3 = 1, b3 = −3. If we set (D/D′) = m, we get the equation as m3 −6m2 + 11m −6 = 0, whose roots are m = 1, 2, 3. Hence, the factors are D −D′, D −2D′, D −3D′. Using Eq. (11), the solution can be written as z = ϕ∗1(−x −y) + ϕ∗2(−2x −y) + ϕ∗3(−3x −y) = ϕ1(x + y) + ϕ2(2x + y) + ϕ3(3x + y). 12 / 16 EV0000456 · EV0000457 · EV0000458

13. Source slide page 13 (extracted text): Distinct Roots (contd.) The aforementioned note does not apply to the following two problems. (iii) We have (3D2+7DD′+2D′2+7D+4D′+2)z = (3D+D′+1) (D+2D′+2)z = 0. For the factor 3D + D′ + 1, we have a1 = 3, b1 = 1, c1 = 1. For the factor D + 2D′ + 2, we have a2 = 1, b2 = 2, c2 = 2. Using Eq. (10), the solution can be written as z = e−x/3ϕ1(x −3y) + e−2xϕ2(2x −y). 13 / 16 EV0000459 · EV0000460 · EV0000461

14. Source slide page 14 (extracted text): Distinct Roots (contd.) (iv) We have (2D2 −DD′ −D′2 −D + D′)z = (D −D′)(2D + D′ −1)z = 0. For the factor D −D′, we have a1 = 1, b1 = −1, c1 = 0. For the factor 2D + D′ −1, we have a2 = 2, b2 = 1, c2 = −1. Using Eqs. (10) and (11), the solution can be written as z = ϕ∗1(−x −y) + ex/2ϕ2(x −2y) = ϕ1(x + y) + ex/2ϕ2(x −2y). 14 / 16 EV0000462 · EV0000463 · EV0000464

15. Source slide page 15 (extracted text): Multiple Factors Let a factor be of multiplicity 2, that is, we have either of the factors (aiD + biD′ + ci)2, or (aiD + ci)2, or (biD′ + ci)2. 1 For the equation (aiD + biD′ + ci)2z = 0, the general solution is given by z = e−(cix)/ai [x ϕi(bix −aiy) + ψi(bix −aiy)] 2 For the equation (aiD + ci)2z = 0, the general solution is given by z = e−(cix)/ai [x ϕi(aiy) + ψi(aiy)] 15 / 16 EV0000465 · EV0000466 · EV0000467

16. Source slide page 16 (extracted text): Multiple Factors (contd.) 1 For the equation (biD′ + ci)2z = 0, the general solution is z = e−(ciy)/bi [y ϕi(bix) + ψi(bix)] 2 If the factor is of multiplicity m, say (a1D + b1D′ + c1)m, then the general solution is z = e−c1xa1 ψ1(b1x −a1y) + x ψ2(b1x −a1y) + . . . + xm−1 ψm(b1x −a1y) Similarly, we can write the general solution in the other cases. 16 / 16 EV0000468 · EV0000469 · EV0000470

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.748] Hello everyone. Welcome back to the course on engineering mathematics one. [00:00:13.572] In this session, we will discuss the solution of higher order linear partial differential equations with constant coefficients. [00:00:29.476] In second chapter, we understood how to solve homogeneous and non-homogeneous higher order linear ordinary differential equity constant coefficients. [00:00:46.340] Those methods can be extended to solve higher order linear partial differential equations with constant coefficients. [00:01:01.316] Let's look at the general linear partial differential equation of order n in two variables. That is aught times dou nz divided by dou x n plus [00:01:20.388] A1 times dou n z divided by dou x n minus 1 dot dou y and so on. And if you see on the right hand side we have f of x y. So in expression 1, a naught a1 a2 and up to a n, they're all constants. [00:01:40.228] So this left hand side can also be expressed in summation form, that is summation from r equal to 0 to infinity a suffix r and these partial derivatives can be expressed as [00:01:56.868] low power n minus r divided by low x power n minus r dot low power r divided by low y r time set equals [00:02:17.476] for x comma y and as we know that this can also be written as b power n minus r

and the second term can be written as b dash [00:02:34.212] Power R of Z equals f of x y. So finally expression one can be expressed in this form. Remember, from now onwards we use these notations. [00:02:52.868] Small p is nothing but douz by dou x, which is same as dz, q same as douz by dou y. This is nothing but d dash up z. And this is r value. [00:03:07.172] And this is S value. And similarly we have a T to be dou square Z divided by dou y square. This is nothing but D dash square of Z. So remember these notations. [00:03:31.204] Okay. We will now understand what is the general solution of this expression number two. Similar to the ordinary differential equations, the general solution of equal can be written as Z equals [00:03:49.572] complementary function that is uh C dot F plus particular integral PI Where complementary function is nothing but the solution of [00:04:03.460] Homogeneous equation that is f of d comma d dash z equals zero, that is left hand side, and the particular integral is a solution satisfying. [00:04:15.716] A non-homogeneous equity that is two and it doesn't contain any arbitrary constants. So remember this uh information. [00:04:27.908] So after understanding the general solution of expression two, we will learn how to compute complementary function for a PDE. [00:04:42.244] Before we understand how to compute complementary function, we need to know there is something called reducible factor and irreducible factor. So what is the meaning of this? [00:04:59.812] Note that if f of d, d dash can be factorized into linear factor in the form AID plus VID dash plus c i, then this f of b, d dash is said to be reducible.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:18.244] Otherwise irreducible. So throughout the chapter from now onwards, we need to look at this concept. Reducible factors. [00:05:33.188] So let us take some of the examples. So before that, remember f of b, d dash may contain some reducible and some irreducible factors. So let's look at the examples 4, 5, 6, 7, 8 equations. [00:05:53.540] Let's look at d square minus a d dash square. As you know, you can use uh a square minus b square formula. So that becomes a plus b times a minus b. So this is going to be d plus d dash and d minus d dash. This exactly looks like [00:06:12.452] AID plus BID dash and of course C0 it is also acceptable. If you look at the second example, this can be written as D minus D dash. [00:06:28.388] D minus 2d dash and d minus 3d dash up zero. So all these are reducible factors. Therefore we say that this f of d comma d dash up z 0 is reducible. [00:06:41.892] And what about the third example? You can clearly see that this is not reducible because you can't express this in the form of EID plus B ID dash plus CI. Even the third one is also. [00:07:00.708] not reducible factor because you can't express this in terms of AID plus BID dash plus ci [00:07:11.556] What about the fifth example? D power three minus d square d dash plus d d d dash minus d dash square can be written as [00:07:24.292] D minus d dash times d square plus d dash square up z zero. It is easy to see that uh from the first two we can take uh d square common and you get uh d minus d dash. [00:07:40.516] From third and fourth term you can take a d dash common, you will get d minus d dash. So that gives d minus d dash times d square plus d dash square z zero. So in these five examples. [00:07:59.268] The first two are reducible factors or reducible forms and equation six and seven are not reducible or irreducible forms. [00:08:11.620] And equation 8, as per the node given above, it has one reducible factor that is uh d minus d dash, and another one is irreducible factor that is d square plus d dash square. So this is how we can convert f of d comma d dash up set 0 to [00:08:30.308] Reducible factors. [00:08:34.084] So once we are able to achieve this, how to write a solution? So that's the next question. So there are two categories. Category number one, we may have distinct rules. Second category, we may have

repeated rules or factors. [00:08:55.300] Of course we call it as existing factors and repeated factors. Okay, assume that we are able to factorize f of p comma d dash z zero into [00:09:13.924] This form that is uh aid plus bid dash plus c i of z zero. So in this context, the solution is written in this form. So z equals e power minus c ix divided by a i times phi i. [00:09:32.804] of BIX minus a i y. So here phi is an arbitrary function. [00:09:40.580] So if C i is zero, obviously the first term becomes one. So let's look at the formulas where E is 0 and B I zero. [00:09:55.460] So as a first case, if C i is zero, this is how we write the solution. In a similar manner, if uh a i is zero, we write the solution in this form.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.692] In a similar manner, if Bi is 0, this is how we write the solution. So after understanding these three, we write the complementary function in this form. That is e power minus c1x by a1. [00:10:32.004] times phi 1 p1x minus a1y and so on. It depends on number of factors present in the problem. [00:10:42.468] So, in order to understand this concept, let's look at some of the problems. Let us take up these four examples. [00:10:56.484] If you look at d square minus d dash square of z zero, as we know from the example, this can be written as d plus d dash and d minus d dash of z zero. So when you compare this with uh [00:11:15.844] A I D plus B I D dash plus CI. It is easy to see that a1 is 1, B1 is minus 1. For the second factor, A2 is 1 and B2 is 1. [00:11:34.820] So in both the examples, CA value is 0. Therefore, we can use equation 11. So by using that, so that formula says yi of b i x minus a i y. [00:11:53.348] So logically speaking, we multiply b by x and and then multiply a by y. So this minus sign is common in the formula. So based on this idea, so we first get z equals phi one. [00:12:13.220] of if we multiply p1 by x we get minus x then minus y. Similarly for the second factor this is phi 2 of x minus y. So remember whenever we have this scenario [00:12:30.052] We write it as some phi one star of minus x minus y. If we take minus common, then that is written as phi one of x plus y. And there's no change in the second term that is phi two of x minus y. And we can call this a general solution because if you look at the problem. [00:12:49.892] The order of the PD is 2, and the number of arbitrary functions are also 2. So therefore we can call this as general solution. So let's look at the second example that is d power tree minus uh 6 d square dash 10 d dash. [00:13:10.244] Plus 11 D D dash square minus 6 d dash 3 of Z 0. So remember there is a shortcut method. So that shortcut method is applicable only when we have scenario like this. So what is that? If you observe [00:13:29.988] The total degree of each term is the same. [00:13:35.076] If you look at uh this power value, this is 3 and 2 plus 1 is 3, and 1 plus 2 is 3, and this is 3. So whenever we have this scenario, what we can do, we divide the equation by highest to power of d dash. So it means so we can divide this expression by [00:13:59.140] As D dash three. So what do we get? So that is same as okay let me solve it here. So this is same as uh divide through [00:14:14.980] by d dash power 3 so the first number is uh d by d dash power three then uh second number term is going to be b by d dash square [00:14:29.764] Then the third term is uh 11 d by d dash and the last one is six so z equals zero. So once we are able to write like this, then we put uh we use the standard substitution. [00:14:48.836] p by d dash equals m then that equation becomes m power three minus uh six m square plus eleven m minus six zero and you can see this is a cubic equation in m

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.436] So when we solve for the roots we see that uh m equal to 1, 2, 3. So once we find m values, we just substitute m value in

this expression. For example, when we write d by d dash equals 1. [00:15:21.956] we get d equal to d dash so it is same as d minus d dash so one factor is d minus d dash similarly when m is 2 we get uh d by d dash equals 2 so that is same as uh d minus 2 d dash so this is second factor [00:15:41.700] At last uh you can write d by d dash equals 3. So this is MS at D minus 3D dash. And you can see all these three are reducible factors. So finally what do we say? [00:15:55.236] So this given problem can be expressed in this form that is d minus d dash times d minus 2d dash and d minus t d dash. So observe carefully there are three factors. So in all the three factors, uh c i is 0. [00:16:13.604] So after finding A1, B one, A2, B2, and A3 C three. [00:16:19.716] We write the solution in this form. And you know very well when the first two terms are negative, we just introduce a star, then we can take minus common that is same as phi one of x plus y, phi two of two x plus y and phi three of the x plus y. So let's go back to the note. [00:16:40.484] If all terms in f of d gamma d dash have the same total degree, then we can set d by d dash equals m and then we find the factors of the equation using m. [00:16:58.340] So I hope this note is clear after looking at this second example. [00:17:07.460] Okay, it is important to remember [00:17:13.348] This uh shortcut method cannot be applicable for the third and fourth examples. Why? In third and fourth, the total degree of each term is not same. So we cannot apply shortcut method. Then how do we solve? [00:17:32.516] So the above mentioned note doesn't applicable to the following two problems because the total degree of each term is not same. Then uh how do you find out the factors? So that should be done by a normal inspection method. So let's see how do we do that? [00:17:52.676] So let's look at uh the problem 3d square plus 7 d dash plus 2d dash square plus 7 d plus 4d dash plus 2. [00:18:06.820] And the total degree is 2, this is 2, and third is 2, and this is 1, this is 1, this is 0. So how do you solve this? So we need to modify this. For example, uh in this case, we write uh the first term same. [00:18:26.340] 7 d d dash can be written as uh D six d d dash plus 1d dash. There's no change in uh two d dash square. Again this 7d can be written as 6d plus d. [00:18:46.564] Again this 4D dash is same as two d dash plus uh two d dash plus two of z right so if you observe there are nine terms [00:19:04.196] So this can be further written as if we take D common, you will get uh 3D plus uh e dash plus one, it means you're combining this, this, and this. And again, if you can take two d dash common. [00:19:24.196] So that leads to same 3D plus D dash plus 1. So finally if you look at 2D dash, 2 and the 6D. So let us take two common, we'll get uh 3D plus uh d dash plus one. [00:19:45.124] Z so from here you can get this again because three D plus D dash plus one. So the remaining terms are uh D plus two D dash plus two which we have here

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:04.516] And it is easier to see that the order of the PD is 2 and there are two factors here after finding a1 b1 c 1 and a2 b2 c2 and substitutes for equation. [00:20:22.372] we get this as a general solution e power minus x pi 3 phi 1 of x minus 3y plus e power minus 2x phi two of two x minus y. Here phi one and phi two are any arbitrary functions [00:20:40.676] Let's take another example. That is uh 2d square minus uh d d dash minus d dash square minus d plus d dash. Again you can observe [00:20:58.852] The total degree of each term is not same. So we cannot apply the shortcut. So this should be solved like the previous example. So what we can do, we keep uh 2d square as it is. [00:21:15.652] We can split dd dash to be minus two plus one and there's no change in the other three terms. So this is what we get in the next step. [00:21:34.308] If you look at odd places, you can take uh d common. So you will get 2d plus uh d dash minus 1. If you look at e1 places, uh take minus t dash common.

[00:21:49.956] We get 2D plus D dash minus 1 of Z 0. So it is easy to see that this is same as this. So there are two more factors. For the first factor, a1 is 1, B1 is minus 1, C1 is 0. [00:22:07.684] For the second factor, uh a2, b2, c2 are 2 1 minus 1 respectively. Because b1 is minus 1, we get a star that is same as phi one of x plus y. Then uh for the second because c is non-zero, we get a power uh x by 2 because. [00:22:26.468] C is minus 1 minus of minus that is plus and a2 is 2, that's why x by 2, by 2 of x minus 2 y. So this is the general solution for this problem. Okay We understood how to solve problems. [00:22:45.828] when we have distinct factors or different factors. So what if we have uh repeated factors or repeated uh [00:22:58.788] Multiple factors. Suppose assume that let a factor be of multiplicity too. You have taken two for uh understanding theory part. It could be three, four, and so on. [00:23:13.476] So we may have these three cases AID plus BID dash plus CI square. If uh BI0 we get this, if AI0 we get this. CI0 is a part of this. So we look at the three formulas. [00:23:31.620] When we have a factor in this form, so this is how we write the resolution. [00:23:39.908] When BI0, this is how we write the solution. Similarly, when BI0, and we have general solution in this one. So it is important to remember these formulae. So finally [00:24:00.164] Uh we write the general solution in this form. Similarly, we can write the general solution in other cases also. So in this session, uh we understood. [00:24:14.468] How to find the factors for the given problem and how to write a general solution based on distinct rules, distinct uh factors, as well as repeated factors. Thank you.

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Full transcript: 2890 normalized words, 89 timestamped segments, 1473.625 seconds duration. Low-confidence segments flagged: 0.

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12 Particular integral of type 1 and 2

This lesson is indexed from the source video titled Particular integral of type 1 and 2. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000138 · EV0000139

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000100. Particular integral of Type 1 and 2 · source slide page 1.

EV0000472

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Contents

1. Particular Integral when $f(x, y) = e^{ax+by}$
2. Particular Integral when $f(x, y) = sin(ax + b)$ or $cos(ax + b)$

Figure G000101. Particular integral of Type 1 and 2 · source slide page 2.

EV0000475

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Particular Integral when $f(x, y) = e^{ax+by}$

Consider the non-homogeneous equation

$$F(D, D')z = f(x, y)$$

The particular integral is now written as

$$z = [F(D, D')^{-1}]f(x, y)$$

Figure G000102. Particular integral of Type 1 and 2 · source slide page 3.

EV0000478

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When $f(x, y) = e^{ax+by}$

The particular integral in this case is

$$z = [F(D, D')^{-1}]e^{ax+by} = \frac{1}{F(a, b)}e^{ax+by}, \quad \text{if } F(a, b) \neq 0.$$

If $F(a, b) = 0$, then we write the particular integral as

$$z = \phi(x, y)e^{ax+by},$$

where

$$\phi(x, y) = [F(D + a, D' + b)]^{-1}(1).$$

We expand the operator $[F(D+a, D'+b)]^{-1}$ in an infinite series symbolically, and determine $\phi(x, y)$. Then, the particular integral is given by $z = \phi(x, y)e^{ax+by}$. Note that D^{-1} and $(D')^{-1}$ mean integral with respect to x and y respectively, keeping the other variable as constant

Figure G000103. Particular integral of Type 1 and 2 · source slide page 4.

EV0000481

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Find the general solution of the partial differential equation

$$[2D^2 - DD' - (D')^2 + D - D']z = e^{2x+3y}.$$

Solution: We write

$$F(D, D')\, z = \left[\, 2D^2 - D\, D' - (D')^2 + D - D' \right] z$$

$$= \left[(2D + D' + 1)(D - D')\right] z = e^{2x+3y}\,.$$

The homogeneous equation is

$$(2D + D' + 1)(D - D')z = 0$$

For the factor $D - D'$, we have $a_1 = 1$, $b_1 = -1$, $c_1 = 0$.
For the factor $2D + D' + 1$, we have $a_2 = 2$, $b_2 = 1$, $c_2 = 1$.
Therefore, the complementary function as

$$z = e^{-x/2}\phi_1(x - 2y) + \phi_2(x + y)$$

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Figure G000104. Particular integral of Type 1 and 2 · source slide page 5.

EV0000484

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Problem (contd.)

Since $F(2,3) = (4+3+1)(2-3) = 8 \times (-1) = -8 \neq 0$, we obtain the particular integral as

$$z = \frac{1}{F(2,3)} e^{2x+3y} = -\frac{1}{8} e^{2x+3y}.$$

Therefore, the general solution of the given differential equation is given by

$$z = e^{-x/2}\phi_1(x-2y) + \phi_2(x+y) - \frac{1}{8}e^{2x+3y}$$

Figure G000105. Particular integral of Type 1 and 2 · source slide page 6.

EV0000487

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Find a particular integral of the differential equation

$$(4D^2 + 3DD' - D'^2 - D - D')z = 3e^{(x+2y)/2}$$

Solution: We have

$$F(D, D') = 4D^2 + 3DD' - D'^2 - D - D'$$

and

$$F\left(\frac{1}{2}, 1\right) = 4\left(\frac{1}{4}\right) + 3\left(\frac{1}{2}\right)(1) - 1 - \frac{1}{2} - 1 = 0.$$

We write the particular integral as $z = \phi(x, y)e^{(x+2y)/2}$, where $\phi(x, y)$ is a function to be determined.
The given equation is: $F(D, D')[z] = 3e^{(x+2y)/2}$
$\Rightarrow F\left(D + \frac{1}{2}, D' + 1\right) \cdot \phi(x, y)e^{(x+2y)/2} = 3e^{(x+2y)/2}$
$\Rightarrow F\left(D + \frac{1}{2}, D' + 1\right) \cdot \phi(x, y) = 3$
Therefore,

$$\phi(x, y) = \left[F\left(D + \frac{1}{2}, D' + 1\right)\right]^{-1} (3)$$

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Figure G000106. Particular integral of Type 1 and 2 · source slide page 7.

EV0000490

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

$$F\left(D+\frac{1}{2},D'+1\right)=4\left(D+\frac{1}{2}\right)^2+3\left(D+\frac{1}{2}\right)(D'+1)-(D'+1)^2-$$
$$(D+\frac{1}{2})-(D'+1)$$

$$=6D-\frac{3}{2}D'+3DD'+4D^2-(D')^2$$
$$=6D\left[1-\frac{1}{4}D'D^{-1}+\frac{1}{6}\{3D'+4D-D'(D')^2\}\right]$$

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Figure G000107. Particular integral of Type 1 and 2 · source slide page 8.

EV0000493

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Problem (contd.)

Therefore,

$$\phi(x,y) = \left[F\left(D + \frac{1}{2}, D' + 1\right)\right]^{-1} (3)$$

becomes

$$\phi(x,y) = \frac{1}{2}D^{-1}\left[1 - \frac{1}{4}D'D^{-1} + \frac{1}{6}\{3D' + 4D - D'(D')^2\}\right]^{-1} (1)$$

$$= \frac{1}{2}D^{-1}(1) = \frac{x}{2}.$$

Hence, the particular integral is given by

$$z = \frac{x}{2}e^{(x+2y)/2}$$

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Figure G000108. Particular integral of Type 1 and 2 · source slide page 9.

EV0000496

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P.I. when $f(x, y) = \sin(ax + by)$, or $f(x, y) = \cos(ax + by)$

The particular integral in this case is

$$z = \frac{\sin(ax + by)}{\left[F\left(D^2,\, DD',\, (D')^2\right)\right]}$$

$$\Rightarrow z = \frac{\sin(ax + by)}{F(-a^2, -ab, -b^2)}, \quad \text{if } F(-a^2, -ab, -b^2) \neq 0.$$

Similarly, if $f(x, y) = \cos(ax + by)$, we get the particular integral as

$$z = \frac{\cos(ax + by)}{F(-a^2, -ab, -b^2)}, \quad \text{if } F(-a^2, -ab, -b^2) \neq 0.$$

If $F(-a^2, -ab, -b^2) = 0$, then the particular integral is found by using Case 4.

Figure G000109. Particular integral of Type 1 and 2 · source slide page 10.

EV0000499

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Find the particular integral of the differential equation

$$2\frac{\partial^2 z}{\partial x^2} - 3\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin(x - 2y)$$

Solution We have

$$F(D^2, DD', (D')^2)z = [2D^2 - 3DD' + (D')^2]z = \sin(x - 2y)$$

We have the right hand side as $\sin(ax + by)$, where $a = 1,\ b = -2$
Hence,

$$F(D^2, DD', (D')^2)\sin(x - 2y) = [2(-1) - 3(2) + (-4)]\sin(x - 2y)$$
$$= -12\sin(x - 2y)$$

The particular integral is given by

$$z = \frac{\sin(x - 2y)}{F(-a^2, -ab, -b^2)} = -\frac{1}{12}\sin(x - 2y)$$

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Figure G000110. Particular integral of Type 1 and 2 · source slide page 11.

EV0000502

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000471 · EV0000472 · EV0000473
2. Source slide page 2 (extracted text): Contents 1 Particular Integral when f(x, y) = eax+by 2 Particular Integral when f(x, y) = sin(ax + b) or cos(ax + b) 2 / 11 EV0000474 · EV0000475 · EV0000476
3. Source slide page 3 (extracted text): Particular Integral when f(x, y) = eax+by Consider the non-homogeneous equation F(D, D′)z = f(x, y) The particular integral is now written as z = [F(D, D′) −1]f(x, y) 3 / 11 EV0000477 · EV0000478 · EV0000479
4. Source slide page 4 (extracted text): When f(x, y) = eax+by The particular integral in this case is 1 z = [F(D, D′)−1]eax+by = if F(a, b) = 0. F(a, b)eax+by, If F(a, b) = 0, then we write the particular integral as z = ϕ(x, y)eax+by, where ϕ(x, y) = [F(D + a, D′ + b)]−1(1). We expand the operator [F(D+a, D′ +b)]−1 in an infinite series symbol- ically, and determine ϕ(x, y). Then, the particular integral is given by z = ϕ(x, y)eax+by. Note that D−1 and (D′)−1 mean integral with respect to x and y respectively, keeping the other variable as constant 4 / 11 EV0000480 · EV0000481 · EV0000482

5. Source slide page 5 (extracted text): Problem Find the general solution of the partial differential equation [2D2 −DD′ −(D′)2 + D −D′]z = e2x+3y. Solution: We write F(D, D′) z = 2D2 −D D′ − (D′)2 + D −D′ z = (2D + D′ + 1)(D −D′) z = e2x+3y . The homogeneous equation is (2D + D′ + 1)(D −D′)z = 0 For the factor D −D′, we have a1 = 1, b1 = −1, c1 = 0. For the factor 2D + D′ + 1, we have a2 = 2, b2 = 1, c2 = 1. Therefore, the complementary function as z = e−x/2ϕ1(x −2y) + ϕ2(x + y) 5 / 11 EV0000483 · EV0000484 · EV0000485

6. Source slide page 6 (extracted text): Problem (contd.) Since F(2, 3) = (4 + 3 + 1)(2 −3) = 8 × (−1) = −8 = 0, we obtain the particular integral as 1 z = = −1 F(2, 3)e2x+3y 8e2x+3y. Therefore, the general solution of the given differential equation is given by z = e−x/2ϕ1(x −2y) + ϕ2(x + y) −1 8e2x+3y 6 / 11 EV0000486 · EV0000487 · EV0000488

7. Source slide page 7 (extracted text): Problem Find a particular integral of the differential equation (4D2 + 3DD′ −D′2 −D −D′)z = 3e(x+2y)/2 Solution: We have F(D, D′) = 4D2 + 3DD′ −D′2 −D −D′ and 1 1 1 1 = 4 + 3 (1) −1 −1 −1 = 0. F 2, 4 2 2 We write the particular integral as z = ϕ(x, y)e(x+2y)/2, where ϕ(x, y) is a function to be determined. The given equation is: F(D, D′)[z] = 3e(x+2y)/2 ⇒F D + 2,1 D′ + 1 · ϕ(x, y)e(x+2y)/2 = 3e(x+2y)/2 ⇒F D + 12, D′ + 1 · ϕ(x, y) = 3 Therefore, −1 1 ϕ(x, y) = F D + 2, D′ + 1 (3) 7 / 11 EV0000489 · EV0000490 · EV0000491

8. Source slide page 8 (extracted text): Problem (contd.) 1 1 2 1 F D + 2, D′ + 1 = 4 D + 2 + 3 D + 2 (D′ + 1) −(D′ + 1)2− 1 (D + 2) −(D′ + 1) = 6D −3 + 3DD′ + 4D2 −(D′)2 2D′ = 6D 1 −1 + 1 + 4D −D′(D′)2} 4D′D−1 6{3D′ 8 / 11 EV0000492 · EV0000493 · EV0000494

9. Source slide page 9 (extracted text): Problem (contd.) Therefore, −1 1 ϕ(x, y) = F D + 2, D′ + 1 (3) becomes −1 1 ϕ(x, y) = 1 −1 + 1 + 4D −D′(D′)2} (1) 2D−1 4D′D−1 6{3D′ 1 x = 2D−1(1) = 2. Hence, the particular integral is given by x z = 2e(x+2y)/2 9 / 11 EV0000495 · EV0000496 · EV0000497

10. Source slide page 10 (extracted text): P.I. when f(x, y) = sin(ax + by), or f(x, y) = cos(ax + by) The particular integral in this case is sin(ax + by) z = F D2, DD′, (D′)2 sin(ax + by) ⇒z = if F(−a2, −ab, −b2) = 0. F(−a2, −ab, −b2), Similarly, if f(x, y) = cos(ax + by), we get the particular integral as cos(ax + by) z = if F(−a2, −ab, −b2) = 0. F(−a2, −ab, −b2), If F(−a2, −ab, −b2) = 0, then the particular integral is found by using Case 4. 10 / 11 EV0000498 · EV0000499 · EV0000500

11. Source slide page 11 (extracted text): Problem Find the particular integral of the differential equation 2∂2z −3 ∂2z + ∂2z = sin(x −2y) ∂x2 ∂x∂y ∂y2 Solution We have F(D2, DD′, (D′)2)z = [2D2 −3DD′ + (D′)2]z = sin(x −2y) We have the right hand side as sin(ax + by), where a = 1, b = −2 Hence, F(D2, DD′, (D′)2) sin(x −2y) = [2(−1) −3(2) + (−4)] sin(x −2y) = −12 sin(x −2y) The particular integral is given by sin(x −2y) z = = −1 sin(x −2y) F(−a2, −ab, −b2) 12 11 / 11 EV0000501 · EV0000502 · EV0000503

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.812] Hello everyone. Welcome back to the course on engineering mathematics one. [00:00:11.748] In the previous class, we understood how to find the general solution for the given partial differential equation where the right hand side is zero. In this session, we will learn [00:00:28.068] How to find the particular integral for the given partial differential equation where the right hand side is non-zero. So let's look at the case when we have f of x comma y to be e power axed b by [00:00:48.036] And f of x comma y equals sine of a x plus b y as well as cos of a x plus b y. [00:01:02.948] Okay.

[00:01:06.020] As we know that the general form of the non-homogeneous partial differential equation is given by f of b, d dash of z equals f of x comma y. It means this is not zero. [00:01:24.900] In this case, the formula for particular integral is given by z equals 1 over f of d comma d dash times f of x gamma y. That is z equals 1 by f of d comma d dash. [00:01:44.420] Then f of x comma y. So this is f of d comma d dash or minus one times f of x comma y. [00:01:56.772] So you need to remember this. So this is the formula for particular integral in the case of partial differential equation. [00:02:07.076] Well let's understand how to find out the particular integral when f of x comma y equals e power ax plus b y. [00:02:19.044] So as per the formula, we write the particular integral to be f of d comma d dash inverse e power x plus b y. So basically what we do is that we need to take up a and b from here and then [00:02:38.596] replace b by a and d dash by b so the particular integral becomes one over f of a comma b tim a power a x plus b y remember we have a condition here f of a comma b [00:02:59.044] Should be non-zero. Otherwise that leads to one by zero case. So when when we have if a comma b to be zero, then we write the particular integral in this manner. That is Z is written as [00:03:16.452] Some foyer of x comma y times e power a plus b y. Where this faya of x comma y is nothing but f of b plus a and d dash plus b. It means we have d and d dash. [00:03:34.788] So replace D by D plus A. So A can be chosen from here and replace D dash by D dash plus B and B can be taken from here. [00:03:47.716] And [00:03:50.500] Once we write foyer of x comma y to be f of b plus a uh d dash plus b we need to expand this as an infinite series. So once we find fo [00:04:06.276] We substitute back to Z equals phi of x comma y times e power a x plus b y. So this is a way to find out the particular integral when we have zero in the denominator. So we need to remember that in this process. [00:04:25.604] D inverse is nothing but integral with respect to x, keeping y constant, and d dash inverse is nothing but integral with respect to y, keeping x constant. So after understanding this process, [00:04:42.596] Let's take up one example. Find the general solution for the partial differential equation 2d square minus dd dash minus d dash square plus d.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.244] Minus D dash equals B power two x plus three y. It is easy to see that a values two and b values three. So let's compute the complementary function. So the given homogeneous. [00:05:20.772] differently uh partially different equation can be written as [00:05:26.532] 2D plus 1 and D minus D dash. So there are two factors here. So for the first factor, we can see this is A1 value. B1 is 1 and C1 is 1. For the second factor, A2 is [00:05:44.804] 1 b2 is minus 1 and c2 is 0. So based on this, we can write the general solution, so the complementary function in the form e power minus x pi 2 times phi one of x minus 2y. So this is from [00:06:03.556] 2D plus d dash plus one and d minus d dash we get uh phi two of x plus y. [00:06:13.988] Let us find out the particular integral. And uh we know that in the divine problem a values two and b value is three, when you substitute b s two and uh d dash to be three. [00:06:30.308] In the given problem that is f of d comma d dash, we see that this is going to be minus 8, so which is non-zero. And hence the particular integral for this particular case is given by minus 1 by 8 times e power 2x plus 3 by. [00:06:49.860] So finally the general solution for the even partial differential equation is given by e power minus x by two times phi one of x minus two y plus phi two of x plus y minus one by eight. [00:07:10.308] [ASR low confidence; check audio] e power 2x plus 3y. [00:07:14.500] Okay. [00:07:17.604] Let us take one more example. So where we find only the particular integral. So let's do that. So find the particular integral for the partial differential equation. [00:07:31.620] 4 d square plus 3 d dash minus d dash square minus d minus d dash of z

equals this. As you can see a value is uh one by two and b values two. And it is easy to see that. [00:07:48.708] When we replace d by half and uh d dash by two, so we get zero. So as you know from theory, when f of d comma d dash become zero, so this is the way we write z. That is phi of x comma y. [00:08:08.932] and e power x plus two y by two. So this is the right hand set of this is some phi of x comma y. So now we need to give of x comma. Okay so normally this is a way to find out [00:08:28.036] So, the one problem is f of d, d' of z equals this right hand side. [00:08:37.028] So as we know that we replace z by d plus a and d dash by d dash plus b so we get f of d plus half and d dash plus one and in place of z we have to substitute phi of x comma y e power x plus two y by two. So that's what we have here. [00:08:55.364] uh equals uh the answer problem that is three times e power x plus two y by two so we can cancel uh e power x plus two y by two and e power x plus two y by two so we end up f of d plus f of d plus half comma d dash plus one [00:09:14.660] times phi of x comma y equal to three and it is easy to see that phi of x comma y is nothing but three divided by this l edges so that becomes minus one inverse okay so before you find out uh phi of x comma y [00:09:34.596] Let us compute f of d plus half and d dash plus one. It means in a given problem replace d by d plus half and d dash by d dash plus one. So that leads to this one. So 6d minus 3 by 2 d dash and so on. [00:09:53.572] So if it takes d common, so that leads to 1 minus 1 by 4 d plus d inverse and so on. So this is the final form of f of d plus half and d dash plus 1. When you substitute this back to

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.004] uh this expression that is comma y so we have p in the numerator and uh we have six d in the denominator so that becomes uh uh one by two one by two d basically so one by two d inverse of [00:10:31.236] this f of d plus up and d dash plus one so when we use uh actual binomial series uh we see that so when we take this one inside so d dash d node that we so this part is zero and even this part is zero [00:10:49.188] So we get only 1 by 2, d inverse of 1. And we know from previous uh slide. So this is going to be simply x by 2. So what is d inverse we differentiate with respect to x. So that's why this is x by 2. [00:11:08.772] So when you substitute this in p x comma y, so this is going to be the particular d particular problem. So whenever f of d comma d dash is zero, we need to use this procedure. [00:11:26.724] Okay, let us take up uh the second case where uh f of x gamma y is either sine of a x plus b y or cos of ax b y. So remember in uh this particular case we expect [00:11:44.068] F comma d dash to contain d square, d d dash and d dash square. Then we take out a and b from the given problem, uh then replace d square by minus a square. [00:12:02.756] D dash by minus a b and a dash square by minus b square. So when you do that, obviously this denominator should be not zero, otherwise that leads to one by zero case. So the same method can be used even for cost of a x plus b y. So [00:12:22.436] In case we get denominator zero, that should be solved by general case, which we will see later. So when f of minus a square minus a b minus b square is zero, then the particular integral is found by using case number four. That is general case. [00:12:42.788] So let's take our example. Find the particular integral for the given partial differential equation. 2 times dough Z by doux square minus 3 dou square z by doux dou y plus dou square z by dou y square. [00:12:59.012] And you know the very well this is same as d square and this is same as the d d dash and this is sam as the dash square. Are the total degree same. So when we try to to find a particular integral [00:13:19.396] We see that. [00:13:23.076] When we replace d square by minus one square, we get minus one. D d dash by exactly minus two, so that become plus two here, then uh d dash square to be minus of minus two square, so that becomes

totally minus four. [00:13:43.076] So finally we see that denominator is non-zero. Is uh minus one by twelve sine of x minus two y. So that is why for this problem, the particular integral is given by minus one by twelve times sine of x minus two y. [00:14:02.276] So in this class uh in the session we learned how to find out the particular integral when the right hand side is e power x plus b y and sine of a x plus b y or cos of a x plus b y. Thank you.

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Full transcript: 1878 normalized words, 54 timestamped segments, 864.128 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:07:10.308

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13 Problems on finding the solution when RHS=0

This lesson is indexed from the source video titled Problems on finding the solution when RHS=0. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000150 · EV0000151

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities

PES UNIVERSITY

Figure G000111. solutions of PDE when RHS 0 · source slide page 1.

EV0000505

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Contents



1 Problems on finding the solution when RHS=0

Figure G000112. solutions of PDE when RHS 0 · source slide page 2.

EV0000508

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Problems on finding the solution when RHS=0

Find the solutions of the following partial differential equations

(i) $(D^4 - 2D^2D'^2 + D'^4)z = 0$,

(ii) $(4D^3 - 3DD'^2 + D'^3)z = 0$.

Solution

(i) We have

$$(D^4 - 2D^2D'^2 + D'^4)z = (D^2 - D'^2)^2 z = (D - D')^2(D + D')^2 z = 0.$$

For the factor $(D - D')^2$, we have $a_1 = 1$, $b_1 = -1$, $c_1 = 0$.
For the factor $(D + D')^2$, we have $a_2 = 1$, $b_2 = 1$, $c_2 = 0$.
If we set $m = D/D'$, we get $m^4 - 2m^2 + 1 = 0$. The roots are $m^2 = 1$, or $m = \pm 1$ which are double roots. Hence, the factors are $(D - D')^2$ and $(D + D')^2$.
Therefore, the general solution as

$$z = [x\phi_1^*(-x - y) + \phi_2^*(-x - y)] + [x\psi_1(x - y) + \psi_2(x - y)]$$

or

$$z = x\phi_1(x + y) + \phi_2(x + y) + x\psi_1(x - y) + \psi_2(x - y).$$

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Figure G000113. solutions of PDE when RHS 0 · source slide page 3.

EV0000511

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Problems (contd.)

(ii) We have

$$(4D^3 - 3DD'^2 + D'^3)z = (D + D')(4D^2 - 4DD' + D'^2)z$$

$$= (D + D')(2D - D')^2 z = 0.$$

For the factor $D + D'$, we have $a_1 = 1$, $b_1 = 1$, $c_1 = 0$.
For the factor $(2D - D')^2$, we have $a_2 = 2$, $b_2 = -1$, $c_2 = 0$.
The general solution as

$$z = \phi_1(x - y) + x\psi_1^*(-x - 2y) + \psi_2^*(-x - 2y)$$

or

$$z = \phi_1(x - y) + x\psi_1(x + 2y) + \psi_2(x + 2y).$$

Figure G000114. solutions of PDE when RHS 0 · source slide page 4.

EV0000514

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Problems (contd.)

3. Find the solutions of the following partial differential equation $(D^2 - DD' - 6D'^2)z = 0$

Solution: Let $m = \frac{D}{D'}$. Then the auxiliary equation becomes:

$$m^2 - m - 6 = 0.$$

Solving:

$$m = \frac{1 \pm \sqrt{1+24}}{2} = \frac{1 \pm 5}{2} \Rightarrow m = 3, -2.$$

- For $m = 3$: $a_1 = 1,\ b_1 = -3 \Rightarrow$ characteristic variable $-3x - y$,
- For $m = -2$: $a_2 = 1,\ b_2 = 2 \Rightarrow$ characteristic variable $2x - y$.

Hence, the general solution is:

$$z = \phi_1^*(-3x - y) + \phi_2(2x - y) = \phi_1(3x + y) + \phi_2(2x - y)$$

Figure G000115. solutions of PDE when RHS 0 · source slide page 5.

EV0000517

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problems (contd.)

4. Find the solutions of the following partial differential equations

(i) $(D^2 - 2DD' + D'^2)^2 z = 0,$

(ii) $(D^3 - 3D^2D' + 3DD'^2 - D'^3)z = 0.$

Solution: (i) We observe that

$$(D^2 - 2DD' + D'^2)^2 z = (D - D')^4 z = 0.$$

So the repeated factor is $(D - D')$, four times.
This corresponds to:

$$a = 1, \quad b = -1, \quad \text{(repeated root)}.$$

Hence, the general solution is:

$$z = x^3\phi_1(x + y) + x^2\phi_2(x + y) + x\phi_3(x + y) + \phi_4(x + y)$$

Figure G000116. solutions of PDE when RHS 0 · source slide page 6.

EV0000520

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Problems (contd.)

(ii) We notice that

$$(D^3 - 3D^2D' + 3DD'^2 - D'^3)z = (D - D')^3 z = 0.$$

So the repeated factor is $(D - D')$, three times.
Hence, the general solution is:

$$z = x^2\phi_1(x + y) + x\phi_2(x + y) + \phi_3(x + y)$$

Figure G000117. solutions of PDE when RHS 0 · source slide page 7.

EV0000523

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000504 · EV0000505 · EV0000506
2. Source slide page 2 (extracted text): Contents 1 Problems on finding the solution when RHS=0 2 / 7 EV0000507 · EV0000508 · EV0000509
3. Source slide page 3 (extracted text): Problems on finding the solution when RHS=0 Find the solutions of the following partial differential equations (i) (D4 −2D2D′2 + D′4)z = 0, (ii) (4D3 −3DD′2 + D′3)z = 0. Solution (i) We have (D4 −2D2D′2 + D′4)z = (D2 −D′2)2z = (D −D′)2(D + D′)2z = 0. For the factor (D −D′)2, we have a1 = 1, b1 = −1, c1 = 0. For the factor (D + D′)2, we have a2 = 1, b2 = 1, c2 = 0. If we set m = D/D′, we get m4 −2m2 + 1 = 0. The roots are m2 = 1, or m = ±1 which are double roots. Hence, the factors are (D −D′)2 and (D + D′)2. Therefore, the general solution as z = [xϕ∗1(−x −y) + ϕ∗2(−x −y)] + [xψ1(x −y) + ψ2(x −y)] or z = xϕ1(x + y) + ϕ2(x + y) + xψ1(x −y) + ψ2(x −y). 3 / 7 EV0000510 · EV0000511 · EV0000512

4. Source slide page 4 (extracted text): Problems (contd.) (ii) We have (4D3 −3DD′2 + D′3)z = (D + D′)(4D2 −4DD′ + D′2)z = (D + D′)(2D −D′)2z = 0. For the factor D + D′, we have a1 = 1, b1 = 1, c1 = 0. For the factor (2D −D′)2, we have a2 = 2, b2 = −1, c2 = 0. The general solution as z = ϕ1(x −y) + xψ∗1(−x −2y) + ψ∗2(−x −2y) or z = ϕ1(x −y) + xψ1(x + 2y) + ψ2(x + 2y). 4 / 7 EV0000513 · EV0000514 · EV0000515

5. Source slide page 5 (extracted text): Problems (contd.) 3. Find the solutions of the following partial differential equation (D2 −DD′ −6D′2)z = 0 Solution: Let m = D . Then the auxiliary equation becomes: D′ m2 −m −6 = 0. Solving: 1 ± √1 + 24 1 ± 5 m = = ⇒m = 3, −2. 2 2 For m = 3: a1 = 1, b1 = −3 ⇒characteristic variable −3x −y, For m = −2: a2 = 1, b2 = 2 ⇒characteristic variable 2x −y. Hence, the general solution is: z = ϕ∗1(−3x −y) + ϕ2(2x −y) = ϕ1(3x + y) + ϕ2(2x −y) 5 / 7 EV0000516 · EV0000517 · EV0000518

6. Source slide page 6 (extracted text): Problems (contd.) 4. Find the solutions of the following partial differential equations (i) (D2 −2DD′ + D′2)2z = 0, (ii) (D3 −3D2D′ + 3DD′2 −D′3)z = 0. Solution: (i) We observe that (D2 −2DD′ + D′2)2z = (D −D′)4z = 0. So the repeated factor is (D −D′), four times. This corresponds to: a = 1, b = −1, (repeated root). Hence, the general solution is: z = x3ϕ1(x + y) + x2ϕ2(x + y) + xϕ3(x + y) + ϕ4(x + y) 6 / 7 EV0000519 · EV0000520 · EV0000521

7. Source slide page 7 (extracted text): Problems (contd.) (ii) We notice that (D3 −3D2D′ + 3DD′2 −D′3)z = (D −D′)3z = 0. So the repeated factor is (D −D′), three times. Hence, the general solution is: z = x2ϕ1(x + y) + xϕ2(x + y) + ϕ3(x + y) 7 / 7 EV0000522 · EV0000523 · EV0000524

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.652] Hello everyone. Welcome back to the course on engineering mathematics one. [00:00:14.404] In this session, we'll solve problems on finding the general solution of homogeneous linear partial differential equations with constant coefficients. In the last class, we understood how to factorize [00:00:36.356] F of D comma D dash Z equals zero into linear factor that is uh AID plus AI D dash C i. [00:00:55.716] Sometimes CI may be zero or B I may be zero or AI may be zero. So based on the requirement, we write the general solution. So we also understood that before we solve the problem. [00:01:12.548] Uh if we get to know that the total degree of each term is uh the same, then we can use this shortcut to make the problem simple. [00:01:25.924] So let's solve some of the problems on finding the general solution when the right hand side is zero. [00:01:37.124] Let's look at the first problem d power four minus uh two d square t dash square plus d dash power four of the d plus zero. [00:01:51.940] And it clearly shows that the total degree of each term is M. Either we can use d by d dash to be m or we may look for [00:02:06.724] A plus b ol square or a minus b whole square that makes the problem even simpler. [00:02:15.364] So it is easy to see that this uh d power 4 can be written as d square square then uh this uh d dash power four is same as uh d dash square and then another square. [00:02:32.772] And this is same as 2ab that is d square and d dash square. So this leads to a minus b whole square, where a is d square and b is d dash square. So this is same as [00:02:52.964] A plus B and A minus B. So we get T minus D dash and D plus G dash. So this square is applicable for both. So we have repeated factor. [00:03:10.724] So for this factor, the coefficient of d is 1, so a1 is 1. The coefficient of d dash is minus 1, so b1 is minus 1, and there is no c. [00:03:24.100] Yeah, so C1 is 0. Similarly for the second factor, A2, B2, C2 are 110 respectively. So based on this, uh we can write the solution. [00:03:37.444] As z equals x of phi 1 x plus y and phi 2 of x plus y. This is for the first factor. And this is for the second factor. We know

a logic when we have uh both the terms to be negative, we take we can take minus common and that is same as this. [00:03:56.708] Before that we introduced our [00:04:03.844] Okay, uh let's look at uh 4d power 3 minus uh 3 d dash square plus t dash power 3 of z equals zero. [00:04:17.060] So what we can do now is if we divide through by d dash power three so that gives uh four then uh d by d dash power three then [00:04:36.804] And now we get the three of D by D dash plus one of Z. So when we put D by D dash as M, so that gives four M power three. [00:04:57.540] Minus 3m plus 1 0. So we need to find out what are the three roots of m because this is a cubic equation in M

EV0000150 · EV0000151 · EV0000152 · EV0000153

Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.460] So when we use synthetic division we see that uh m equals minus 1 is the first root by inspection and then uh we get to know that we get uh 4m square minus uh 4m plus 10. [00:05:30.468] And this is same as uh 2m minus 1 whole square zero and uh we see m equal to 1 by 2 two times. [00:05:42.820] So how to find out a factor? Let us substitute m as 1 minus 1 and 1 by 2 respectively in d by d dash. So d by d dash equals m becomes minus 1. So this is same as a d plus d dash that we can see here. [00:06:03.588] and when we put m equal to half we get uh d by d dash equals one by two this is same as uh 2d equal to d dash so this is same as uh 2d minus d dash and this square because uh m equals half repeated two times from here [00:06:25.668] So for this problem we need to use uh both distinct factor case as well as repeated factor. So let's first find out a i, bi and c i values. But the first [00:06:39.076] D plus d dash A B C R 110 respectively and for the second ABC R 2 minus 1 and 0 respectively. Whenever we see B is negative, we need to go for star concept, what we know from earlier. [00:06:58.692] So for the first factor this is the solution for the second factor this is the solution. So finally the general solution for this problem is going to be this much. And you can see the order of equation is three. [00:07:14.180] And the number of orbital functions are also three, therefore this is a general solution written as Z equals this. [00:07:25.124] Okay. [00:07:28.228] Let's look at the third problem that is uh d squared minus d d dash minus six d dash square. So when we use the shortcut method [00:07:42.564] we see that uh the right equation in m is uh m square minus m minus x zero so m values uh three and minus two so obviously when we say a by d dash equals uh three the first factor will be d minus uh three d dash [00:08:02.532] And the second factor is d plus 2d dash. And these are the abc value C0. Then we get the solution to be phi one of 3x plus y, phi 2 of 32x minus y. [00:08:21.316] In a similar manner uh if we look at the port problem, it is easy to see that and this part is uh nothing but uh a square sorry a minus b whole square. [00:08:39.652] That is a d minus d dash square and we have another square outside. So that leads to b minus d dash power four. So this is repeated four times. [00:08:58.628] So when we use the formula by choosing a1 b1 to be one minus one respectively, uh we get this answer. So before that we should have got uh minus x minus y and then we write it as x plus y. [00:09:16.548] So what about this problem? So this leads to our uh a minus b formula again a b are one and minus one respectively. So when we use the formula we get this as a general solution, order is three, arbitrary functions are also three. [00:09:36.164] So in this session uh we sold for the general solution where the right hand side is zero. Thank you.

EV0000154 · EV0000155 · EV0000156 · EV0000157

Full transcript: 1145 normalized words, 37 timestamped segments, 590.272 seconds duration. Low-confidence segments flagged: 0.

EV0000150 · EV0000151 · EV0000152 · EV0000153

14 Particular integral -Type 3 and Type 4

This lesson is indexed from the source video titled Particular integral -Type 3 and Type 4. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000158 · EV0000159

Graphs and diagrams

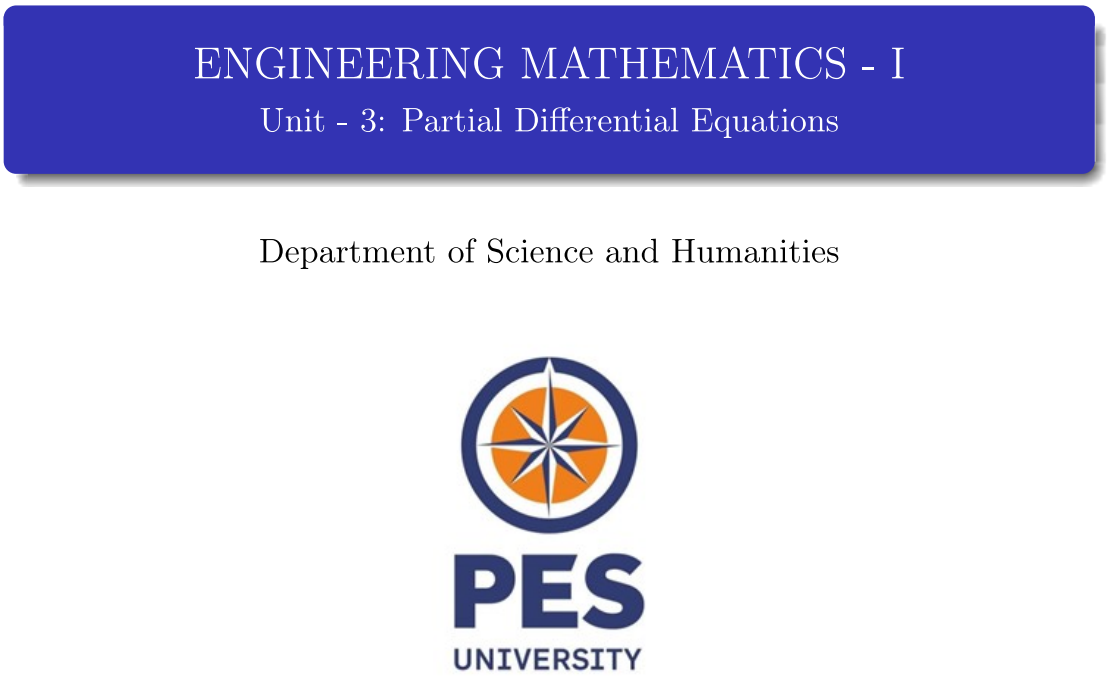


Figure G000131. Particular integral of type 3 and 4 · source slide page 1.

EV0000565

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Contents

1. P.I. when $f(x,y) = x^m y^n$, or a polynomial in x, y
2. P.I. when $f(x,y)$ is not of the form e^{ax+by} or $sin(ax+b)$ or $cos(ax+b)$ or $x^m y^n$ (a polynomial in x, y)

Figure G000132. Particular integral of type 3 and 4 · source slide page 2.

EV0000568

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

P.I. when $f(x, y) = x^m y^n$, or a polynomial in x, y

We write the particular integral as

$$z = [F(D, D')]^{-1} x^m y^n$$

We expand $[F(D, D')]^{-1}$ as an infinite series and operate on $x^m y^n$. If $F(D, D')$ does not contain the constant term, that is, a term independent of D and D', then we expand $[F(D, D')]^{-1}$ in powers of $(D')^{-1}D$ if $m < n$ and in powers of $D^{-1}D'$ if $m > n$

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Figure G000133. Particular integral of type 3 and 4 · source slide page 3.

EV0000571

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Problem

Find the particular integral of the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = x^2 + y^2$$

Solution: We have $[D^2 - (D')^2]z = x^2 + y^2$
We write $[D^2 - (D')^2] = D^2\left[1 - (D^{-1})^2(D')^2\right]$

$$\therefore [D^2 - (D')^2]^{-1} = D^{-2}\left[1 - (D^{-1})^2(D')^2\right]^{-1}$$
$$\Rightarrow [D^2 - (D')^2]^{-1} = D^{-2}\left[1 + (D^{-1})^2(D')^2 + (D^{-1})^4(D')^4 + \ldots\right]$$

Figure G000134. Particular integral of type 3 and 4 · source slide page 4.

EV0000574

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Problem (contd.)

Therefore, the particular integral is given by

$$\begin{aligned} PI &= [D^2 - (D')^2]^{-1}(x^2 + y^2) \\ &= D^{-2}\left[1 + (D^{-1})^2(D')^2 + (D^{-1})^4(D')^4 + \ldots\right](x^2 + y^2) \\ &= D^{-2}\left[(x^2 + y^2) + (D^{-1})^2(D')^2(x^2 + y^2)\right] \\ &= D^{-2}\left[(x^2 + y^2) + (D^{-1})^2(2)\right] \\ &= D^{-2}[x^2 + y^2 + x^2] \\ &= D^{-2}[2x^2 + y^2] \\ &= \frac{x^4}{6} + \frac{x^2y^2}{2}. \quad \textbf{On integrating twice w.r.t. } x \end{aligned}$$

We may also write $[D^2 - (D')^2] = -(D')^2[1 - (D')^{-2}D^2]$.
In this case, we obtain a different form of the particular integral as

$$z = -\frac{y^2}{6}[3x^2 + y^2]$$

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Figure G000135. Particular integral of type 3 and 4 · source slide page 5.

EV0000577

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Find the particular integral of the partial differential equation

$$[D^2 + 2DD' + (D')^2]z = 3x + 2y, \quad \text{or} \quad [D + D']^2 z = 3x + 2y$$

We write

$$[D + D']^2 = [D(1 + D'D^{-1})]^2$$

and

$$[D + D']^2 = D^2(1 + D'D^{-1})^2$$

Figure G000136. Particular integral of type 3 and 4 · source slide page 6.

EV0000580

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Problem (contd.)

The particular integral is given by

$$
\begin{aligned}
z &= D^{-2}(1 + D'D^{-1})^{-2}(3x + 2y) \\
&= D^{-2}[1 - 2D'D^{-1} + 3(D'D^{-1})^2 - \ldots](3x + 2y) \\
&= D^{-2}[3x + 2y - 2D'D^{-1}(3x + 2y)] \\
&= D^{-2}[3x + 2y - 2D'(\frac{3x^2}{2} + 2xy)] \\
&= D^{-2}[3x + 2y - 4x] \\
&= D^{-2}[-x + 2y] \\
&= -\frac{x^3}{6} + x^2y
\end{aligned}
$$

Figure G000137. Particular integral of type 3 and 4 · source slide page 7.

EV0000583

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Problem (contd.)

We may also write

$$[D + D']^2 = [D'(1 + (D')^{-1}D)]^2$$

The particular integral is given by

$$\begin{aligned} PI &= (D')^{-2}[1 + (D')^{-1}D]^{-2}(3x + 2y) \\ &= (D')^{-2}[1 - 2(D')^{-1}D + 3((D')^{-1}D)^2 - \ldots](3x + 2y) \\ &= (D')^{-2}[3x + 2y - 2(D')^{-1}D(3x + 2y)] \\ &= (D')^{-2}[3x + 2y - 2(D')^{-1}(3)] \\ &= (D')^{-2}[3x + 2y - 6y] \\ &= (D')^{-2}[3x - 4y] \quad \textbf{On integrating twice w.r.t. } y \\ &= \frac{3xy^2}{2} - \frac{2y^3}{3} \end{aligned}$$

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Figure G000138. Particular integral of type 3 and 4 · source slide page 8.

EV0000586

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Let $f(x, y)$ be not of any one of the forms as given in the above cases or the case of failure in Case 2

We use the following procedure:
Assume that $F(D, D')$ is reducible, that is, it can be factorized. Now consider one of the factors as

$$(a_1 D + b_1 D')\, z = a_1 p + b_1 q = f(x, y).$$

Since this is a Lagrange's equation, the auxiliary equations are

$$\frac{dx}{a_1} = \frac{dy}{b_1} = \frac{dz}{f(x, y)}.$$

The first two ratios give the solution

$$b_1\, x - a_1\, y = c, \quad c \text{ arbitrary constant.} \tag{1}$$

Figure G000139. Particular integral of type 3 and 4 · source slide page 9.

EV0000589

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

(Contd.)

Consider now the first and third ratios:

$$\frac{dx}{a_1} = \frac{dz}{f(x,y)} = \frac{dz}{f\big[x,\,(b_1x-c)/a_1\big]},$$

or equivalently

$$a_1\,dz = f\Big[x,\,(b_1x-c)/a_1\Big]\,dx.$$

Integrating, we get

$$a_1\,z = \int f\Big(x,\,\tfrac{b_1x-c}{a_1}\Big)\,dx + c_1 = F(x,c) + c_1,$$

where c_1 is an arbitrary constant. After integration, we replace c by $b_1x - a_1y$ as given in Eq. 1. Since a particular integral is required, we set $c_1 = 0$.

Figure G000140. Particular integral of type 3 and 4 · source slide page 10.

EV0000592

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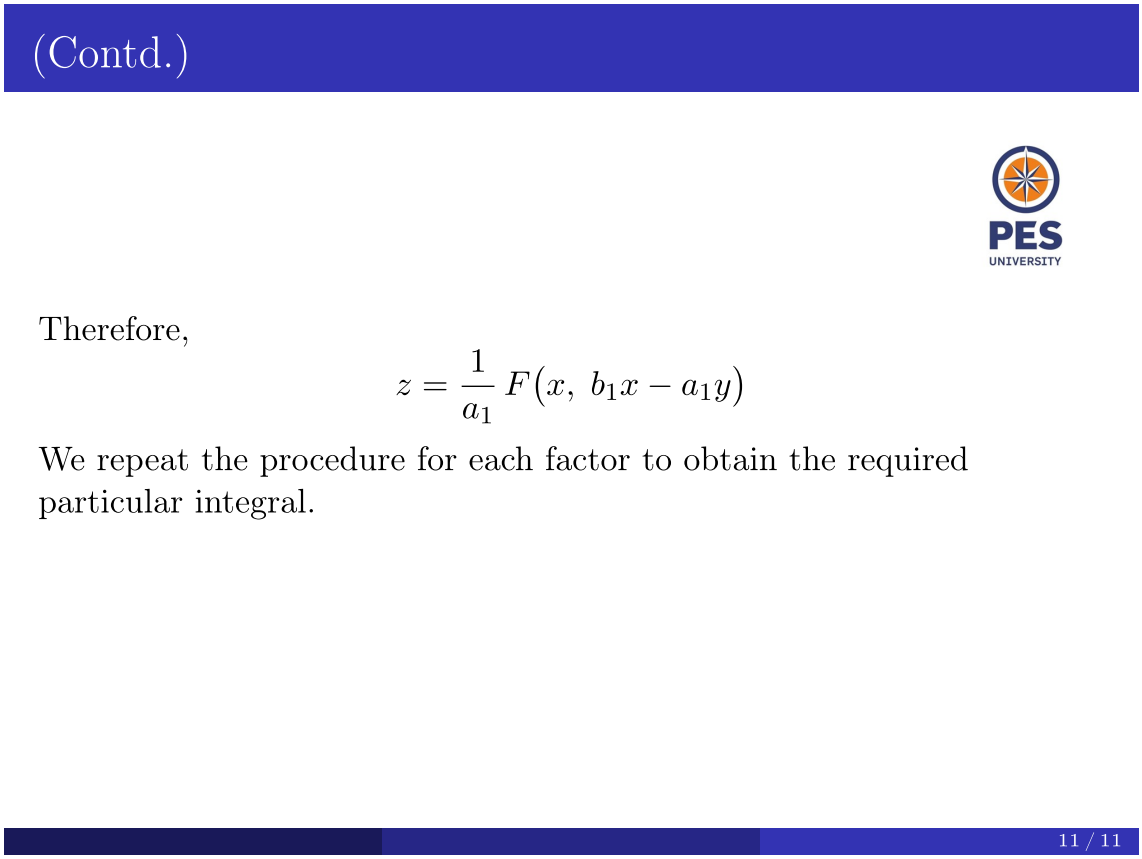


Figure G000141. Particular integral of type 3 and 4 · source slide page 11.

EV0000595

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000564 · EV0000565 · EV0000566
2. Source slide page 2 (extracted text): Contents 1 P.I. when f(x, y) = xmyn, or a polynomial in x, y 2 P.I. when f(x, y) is not of the form eax+by or sin(ax + b) or cos(ax + b) or xmyn (a polynomial in x, y) 2 / 11 EV0000567 · EV0000568 · EV0000569
3. Source slide page 3 (extracted text): P.I. when f(x, y) = xmyn, or a polynomial in x, y We write the particular integral as z = [F(D, D′)]−1xmyn We expand [F(D, D′)]−1 as an infinite series and operate on xmyn. If F(D, D′) does not contain the constant term, that is, a term independent of D and D′, then we expand [F(D, D′)]−1 in powers of (D′)−1D if m < n and in powers of D−1D′ if m > n 3 / 11 EV0000570 · EV0000571 · EV0000572
4. Source slide page 4 (extracted text): Problem Find the particular integral of the partial differential equation ∂2z −∂2z = x2 + y2 ∂x2 ∂y2 Solution: We have [D2 −(D′)2]z = x2 + y2 We write [D2 −(D′)2] = D2 1 −(D−1)2(D′)2 ∴[D2 −(D′)2]−1 = D−2 1 −(D−1)2(D′)2 −1 ⇒[D2 −(D′)2]−1 = D−2 1 + (D−1)2(D′)2 + (D−1)4(D′)4 + . . . 4 / 11 EV0000573 · EV0000574 · EV0000575

5. Source slide page 5 (extracted text): Problem (contd.) Therefore, the particular integral is given by PI = [D2 −(D′)2]−1(x2 + y2) = D−2 1 + (D−1)2(D′)2 + (D−1)4(D′)4 + . . . (x2 + y2) = D−2 (x2 + y2) + (D−1)2(D′)2(x2 + y2) = D−2 (x2 + y2) + (D−1)2(2) = D−2[x2 + y2 + x2] = D−2[2x2 + y2] x4 x2y2 = + . On integrating twice w.r.t. x 6 2 We may also write [D2 −(D′)2] = −(D′)2[1 −(D′)−2D2]. In this case, we obtain a different form of the particular integral as z = −y2 [3x2 + y2] 6 5 / 11 EV0000576 · EV0000577 · EV0000578
6. Source slide page 6 (extracted text): Problem Find the particular integral of the partial differential equation [D2 + 2DD′ + (D′)2]z = 3x + 2y, or [D + D′]2z = 3x + 2y We write [D + D′]2 = [D(1 + D′D−1)]2 and [D + D′]2 = D2(1 + D′D−1)2 6 / 11 EV0000579 · EV0000580 · EV0000581
7. Source slide page 7 (extracted text): Problem (contd.) The particular integral is given by z = D−2(1 + D′D−1)−2(3x + 2y) = D−2[1 −2D′D−1 + 3(D′D−1)2 −. . .](3x + 2y) = D−2[3x + 2y −2D′D−1(3x + 2y)] = D−2[3x + 2y −2D′(3x2 + 2xy)] 2 = D−2[3x + 2y −4x] = D−2[−x + 2y] = −x3 + x2y 6 7 / 11 EV0000582 · EV0000583 · EV0000584
8. Source slide page 8 (extracted text): Problem (contd.) We may also write [D + D′]2 = [D′(1 + (D′)−1D)]2 The particular integral is given by PI = (D′)−2[1 + (D′)−1D]−2(3x + 2y) = (D′)−2[1 −2(D′)−1D + 3((D′)−1D)2 −. . .](3x + 2y) = (D′)−2[3x + 2y −2(D′)−1D(3x + 2y)] = (D′)−2[3x + 2y −2(D′)−1(3)] = (D′)−2[3x + 2y −6y] = (D′)−2[3x −4y] On integrating twice w.r.t. y 3xy2 −2y3 = 2 3 8 / 11 EV0000585 · EV0000586 · EV0000587
9. Source slide page 9 (extracted text): Let f(x, y) be not of any one of the forms as given in the above cases or the case of failure in Case 2 We use the following procedure: Assume that F(D, D′) is reducible, that is, it can be factorized. Now consider one of the factors as (a1D + b1D′) z = a1p + b1q = f(x, y). Since this is a Lagrange's equation, the auxiliary equations are dx dy dz = = a1 b1 f(x, y). The first two ratios give the solution b1 x −a1 y = c, c arbitrary constant. (1) 9 / 11 EV0000588 · EV0000589 · EV0000590
10. Source slide page 10 (extracted text): (Contd.) Consider now the first and third ratios: dx dz dz = = , a1 f x, y f x, (b1x −c)/a1 or equivalently h i a1 dz = f x, (b1x −c)/a1 dx. Integrating, we get Z b1x−c a1 z = f x, a1 dx + c1 = F(x, c) + c1, where c1 is an arbitrary constant. After integration, we replace c by b1x −a1y as given in Eq. 1. Since a particular integral is required, we set c1 = 0. 10 / 11 EV0000591 · EV0000592 · EV0000593
11. Source slide page 11 (extracted text): (Contd.) Therefore, 1 z = F x, b1x −a1y a1 We repeat the procedure for each factor to obtain the required particular integral. 11 / 11 EV0000594 · EV0000595 · EV0000596

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:06.340] Hello everyone. Welcome back to the course on engineering mathematics one. In previous session, we learned how to find the particular integral where [00:00:25.956] f of x comma y equals sine e but x plus b y as well as sine of x plus b y or cos of x plus b y in this session [00:00:43.140] We will uh take up two more cases. That is finding Pi when f of x comma y is a polynomial in x and y. That is f of x comma y equals x power m times y power n. [00:01:03.524] Also, we look at the general case. So normally when f of x comma y is not e ball axes b y or when f of x comma y is not sine of x plus b y or cos of x plus b y [00:01:22.788] Or when f of x comma y is not is not a polynomial in x and y. [00:01:30.308] So let's understand how to find the particular integral in these cases. Okay. Let's start with case number

three. That is when f of x comma y equals [00:01:47.460] x power m dot y power n or at polynomial in x and y. How to how do we find out particular integral? As you know that the formula for particular integral is going to be one divided by [00:02:06.660] f of d, d dash of f of f of x, y. So, in this case, f of x, y is x power m dot y power n. [00:02:24.100] As we have seen in ordinary differential equation, here also we bring f of b, comma d dash to the curator so that it's going to be z equals f of b, d dash inverse. [00:02:42.404] X power m dot by power n. Then we need to expand f of b, d dash inverse as an infinite series, and then operate by x power m dot by power n. As a special case. [00:03:01.924] suppose f of d gamma d dash doesn't contain the constant term that is the term which is independent of d and d dash then we have two cases we expand f of d gamma d dash inverse [00:03:21.284] power minus one in power of d dash inverse dot d if m is less than n or in power sub d inverse dash dot d dash if m is greater than n. So these are two special cases [00:03:42.788] Okay, let's take an example. How do you find out the particular integral when we have a polynomial on the right hand side? [00:03:56.004] So the given PD here is 2 square z by 2 x square which is nothing but d square z minus d dash square z equals x square plus y square, which is nothing but d square minus d dash square of z equals x square. [00:04:15.300] Square plus y square. We can easily observe that in this example, both m and n are equal to two. Okay. [00:04:26.916] So basically we have two options here. Either we can take out d square common or d square out common because m and n are equal here. In case we take out d square common, what happens? So let's understand why do we take d square common. [00:04:44.804] Because in the polynomial case, so it is convenient to have either one plus something or one minus something before we apply binomial series. So for that, we'll take out d square common. So that leads to the innovation.

EV0000158 · EV0000159 · EV0000160 · EV0000161

Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.676] minus square and d da square which is nothing but d square of one minus d da square divided by d square so this is going to be uh d na square [00:05:19.972] And you can easily see that this is like one minus x or one minus something that something is this b inverse square dot b dash square. Okay. Moreover, all these things are in the denominator. So when you're [00:05:39.844] the particular integral this is something like 1 by t square of 1 minus e in the square and then beta square of x square plus y square [00:05:55.492] So in the next time we need to bring it up. So that's going to be this one. D square minus theta square inverse of x square plus y square. So when we use the series, so that's going to be [00:06:15.748] the minus two of one minus the inverse square then uh d dash square of x square plus y square because m and n are equal to two so we stop [00:06:33.828] the series so when we reach d square as well as d data square [00:06:41.540] So again as we know from binomial series, so 1 minus x minus 1 is going to be 1 plus x plus x square plus x power 3 and so on. So let's write that and so on. [00:06:59.812] So because the highest power of uh m and n is two, we stop the series at uh t square and d ta square. So that is why if we look at the formula for binomial series, these terms are not considered because anyway. [00:07:14.756] D inverse power four and so on of this x square plus y square become zero. [00:07:22.628] So when you write the formula we are considering only one as well as uh x because x is uh t square and uh d naught square and d da square. So it means uh we are getting power minus two of one plus. [00:07:43.716] B inverse square. Then we have our d dash square. Other terms are neglected. Outside we have x square plus y square. Then we should take it inside. So that leads to x square plus y square plus [00:08:03.844] D inverse square dot d dash square of x square plus y square. Next step we just read an as it is. So

what is a d inverse square dot d dash square? So [00:08:19.300] d dash square means uh we need to find out the partial derivative of this with respect to y. So x square becomes 0, y square becomes 2y and then second time it's going to be simply y. So d dash square square of this is going to be simply 2. [00:08:39.268] Then what is uh t inverse square of 2? So this is going to be integration of this with respect to x2 times. So first time this is 2x, second time this is 2x square by 2, but cancellation 2 on 2 we get x square. [00:08:59.876] So finally we get uh 2x square plus y square. Then what is uh d power minus two? So we have to integrate this with respect to x two times, so that leads to x power four by six. [00:09:17.604] plus x square dot y square by two. So this is the particular integral for this particular problem. So as we know from the earlier concept, because m and n are equal, we have two choices. [00:09:34.148] Either we can take d square common or we can take d dash square common. So in case we take d dash common, so for that particular case, so this is going to be the particular integral. So obviously we get two different answers for this problem. [00:09:53.540] Uh for a PD there are chances of getting different answers.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.212] Okay, let's take a another example. Find the particular integral for the partial differential equation. So d square plus uh two dd dash plus d dash square of z equals three x plus two y. So [00:10:20.580] So it is easy to observe that m and n r equals one. So again we have two choices here. Either we can take a t outside or t dash outside. [00:10:34.500] Assume we take D outside, so that leads to this particular step, and we get uh one plus x, then that series formula minus two. [00:10:47.844] So finally we get uh d power minus two of uh one plus x minus two. So when you use the binomial formula that leads to one minus two x plus three x square and so on. [00:11:05.060] Because the highest power of uh x uh m and n or the values of m and n r equals one, it is enough to go up to d and d dash. So based on that, we go only up to uh two. [00:11:20.260] two d dash and d power minus one in it these things are zeros so once we take three x plus two y inside so that leads to three x plus two y and two d dash d power minus one three x plus two y [00:11:39.012] As we know, this is nothing but uh integration of x with respect to x, integration with respect to x, uh keeping y constant. So 3x becomes 3x square by 2, so y becomes 2xy because 2y is constant and dx become x. [00:11:58.372] Then what is uh d dash of uh this part? So derivative of this with respect to y keeping it constant. Of course partial derivative. So 3x square by two becomes zero. So 2xy with respect to y becomes only 2x. [00:12:17.924] Outside we have one more 2, therefore we get minus 4x. So that leads to minus x plus 2y. So then what is t power minus 2? We need to integrate with respect to x two times. So that leads to minus x power 3 by 6. [00:12:36.420] plus x square y. So there is one more way to find out a particular integral. In case we take out the d dash common, so that leads to this particular answer. You can clearly see that [00:12:55.140] Both answers are different and it is valid also. Okay, let's move on to another case. When a form x comma y is not any of the forms as given in the above cases or [00:13:15.428] If you remember uh when we discuss case number two, it was said that when uh f of uh minus d square minus uh d d dash and minus d dash square become zero. [00:13:33.796] So this is a d square because uh we replace d square by minus a square, so it is a d square minus a d d dash and d dash square zero. So we use uh case number four, so that's what we are discussing now. So [00:13:54.436] when f of x comma y equals sine of x plus b y or so of x plus b y in this case in the case denominator is zero [00:14:11.876] How do you find out P PI? So even that case can be solved by using this method. Okay. So we have a a procedure to find out PI under this category. [00:14:31.172] So let's see what is that. Assume that f

of b, d dash is reducible. As you know from the beginning of this chapter, so if we can express f of b comma d dash to be something like AI. [00:14:47.268] D plus BI D dash plus CI. If F of B gamma D dash can be reduced to this form, then we say it is reducible. So suppose F of D comma d dash is reducible. Then how do we apply the procedure here?

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.980] Remember uh in this case uh C is taken zero. So C is not in the picture, so C I zero. So we have only AID plus B I D dash. So if we have A1 D plus uh B one D dash up Z equals [00:15:27.236] So as we know from standard notation, so Dz. So DZ stands for uh dou z by doux which is same as P. So that's why this is same as uh [00:15:41.188] A1P and uh D dash Z is nothing but douz by dou y which is same as Q so therefore we get B1 Q equals F of X com Y. If you remember Lagrange's equation. [00:15:57.156] is exactly the same so what is uh Lagrangius linear equation capital p of x y z times all p capital q of x y z small q equals r so it we learned that this is how lagrangius linear equation looks like so here also we have the same thing [00:16:16.612] So in this case capital P is simply A1, capital Q is B1, and R is going to be above X kap of Y. So when read the X in the equation, so that is uh Dx capital P and Dy. [00:16:34.084] by q and dz by r so that leads to this particular step that is d x by a1 d y by b1 and z by f of x comma y okay so what we'll do from now on so let us take [00:16:54.596] First and second fractions and first and third fractions. So what do we get? So from the first and second ratios, uh we can get uh when we have variable separable that is uh b1 dx equals a one dy. [00:17:10.404] After integration that leads to B1 minus A1 to Y equal to C where A C is arbitrary constant. Remember in later stage we need to solve for y. Let's see. Okay, when we take the first and third fractions or first and third ratios. [00:17:28.740] We get dx by a1 equals dz by f of x comma y. So we need to take care of y, otherwise we can't get variable separable. So we replace y to b or we take out y from here. So that is b1x minus a divided by a1. [00:17:48.932] So that leads to this particular step. Now we can go for variable separable. That is a1 dz equals f of x karma b1x minus c by a1 dot bx. So after integration, so that [00:18:06.276] gives a1x a1 z so we get f of x comma c plus c one remember after integration we need to replace c y p1x minus a1 y as equation one [00:18:22.084] Since we are looking for a particular integral, we need to put C1 to be zero. So once we do that, this is how we get Z to B that is 1 by A1. F of x comma b1 x minus a1. Remember [00:18:41.924] This procedure is repeated based on how many factors are involved in the problem. So in this session we understood how to find out the particular integral when f of x gamma y is a polynomial that is x power m dot y power n. Also we looked at the general [00:19:01.668] That is when Mf of X comma y is not e pous b y not sine of x plus b y or cos of x plus b y or not a polynomial in x and y. Thank you.

EV0000170 · EV0000171 · EV0000172 · EV0000173

Full transcript: 2452 normalized words, 66 timestamped segments, 1157.461 seconds duration. Low-confidence segments flagged: 0.

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15 Source slide deck: Lagrange s Linear Equation

This entry preserves the complete slide deck titled Lagrange s Linear Equation as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_03, to avoid an unsupported match. EV0000261 · EV0000262

Graphs and diagrams

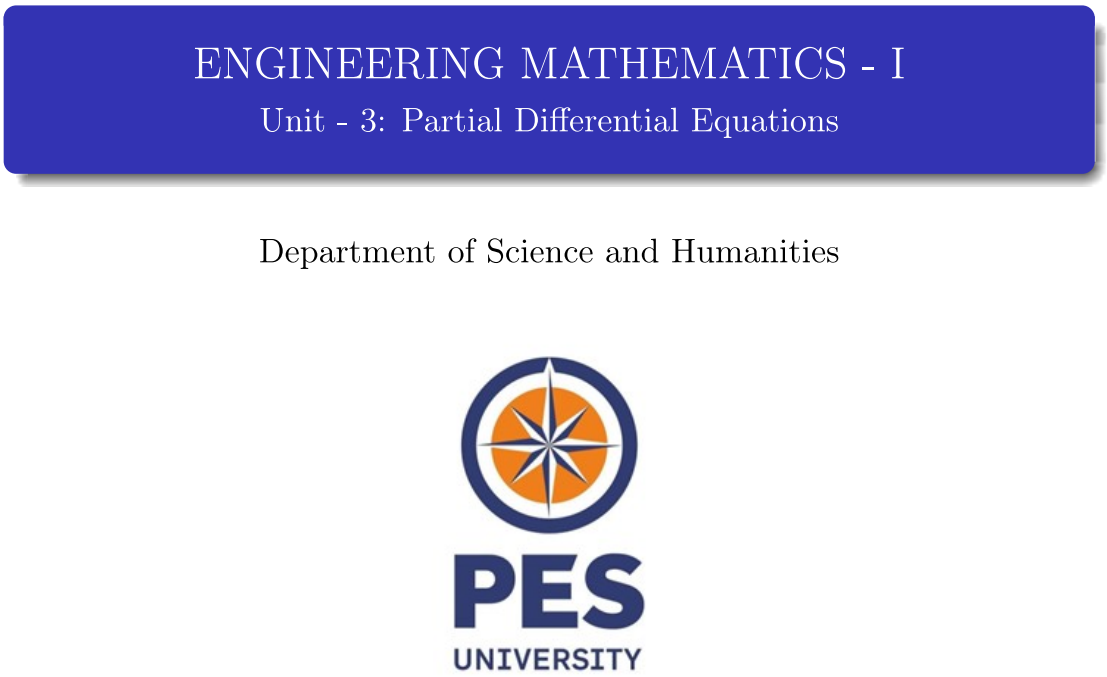


Figure G000030. Lagrange s Linear Equation · source slide page 1.

EV0000262

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Contents

1 Lagrange's Linear Equation

2 Working Rule to solve Lagrange's Linear Equation

Figure G000031. Lagrange s Linear Equation · source slide page 2.

EV0000265

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Lagrange's Linear Equation

1. The linear first-order partial differential equation of the form
$$P(x, y, z)\, p + Q(x, y, z)\, q = R(x, y, z) \tag{1}$$
is called the *Lagrange's equation* in two independent variables x, y
2. **Theorem:** The general solution of the equation $Pp + Qq = R$ is given by $\phi(u, v) = 0$, where ϕ is an arbitrary function and $u(x, y, z) = c_1$, $v(x, y, z) = c_2$ are two linearly independent solutions of the equations
$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \tag{2}$$
(These equations are called the *auxiliary* or *subsidiary equations.*)

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Figure G000032. Lagrange s Linear Equation · source slide page 3.

EV0000268

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Working Rule to Solve Lagrange's Linear Equation

To solve the equation

$$Pp + Qq = R$$

1. Form the auxiliary equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

2. Solve the auxiliary equations by the method of grouping or the method of multipliers or both to get two independent solutions $u = a$ and $v = b$, where a, b are arbitrary constants
3. $\phi(u, v) = 0$ is the general solution of the equation $Pp + Qq = R$

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Figure G000033. Lagrange s Linear Equation · source slide page 4.

EV0000271

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Method of Grouping in Lagrange's Equation

1. The method of grouping involves pairing two of these ratios at a time to form ordinary differential equations which can be integrated to find the solution
2. Group two ratios at a time:

$$\frac{dx}{P} = \frac{dy}{Q}$$
$$\frac{dy}{Q} = \frac{dz}{R}$$
$$\frac{dx}{P} = \frac{dz}{R}$$

3. Integrate each group to obtain two independent solutions
4. The general solution is given by $\phi(u, v) = 0$, where u and v are the two independent solutions obtained

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Figure G000034. Lagrange s Linear Equation · source slide page 5.

EV0000274

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Example:

Suppose the auxiliary equations are:

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$$

Grouping $\frac{dx}{x} = \frac{dy}{y}$:

$$\int \frac{dx}{x} = \int \frac{dy}{y} \implies \ln x = \ln y + c_1 \implies \frac{x}{y} = c_2$$

Grouping $\frac{dx}{x} = \frac{dz}{z}$:

$$\int \frac{dx}{x} = \int \frac{dz}{z} \implies \ln x = \ln z + c_3 \implies \frac{x}{z} = c_4$$

The general solution is:

$$F\left(\frac{x}{y}, \frac{x}{z}\right) = 0$$

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Figure G000035. Lagrange s Linear Equation · source slide page 6.

EV0000277

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Method of Multipliers in Lagrange's Equation

1. Let the auxiliary equations be $\frac{dx}{p} = \frac{dy}{Q} = \frac{dz}{R}$
2. Let l, m, n be constants or functions of x, y, z. Then by the property of ratio and proportion, we have

$$Each\ Ratio = \frac{dx}{p} = \frac{dy}{Q} = \frac{dz}{R} = \frac{ldx + mdy + ndz}{lp + mQ + nR}$$

3. l, m, n are chosen in such a way that $lP + mQ + nR = 0$. Thus

$$ldx + mdy + ndz = 0$$

4. Solve this differential equation, and the solution is $u = c_1$
5. Similarly, choose another set of multipliers (l_1, m_1, n_1) and the second solution is $v = C_2$
6. Required general solution is $\phi(u, v) = 0$, where u and v are the two independent solutions

Figure G000036. Lagrange s Linear Equation · source slide page 7.

EV0000280

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Example:

Suppose the auxiliary equations are:

$$\frac{dx}{y-z} = \frac{dy}{z-x} = \frac{dz}{x-y}$$

Choose multipliers $l = 1, m = 1, n = 1$. Then,

$$Each\ Ratio = \frac{dx + dy + dz}{y - z + z - x + x - y} = \frac{dx + dy + dz}{0}$$

This gives $dx + dy + dz = 0$. On integration, it gives $x + y + z = C_1$.

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Figure G000037. Lagrange s Linear Equation · source slide page 8.

EV0000283

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Differential Concept in Lagrange's Equation

1. Given the auxiliary equations: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$
2. Sometimes, instead of grouping or using multipliers directly, we use the **differential concept** to form a linear combination of differentials that is easier to integrate
3. It is essentially the same as the method of multipliers, but emphasizes the use of differentials and their integration
4. Multipliers may be chosen (more than once) such that the numerator $ldx + mdy + ndz$ is an exact differential of the denominator $lP + mQ + nR$
5. Finally, $\frac{ldx+mdy+ndz}{lP+mQ+nR}$ is combined with a fraction of $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ to get an integral

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Figure G000038. Lagrange s Linear Equation · source slide page 9.

EV0000286

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000261 · EV0000262 · EV0000263
2. Source slide page 2 (extracted text): Contents 1 Lagrange's Linear Equation 2 Working Rule to solve Lagrange's Linear Equation 2 / 9 EV0000264 · EV0000265 · EV0000266
3. Source slide page 3 (extracted text): Lagrange's Linear Equation 1 The linear first-order partial differential equation of the form P(x, y, z) p + Q(x, y, z) q = R(x, y, z) (1) is called the Lagrange's equation in two independent variables x, y 2 Theorem: The general solution of the equation Pp + Qq = R is given by ϕ(u, v) = 0, where ϕ is an arbitrary function and u(x, y, z) = c1, v(x, y, z) = c2 are two linearly independent solutions of the equa- tions dx dy dz = = (2) P Q R (These equations are called the auxiliary or subsidiary equations.) 3 / 9 EV0000267 · EV0000268 · EV0000269
4. Source slide page 4 (extracted text): Working Rule to Solve Lagrange's Linear Equation To solve the equation Pp + Qq = R 1 Form the auxiliary equations dx dy dz = = P Q R 2 Solve the auxiliary equations by the method of grouping or the method of multipliers or both to get two independent solutions u = a and v = b, where a, b are arbitrary constants 3 ϕ(u, v) = 0 is the general solution of the equation Pp + Qq = R 4 / 9 EV0000270 · EV0000271 · EV0000272

5. Source slide page 5 (extracted text): Method of Grouping in Lagrange's Equation 1 The method of grouping involves pairing two of these ratios at a time to form ordinary differential equations which can be integrated to find the solution 2 Group two ratios at a time: dx dy = P Q dy dz = Q R dx dz = P R 3 Integrate each group to obtain two independent solutions 4 The general solution is given by $\phi(u, v) = 0$, where u and v are the two independent solutions obtained 5 / 9 EV0000273 · EV0000274 · EV0000275
6. Source slide page 6 (extracted text): Example: Suppose the auxiliary equations are: dx dy dz = = x y z Grouping dx = dy : x y Z dx Z dy = =⇒ln x = ln y + c1 =⇒x = c2 x y y Grouping dx = dz : x z Z dx Z dz = =⇒ln x = ln z + c3 =⇒x = c4 x z z The general solution is: x x F , = 0 y z 6 / 9 EV0000276 · EV0000277 · EV0000278
7. Source slide page 7 (extracted text): Method of Multipliers in Lagrange's Equation 1 Let the auxiliary equations be dx = dy = dz p Q R 2 Let l, m, n be constants or functions of x, y, z. Then by the property of ratio and proportion, we have dx dy dz ldx + mdy + ndz Each Ratio = = = = p Q R lp + mQ + nR 3 l, m, n are chosen in such a way that lP + mQ + nR = 0. Thus ldx + mdy + ndz = 0 4 Solve this differential equation, and the solution is u = c1 5 Similarly, choose another set of multipliers (l1, m1, n1) and the second solution is v = C2 6 Required general solution is $\phi(u, v) = 0$, where u and v are the two independent solutions 7 / 9 EV0000279 · EV0000280 · EV0000281
8. Source slide page 8 (extracted text): Example: Suppose the auxiliary equations are: dx dy dz = = y −z z −x x −y Choose multipliers l = 1, m = 1, n = 1. Then, dx + dy + dz dx + dy + dz Each Ratio = = y −z + z −x + x −y 0 This gives dx + dy + dz = 0. On integration, it gives x + y + z = C1. 8 / 9 EV0000282 · EV0000283 · EV0000284
9. Source slide page 9 (extracted text): Differential Concept in Lagrange's Equation 1 Given the auxiliary equations: dx = dy = dz P Q R 2 Sometimes, instead of grouping or using multipliers directly, we use the differential concept to form a linear combination of differentials that is easier to integrate 3 It is essentially the same as the method of multipliers, but empha- sizes the use of differentials and their integration 4 Multipliers may be chosen (more than once) such that the numerator ldx + mdy + ndz is an exact differential of the denominator lP + mQ + nR 5 Finally, ldx+mdy+ndz is combined with a fraction of dx = dy = dz to lP+mQ+nR P Q R get an integral 9 / 9 EV0000285 · EV0000286 · EV0000287

SHORT-NOTE RECAP

Source deck preserved: Lagrange s Linear Equation. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000261

16 Source slide deck: Problems of Lagranges equation

This entry preserves the complete slide deck titled Problems of Lagranges equation as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_03, to avoid an unsupported match. EV0000288 · EV0000289

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000039. Problems of Lagranges equation · source slide page 1.

EV0000289

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Contents



1 Problems on Lagrange's Linear Equation

Figure G000040. Problems of Lagranges equation · source slide page 2.

EV0000292

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $\frac{y^2 z}{x} p + xz\, q = y^2$

Given:

$$\frac{y^2 z}{x} p + xz\, q = y^2$$

This is Lagrange's equation $Pp + Qq = R$.
Here,

$$P = \frac{y^2 z}{x}, \quad Q = xz, \quad R = y^2$$

The subsidiary equations are: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{y^2 z/x} = \frac{dy}{xz} = \frac{dz}{y^2} \implies \frac{x\, dx}{y^2 z} = \frac{dy}{xz} = \frac{dz}{y^2}$$

Considering the first two ratios:

$$\frac{x\, dx}{y^2 z} = \frac{dy}{xz} \implies x^2 dx = y^2 dy$$

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Figure G000041. Problems of Lagranges equation · source slide page 3.

EV0000295

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $\frac{y^2 z}{x} p + x z\, q = y^2$ (contd.)

Integrating,

$$\int x^2 dx = \int y^2 dy \implies \frac{x^3}{3} = \frac{y^3}{3} + c \implies x^3 - y^3 = 3c$$

$$\implies x^3 - y^3 = a \tag{1}$$

Considering the first and last ratios:

$$\frac{x\, dx}{y^2 z} = \frac{dz}{y^2} \implies x\, dx = z\, dz$$

Integrating,

$$\int x\, dx = \int z\, dz \implies \frac{x^2}{2} = \frac{z^2}{2} + c \implies x^2 - z^2 = 2c \implies x^2 - z^2 = b \tag{2}$$

General solution:

$$\therefore \quad \Phi(x^3 - y^3,\, x^2 - z^2) = 0$$

where Φ is arbitrary.

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Figure G000042. Problems of Lagranges equation · source slide page 4.

EV0000298

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $yzp - xzq = xy$

Solution:

Auxiliary equations are

$$\frac{dx}{yz} = \frac{dy}{-xz} = \frac{dz}{xy}$$

From first and second fractions, we get

$$\frac{dx}{y} = \frac{dy}{-x}$$

or

$$x\,dx + y\,dy = 0$$

Integrating,

$$x^2 + y^2 = c_1$$

From first and third fractions,

$$\frac{dx}{yz} = \frac{dz}{xy}$$

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Figure G000043. Problems of Lagranges equation · source slide page 5.

EV0000301

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $yzp - xzq = xy$ (contd.)

or

$$\frac{dx}{z} = \frac{dz}{x}$$

Integrating,

$$x^2 - z^2 = c_2$$

Thus the general solution is

$$\phi(x^2 + y^2,\ x^2 - z^2) = 0$$

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Figure G000044. Problems of Lagranges equation · source slide page 6.

EV0000304

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Solve $z(z^2 + xy)(px - qy) = x^4$

Solution:
Auxiliary equations are

$$\frac{dx}{xz(z^2+xy)} = \frac{dy}{-yz(z^2+xy)} = \frac{dz}{x^4}$$

From first and second fractions, we get

$$\frac{dx}{x} = \frac{dy}{-y}$$

On integration, $xy = c_1$

Figure G000045. Problems of Lagranges equation · source slide page 7.

EV0000307

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Solve $z(z^2+xy)(px-qy)=x^4$ (contd.)

From first and third fractions,

$$x^3dx=(z^3+xyz)dz$$

Using $xy=c_1$,

$$x^3dx=(z^3+c_1z)dz$$

Integrating,

$$\int x^3dx=\int(z^3+c_1z)dz$$

$$\frac{x^4}{4}=\frac{z^4}{4}+c_1\frac{z^2}{2}+c_2$$

or

$$x^4-z^4-2c_1z^2=c_2$$

Substituting for c_1,

$$x^4-z^4-2(xy)z^2=c_2$$

The general solution is

$$\phi(xy,\ x^4-z^4-2xyz^2)=0$$

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Figure G000046. Problems of Lagranges equation · source slide page 8.

EV0000310

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000288 · EV0000289 · EV0000290
2. Source slide page 2 (extracted text): Contents 1 Problems on Lagrange's Linear Equation 2 / 8 EV0000291 · EV0000292 · EV0000293
3. Source slide page 3 (extracted text): y2z Solve p + xz q = y2 x Given: y2z p + xz q = y2 x This is Lagrange's equation Pp + Qq = R. Here, y2z P = , Q = xz, R = y2 x The subsidiary equations are: dx = dy = dz P Q R dx dy dz dx dy dz = = =⇒x = = y2z/x xz y2 y2z xz y2 Considering the first two ratios: x dx dy = =⇒x2dx = y2dy y2z xz 3 / 8 EV0000294 · EV0000295 · EV0000296
4. Source slide page 4 (extracted text): y2z Solve p + xz q = y2 (contd.) x Integrating, Z Z y3 x2dx = y2dy =⇒x3 = + c =⇒x3 −y3 = 3c 3 3 =⇒x3 −y3 = a (1) Considering the first and last ratios: x dx dz = =⇒x dx = z dz y2z y2 Integrating, Z Z z2 x dx = z dz =⇒x2 = + c =⇒x2 −z2 = 2c =⇒x2 −z2 = b 2 2 (2) General solution: ∴ Φ(x3 −y3, x2 −z2) = 0 where Φ is arbitrary. 4 / 8 EV0000297 · EV0000298 · EV0000299

5. Source slide page 5 (extracted text): Solve yzp −xzq = xy Solution: Auxiliary equations are dx dy dz = = yz −xz xy From first and second fractions, we get dx dy = y −x or x dx + y dy = 0 Integrating, x2 + y2 = c1 From first and third fractions, dx dz = yz xy 5 / 8 EV0000300 · EV0000301 · EV0000302
6. Source slide page 6 (extracted text): Solve yzp −xzq = xy (contd.) or dx dz = z x Integrating, x2 −z2 = c2 Thus the general solution is ϕ(x2 + y2, x2 −z2) = 0 6 / 8 EV0000303 · EV0000304 · EV0000305
7. Source slide page 7 (extracted text): Solve z(z2 + xy)(px −qy) = x4 Solution: Auxiliary equations are dx dy dz = = xz(z2 + xy) −yz(z2 + xy) x4 From first and second fractions, we get dx dy = x −y On integration, xy = c1 7 / 8 EV0000306 · EV0000307 · EV0000308
8. Source slide page 8 (extracted text): Solve z(z2 + xy)(px −qy) = x4 (contd.) From first and third fractions, x3dx = (z3 + xyz)dz Using xy = c1, x3dx = (z3 + c1z)dz Integrating, Z Z x3dx = (z3 + c1z)dz x4 z4 z2 = + c1 + c2 4 4 2 or x4 −z4 −2c1z2 = c2 Substituting for c1, x4 −z4 −2(xy)z2 = c2 The general solution is ϕ(xy, x4 −z4 −2xyz2) = 0 8 / 8 EV0000309 · EV0000310 · EV0000311

SHORT-NOTE RECAP

Source deck preserved: Problems of Lagranges equation. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000288

17 Source slide deck: Method of separation of variables continued

This entry preserves the complete slide deck titled Method of separation of variables continued as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_03, to avoid an unsupported match. EV0000363 · EV0000364

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000064. Method of separation of variables continued · source slide page 1.

EV0000364

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Contents



1 Method of Separation of Variables

Figure G000065. Method of separation of variables continued · source slide page 2.

EV0000367

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Method of Separation of Variables

- The method of separation of variables is a powerful and widely used technique to solve certain types of Partial Differential Equations (PDEs), especially when the PDE and boundary conditions are defined over a rectangular or regular domain
- It is based on the idea that a solution can be written as a product of functions, each depending on only one independent variable
- Separation of variables provides closed-form solutions for many fundamental partial differential equations—the one-dimensional heat equation, the wave equation, and Laplace's equation—by reducing each PDE to a set of ordinary differential equations

Figure G000066. Method of separation of variables continued · source slide page 3.

EV0000370

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Working Rule

1. For a P.D.E. in the function u of two independent variables x and y, assume that the required solution is separable, i.e.,
$$u(x, y) = X(x)Y(y), \tag{1}$$
where $X(x)$ is a function of x alone and $Y(y)$ is a function of y alone
2. Substitution of u from (1) and its derivatives reduces the P.D.E. to the form
$$f(X, X', X'', \ldots) = g(Y, Y', Y'', \ldots) \tag{2}$$
which is separable in X and Y
3. Since the L.H.S. of (2) is a function of x alone and the R.H.S. of (2) is a function of y alone, then (2) must be equal to a common constant, say k

Figure G000067. Method of separation of variables continued · source slide page 4.

EV0000373

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Working Rule (contd.)



1. Thus (2) reduces to

$$f(X, X', X'', \ldots) = k \tag{3}$$

$$g(Y, Y', Y'', \ldots) = k \tag{4}$$

2. The determination of a solution to P.D.E. reduces to the determination of solutions to two O.D.E.s (with appropriate conditions).

Figure G000068. Method of separation of variables continued · source slide page 5.

EV0000376

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Example

Use the separation of variables technique to solve

$$3u_x + 2u_y = 0 \quad \text{with} \quad u(x,0) = 4e^{-x}.$$

Solution: Assume $u(x,y) = X(x)Y(y)$.
Then the P.D.E. becomes $3X'Y + 2XY' = 0$ or

$$\frac{X'}{X} = -\frac{2}{3}\frac{Y'}{Y} = k \quad \text{(constant)}$$

Solving $X' - kX = 0$, we get $X(x) = c_1 e^{kx}$.
Similarly, $Y' + \frac{3}{2}kY = 0$, so $Y(y) = c_2 e^{-\frac{3}{2}ky}$.
So $u(x,y) = X(x)Y(y) = c_1 e^{kx} \cdot c_2 e^{-\frac{3}{2}ky} = ce^{kx}e^{-\frac{3}{2}ky} = ce^{k(x-\frac{3}{2}y)}$.
Given that $4e^{-x} = u(x,0) = X(x)Y(0) = ce^{kx}$,
Thus $c = 4$, $k = -1$.
Hence, the required solution is

$$u(x,y) = 4e^{-\frac{1}{2}(2x-3y)}$$

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Figure G000069. Method of separation of variables continued · source slide page 6.

EV0000379

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Example

Use separation of variables to solve

$$5\,u_x \;-\; 4\,u_y \;=\; 0, \qquad u(x,0) = 3e^{2x}.$$

Solution. Assume $u(x,y) = X(x)\,Y(y)$. Then,
$u_x = X'(x)Y(y), \quad u_y = X(x)Y'(y)$, so $5X'Y \;-\; 4X\,Y' = 0$
$\Longrightarrow \; \frac{X'}{X} = \frac{4}{5}\,\frac{Y'}{Y} = k.$
Solve ODEs.

$$X'/X = k \;\Rightarrow\; X(x) = A\,e^{kx},$$
$$\frac{Y'}{Y} = \frac{5}{4}k \;\Rightarrow\; Y(y) = B\,e^{\frac{5}{4}k\,y}.$$

Combine.

$$u(x,y) = AB\,e^{kx}\,e^{\frac{5}{4}k\,y} = C\,e^{k\,(x+\frac{5}{4}y)}.$$

Initial condition.

$$u(x,0) = C\,e^{kx} = 3e^{2x} \;\Longrightarrow\; C = 3,\; k = 2.$$

Final answer. $u(x,y) = 3\,e^{2x+\frac{5}{2}y}$.

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Figure G000070. Method of separation of variables continued · source slide page 7.

EV0000382

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000363 · EV0000364 · EV0000365
2. Source slide page 2 (extracted text): Contents 1 Method of Separation of Variables 2 / 7 EV0000366 · EV0000367 · EV0000368
3. Source slide page 3 (extracted text): Method of Separation of Variables The method of separation of variables is a powerful and widely used technique to solve certain types of Partial Differential Equations (PDEs), especially when the PDE and boundary conditions are defined over a rectangular or regular domain It is based on the idea that a solution can be written as a product of functions, each depending on only one independent variable Separation of variables provides closed-form solutions for many fun- damental partial differential equations—the one-dimensional heat equation, the wave equation, and Laplace's equation—by reducing each PDE to a set of ordinary differential equations 3 / 7 EV0000369 · EV0000370 · EV0000371

4. Source slide page 4 (extracted text): Working Rule 1 For a P.D.E. in the function u of two independent variables x and y, assume that the required solution is separable, i.e., u(x, y) = X(x)Y (y), (1) where X(x) is a function of x alone and Y (y) is a function of y alone 2 Substitution of u from (1) and its derivatives reduces the P.D.E. to the form f(X, X′, X″, . . .) = g(Y, Y ′, Y ″, . . .) (2) which is separable in X and Y 3 Since the L.H.S. of (2) is a function of x alone and the R.H.S. of (2) is a function of y alone, then (2) must be equal to a common constant, say k 4 / 7 EV0000372 · EV0000373 · EV0000374

5. Source slide page 5 (extracted text): Working Rule (contd.) 1 Thus (2) reduces to f(X, X′, X″, . . .) = k (3) g(Y, Y ′, Y ″, . . .) = k (4) 2 The determination of a solution to P.D.E. reduces to the determi- nation of solutions to two O.D.E.s (with appropriate conditions). 5 / 7 EV0000375 · EV0000376 · EV0000377

6. Source slide page 6 (extracted text): Example Use the separation of variables technique to solve 3ux + 2uy = 0 with u(x, 0) = 4e−x. Solution: Assume u(x, y) = X(x)Y (y). Then the P.D.E. becomes 3X′Y + 2XY ′ = 0 or X′ = −2 Y ′ = k (constant) X 3 Y Solving X′ −kX = 0, we get X(x) = c1ekx. 2 ky. Similarly, Y ′ + 32kY = 0, so Y (y) = c2e−3 2 ky = cekxe−32 ky = cek(x−32 y).So u(x, y) = X(x)Y (y) = c1ekx · c2e−3 Given that 4e−x = u(x, 0) = X(x)Y (0) = cekx, Thus c = 4, k = −1. Hence, the required solution is u(x, y) = 4e−12 (2x−3y) 6 / 7 EV0000378 · EV0000379 · EV0000380

7. Source slide page 7 (extracted text): Example Use separation of variables to solve 5 ux −4 uy = 0, u(x, 0) = 3e2x. Solution. Assume u(x, y) = X(x) Y (y). Then, ux = X′(x)Y (y), uy = X(x)Y ′(y), so 5X′Y −4X Y ′ = 0 X′ 4 Y ′=⇒ = = k. X 5 Y Solve ODEs. X′/X = k ⇒X(x) = A ekx, Y ′ 5 5 k = ⇒Y (y) = B e 4 y. Y 4k Combine. 5 5 u(x, y) = AB ekx e 4 k y = C ek (x+ 4 y). Initial condition. u(x, 0) = C ekx = 3e2x =⇒C = 3, k = 2. 5 Final answer. u(x, y) = 3 e2x+ 2 y. 7 / 7 EV0000381 · EV0000382 · EV0000383

SHORT-NOTE RECAP

Source deck preserved: Method of separation of variables continued. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000363

18 Source slide deck: Particular integral of Type 1 and 2 Continued

This entry preserves the complete slide deck titled Particular integral of Type 1 and 2 Continued as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_03, to avoid an unsupported match. EV0000525 · EV0000526

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities



Figure G000118. Particular integral of Type 1 and 2 Continued · source slide page 1.

EV0000526

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Contents



1 Problems when $f(x,y) = e^{ax+by}$, or $f(x,y) = sin(ax+by) = cos(ax+by)$

Figure G000119. Particular integral of Type 1 and 2 Continued · source slide page 2.

EV0000529

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Solve the partial differential equation:

$$\left[D^2 + DD' - 2(D')^2\right] z = 5e^{x+2y}$$

To find CF: The factors are:

$$D^2 + DD' - 2(D')^2 = (D + 2D')(D - D')$$

For

$$(D + 2D'), a = 1, b = 2$$

$$(D - D') = 0, a = 1, b = -1$$

Therefore, $CF = \phi_1(2x - y) + \phi_2(-x - y) = \phi_1(2x - y) + \phi_2(x + y)$

Figure G000120. Particular integral of Type 1 and 2 Continued · source slide page 3.

EV0000532

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Problem (contd.)

We now solve:

$$\text{P.I.} = \frac{1}{D^2 + DD' - 2(D')^2} e^{x+2y}$$

$$\Rightarrow P.I. = 5\frac{1}{1^2 + (1)(2) - 2(2)^2} e^{x+2y}$$

Hence, the particular integral is:

$$\text{PI} = -e^{x+2y}$$

General Solution is: $z = \phi_1(2x - y) + \phi_2(x + y) - e^{x+2y}$

Figure G000121. Particular integral of Type 1 and 2 Continued · source slide page 4.

EV0000535

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Problem

Solve the partial differential equation

$$(3D^2 - 2DD' - D'^2 + 4D' - 3D)\, z \;=\; 5\, e^{(2x+y)/3}$$

To find CF: The factors are:

$$3D^2 - 2DD' - D'^2 + 4D' - 3D = (3D + D' - 3)\,(D - D')$$

For $D - D'$, $a = 1, b = -1, c = 0$. For $3D + D' - 3$, $a = 3, b = 1, c = -1$
Thus, CF is $z = \phi_1(-x - y) + e^{-\frac{x}{3}}\phi_2(x - 3y)$
$\Rightarrow z = \phi_1(x + y) + e^{-\frac{x}{3}}\phi_2(x - 3y)$

Figure G000122. Particular integral of Type 1 and 2 Continued · source slide page 5.

EV0000538

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Problem (contd.)

We set

$$F(D, D') \;=\; 3D^2 \;-\; 2\,DD' \;-\; D'^2 \;+\; 4D' \;-\; 3D.$$

Since the RHS is $5e^{(2x+y)/3}$, we evaluate

$$F\left(\tfrac{2}{3},\, \tfrac{1}{3}\right) = 3\left(\tfrac{2}{3}\right)^2 - 2\left(\tfrac{2}{3}\right)\left(\tfrac{1}{3}\right) - \left(\tfrac{1}{3}\right)^2 + 4\left(\tfrac{1}{3}\right) - 3\left(\tfrac{2}{3}\right) = \frac{1}{9}$$

Figure G000123. Particular integral of Type 1 and 2 Continued · source slide page 6.

EV0000541

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Problem (contd.)

$$\text{PI} = \frac{1}{F\left(\frac{2}{3}, \frac{1}{3}\right)} \cdot 5e^{(2x+y)/3} = \frac{1}{\frac{1}{9}} \cdot 5e^{(2x+y)/3} = 45e^{(2x+y)/3}$$

Figure G000124. Particular integral of Type 1 and 2 Continued · source slide page 7.

EV0000544

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Problem

Solve the PDE

$$(4D^2 + 3DD' - D'^2 - D - D')z = 3e^{(x+2y)/2}$$

Solution: To find DF: The factors are:

$$4D^2 + 3DD' - D'^2 - D - D' = \left(D + D'\right)\left(4D - D' - 1\right)$$

For $D + D'$, $a = b = 1$. For $4D - D' - 1$, $a = 4, b = -1, c = -1$.
Hence, CF is

$$z = \phi_1(x - y) + e^{\frac{x}{4}}\,\phi_2(-x - 4y) = \phi_1(x - y) + e^{\frac{x}{4}}\,\phi_2(x + 4y)$$

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Figure G000125. Particular integral of Type 1 and 2 Continued · source slide page 8.

EV0000547

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

We have

$$F(D, D') = 4D^2 + 3DD' - D'^2 - D - D'$$

and

$$F\left(\frac{1}{2}, 1\right) = 4\left(\frac{1}{4}\right) + 3\left(\frac{1}{2}\right)(1) - 1 - \frac{1}{2} - 1 = 0.$$

We write the particular integral as $z = \phi(x, y)e^{(x+2y)/2}$, where $\phi(x, y)$ is a function to be determined.
The given equation is: $F(D, D')[z] = 3e^{(x+2y)/2}$
$\Rightarrow F\left(D + \frac{1}{2}, D' + 1\right) \cdot \phi(x, y)e^{(x+2y)/2} = 3e^{(x+2y)/2}$
$\Rightarrow F\left(D + \frac{1}{2}, D' + 1\right) \cdot \phi(x, y) = 3$
Therefore,

$$\phi(x, y) = \left[F\left(D + \frac{1}{2}, D' + 1\right)\right]^{-1} (3)$$

Figure G000126. Particular integral of Type 1 and 2 Continued · source slide page 9.

EV0000550

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Problem (contd.)

$$F\left(D+\frac{1}{2},D'+1\right)=4\left(D+\frac{1}{2}\right)^2+3\left(D+\frac{1}{2}\right)(D'+1)-(D'+1)^2-$$
$$(D+\frac{1}{2})-(D'+1)$$

$$=6D-\frac{3}{2}D'+3DD'+4D^2-(D')^2$$
$$=6D\left[1-\frac{1}{4}D'D^{-1}+\frac{1}{6}\{3D'+4D-D'(D')^2\}\right]$$

Figure G000127. Particular integral of Type 1 and 2 Continued · source slide page 10.

EV0000553

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Problem (contd.)

Therefore,

$$\phi(x,y) = \left[F\left(D+\frac{1}{2}, D'+1\right)\right]^{-1} (3)$$

becomes

$$\phi(x,y) = \frac{1}{2}D^{-1}\left[1 - \frac{1}{4}D'D^{-1} + \frac{1}{6}\{3D' + 4D - D'(D')^2\}\right]^{-1} (1)$$

$$= \frac{1}{2}D^{-1}(1) = \frac{x}{2}.$$

Hence, the particular integral is given by

$$z = \frac{x}{2}e^{(x+2y)/2}$$

Figure G000128. Particular integral of Type 1 and 2 Continued · source slide page 11.

EV0000556

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Solve the PDE

$$(2D^2 + 5DD' - 3D'^2)\, z \;=\; \sin\big(2x - y\big).$$

To find CF: The factors are
$2D^2 + 5DD' - 3D'^2 \;=\; \big(2D - D'\big)\,\big(D + 3D'\big)$.
For $2D - D'$, $a = 2, b = -1$. For $D + 3D'$, $a = 1, b = 3$.
Therefore, CF is
$z = \phi_1(-x - 2y) + \phi_2(3x - y) = \phi_1(x + 2y) + \phi_2(3x - y)$

Figure G000129. Particular integral of Type 1 and 2 Continued · source slide page 12.

EV0000559

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

We have the right hand side as $\sin(ax+by)$,
where a = 2, b = -1
Hence,

$$F(D^2, DD', (D')^2)\sin(2x-y) = [2(-4)+5(2)-3(-1)]\sin(2x-y)$$
$$= 5\sin(2x-y)$$

The particular integral is given by

$$z = \frac{\sin(2x-y)}{F(-a^2,-ab,-b^2)} = \frac{1}{5}\sin(2x-y)$$

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Figure G000130. Particular integral of Type 1 and 2 Continued · source slide page 13.

EV0000562

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000525 · EV0000526 · EV0000527
2. Source slide page 2 (extracted text): Contents 1 Problems when f(x, y) = eax+by, or f(x, y) = sin(ax + by) = cos(ax + by) 2 / 13 EV0000528 · EV0000529 · EV0000530
3. Source slide page 3 (extracted text): Problem Solve the partial differential equation: D2 + DD′ −2(D′)2 z = 5ex+2y To find CF: The factors are: D2 + DD′ −2(D′)2 = (D + 2D′)(D −D′) For (D + 2D′), a = 1, b = 2 (D −D′) = 0, a = 1, b = −1 Therefore, CF = ϕ1(2x −y) + ϕ2(−x −y) = ϕ1(2x −y) + ϕ2(x + y) 3 / 13 EV0000531 · EV0000532 · EV0000533
4. Source slide page 4 (extracted text): Problem (contd.) We now solve: 1 P.I. = ex+2y D2 + DD′ −2(D′)2 1 ⇒P.I. = 5 ex+2y 12 + (1)(2) −2(2)2 Hence, the particular integral is: PI = −ex+2y General Solution is: z = ϕ1(2x −y) + ϕ2(x + y) −ex+2y 4 / 13 EV0000534 · EV0000535 · EV0000536

5. Source slide page 5 (extracted text): Problem Solve the partial differential equation (3D2 −2DD′ −D′2 + 4D′ −3D) z = 5 e(2x+y)/3 To find CF: The factors are: 3D2 −2DD′ −D′2 + 4D′ −3D = (3D + D′ −3) (D −D′) For D −D′, a = 1, b = −1, c = 0. For 3D + D′ −3, a = 3, b = 1, c = −1 Thus, CF is z = ϕ1(−x −y) + e−x 3 ϕ2(x −3y) ⇒z = ϕ1(x + y) + e−x 3 ϕ2(x −3y) 5 / 13 EV0000537 · EV0000538 · EV0000539
6. Source slide page 6 (extracted text): Problem (contd.) We set F(D, D′) = 3D2 −2 DD′ −D′2 + 4D′ −3D. Since the RHS is 5e(2x+y)/3, we evaluate 2 2 2 1 2 2 1 1 1 2 1 F 3, 3 = 3 3 −2 3 3 − 3 + 4 3 −3 3 = 9 6 / 13 EV0000540 · EV0000541 · EV0000542
7. Source slide page 7 (extracted text): Problem (contd.) 1 1 PI = · 5e(2x+y)/3 = · 5e(2x+y)/3 = 45e(2x+y)/3 F 23, 13 19 7 / 13 EV0000543 · EV0000544 · EV0000545
8. Source slide page 8 (extracted text): Problem Solve the PDE (4D2 + 3DD′ −D′2 −D −D′)z = 3e(x+2y)/2 Solution: To find DF: The factors are: 4D2 + 3DD′ −D′2 −D −D′ = D + D′ 4D −D′ −1 For D + D′, a = b = 1. For 4D −D′ −1, a = 4, b = −1, c = −1. Hence, CF is x x z = ϕ1(x −y) + e 4 ϕ2(−x −4y) = ϕ1(x −y) + e 4 ϕ2(x + 4y) 8 / 13 EV0000546 · EV0000547 · EV0000548
9. Source slide page 9 (extracted text): Problem (contd.) We have F(D, D′) = 4D2 + 3DD′ −D′2 −D −D′ and 1 1 1 F 1 = 4 + 3 (1) −1 −1 −1 = 0. 2, 4 2 2 We write the particular integral as z = ϕ(x, y)e(x+2y)/2, where ϕ(x, y) is a function to be determined. The given equation is: F(D, D′)[z] = 3e(x+2y)/2 ⇒F D + 2,1 D′ + 1 · ϕ(x, y)e(x+2y)/2 = 3e(x+2y)/2 ⇒F D + 12, D′ + 1 · ϕ(x, y) = 3 Therefore, −1 1 ϕ(x, y) = F D + 2, D′ + 1 (3) 9 / 13 EV0000549 · EV0000550 · EV0000551
10. Source slide page 10 (extracted text): Problem (contd.) 1 1 2 1 F D + 2, D′ + 1 = 4 D + 2 + 3 D + 2 (D′ + 1) −(D′ + 1)2− 1 (D + 2) −(D′ + 1) = 6D −3 + 3DD′ + 4D2 −(D′)2 2D′ = 6D 1 −1 + 1 + 4D −D′(D′)2} 4D′D−1 6{3D′ 10 / 13 EV0000552 · EV0000553 · EV0000554
11. Source slide page 11 (extracted text): Problem (contd.) Therefore, −1 1 ϕ(x, y) = F D + 2, D′ + 1 (3) becomes −1 1 ϕ(x, y) = 1 −1 + 1 + 4D −D′(D′)2} (1) 2D−1 4D′D−1 6{3D′ 1 x = 2D−1(1) = 2. Hence, the particular integral is given by x z = 2e(x+2y)/2 11 / 13 EV0000555 · EV0000556 · EV0000557
12. Source slide page 12 (extracted text): Problem Solve the PDE (2D2 + 5DD′ −3D′2) z = sin 2x −y . To find CF: The factors are 2D2 + 5DD′ −3D′2 = 2D −D′ D + 3D′ . For 2D −D′, a = 2, b = −1. For D + 3D′, a = 1, b = 3. Therefore, CF is z = ϕ1(−x −2y) + ϕ2(3x −y) = ϕ1(x + 2y) + ϕ2(3x −y) 12 / 13 EV0000558 · EV0000559 · EV0000560
13. Source slide page 13 (extracted text): Problem (contd.) We have the right hand side as sin(ax + by), where a = 2, b = -1 Hence, F(D2, DD′, (D′)2) sin(2x −y) = [2(−4) + 5(2) −3(−1)] sin(2x −y) = 5 sin(2x −y) The particular integral is given by sin(2x −y) 1 z = = sin(2x −y) F(−a2, −ab, −b2) 5 13 / 13 EV0000561 · EV0000562 · EV0000563

SHORT-NOTE RECAP

Source deck preserved: Particular integral of Type 1 and 2 Continued. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000525

19 Source slide deck: Particular integral of type 3 and 4 continued

This entry preserves the complete slide deck titled Particular integral of type 3 and 4 continued as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_03, to avoid an unsupported match. EV0000597 · EV0000598

Graphs and diagrams

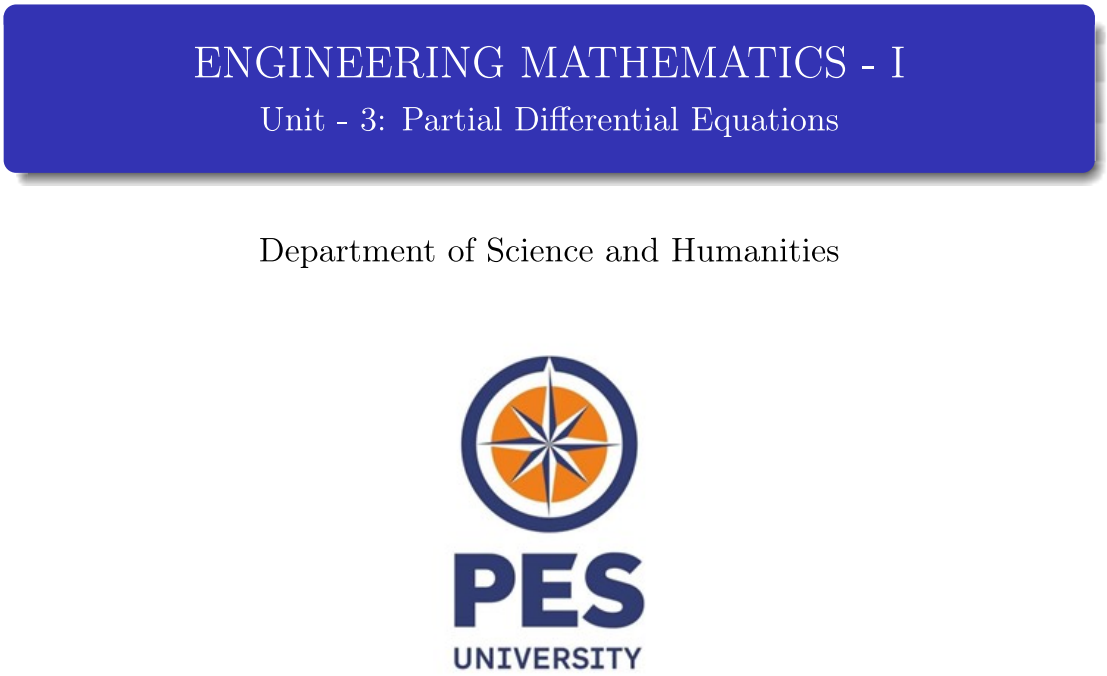


Figure G000142. Particular integral of type 3 and 4 continued · source slide page 1.

EV0000598

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Contents



1 Problems on Case 3 and Case 4

Figure G000143. Particular integral of type 3 and 4 continued · source slide page 2.

EV0000601

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Problem

Solve the PDE

$$4\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x\,\partial y^2} + \frac{\partial^3 z}{\partial y^3} = 6x^2y^2$$

To find CF: The factors are
$4D^3 - 3D(D')^2 + (D')^3 = (D - 2D')^2(D + D')$.
For $D + D'$, $a = b = 1$. For $D - 2D'$, $a = 1, b = -2$.
Hence, the complementary function is
$z = \phi_1(x - y) + \phi_2(2x + y) + x\,\phi_3(2x + y)$

Figure G000144. Particular integral of type 3 and 4 continued · source slide page 3.

EV0000604

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Problem (contd.)

We write $4D^3-3D(D')^2+(D')^3 = 4D^3\left[1-\frac{3}{4}D^{-2}(D')^2+\frac{1}{4}D^{-3}(D')^3\right]$

The PI is given by $z = \frac{1}{4}D^{-3}\left[1-\left\{\frac{3}{4}D^{-2}(D')^2-\frac{1}{4}D^{-3}(D')^3\right\}\right]^{-1}(6x^2y^2)$

$$= \frac{3}{2}D^{-3}\left[1+\frac{3}{4}D^{-2}(D')^2-\frac{1}{4}D^{-3}(D')^3+\cdots\right](x^2y^2)$$

$$= \frac{3}{2}D^{-3}\left[x^2y^2+\frac{3}{2}D^{-2}(x^2)\right] = \frac{3}{2}D^{-3}\left[x^2y^2+\frac{1}{8}x^4\right] = \frac{1}{80}x^5y^2+\frac{1}{1120}x^7$$

Finally, the general solution is $z = C.F + P.I$

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Figure G000145. Particular integral of type 3 and 4 continued · source slide page 4.

EV0000607

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Solve the PDE:

$$2\frac{\partial^2 z}{\partial x^2} + 3\frac{\partial^2 z}{\partial x\,\partial y} + \frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = x - y$$

To find CF: The factors are

$$\begin{aligned} 2D^2 + 3DD' + D'^2 + D + D' &= (2D^2 + 3DD' + D'^2) + (D + D') \\ &= (D + D')(2D + D') + (D + D') \\ &= (D + D')\left[(2D + D') + 1\right] \\ &= (D + D')(2D + D' + 1) \end{aligned}$$

For $(D + D')$, $a = b = 1$. For $2D + D' + 1$, $a = 2, b = 1, c = 1$. Hence, the complementary function is

$$z = \phi_1(x - y) + e^{\frac{-x}{2}}\phi_2(x - 2y)$$

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Figure G000146. Particular integral of type 3 and 4 continued · source slide page 5.

EV0000610

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Problem (contd.)

We write

$$\begin{aligned}&\left[2D^2+3D\,D'+(D')^2+D+D'\right]\\ &=D\Big[1+D^{-1}D'+3D'+2D+D^{-1}(D')^2\Big].\end{aligned}$$

The particular integral is given by

$$\begin{aligned}z&=D^{-1}\Big[1+\{D^{-1}D'+3D'+2D+D^{-1}(D')^2\}\Big]^{-1}(x-y)\\ &=D^{-1}\big[1-\{D^{-1}D'+3D'+2D+D^{-1}(D')^2\}+\cdots\big](x-y)\\ &=D^{-1}\big[x-y-\{\,D^{-1}(-1)+3(-1)+2(1)\}\big]\\ &=\;D^{-1}[x-y+x+1]\;=\;x^2+(1-y)\,x\end{aligned}$$

Finally, the general solution is $z=C.F+P.I$

Figure G000147. Particular integral of type 3 and 4 continued · source slide page 6.

EV0000613

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Solve the PDE

$$\left[D^2 - DD' - 2(D')^2\right] z = 16x\, e^{2y}$$

To find CF: The factors are

$$D^2 \;-\; D\,D' \;-\; 2\,(D')^2 \;=\; \left(D - 2\,D'\right)\left(D + D'\right)$$

For $D - 2\,D'$, $a = 1, b = -2$. For $D + D'$, $a = b = 1$.
Hence, the complementary function is

$$z = \phi_1(-2x - y) + \phi_2(x - y) = \phi_1(2x + y) + \phi_2(x - y)$$

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Figure G000148. Particular integral of type 3 and 4 continued · source slide page 7.

EV0000616

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Problem (contd.)

The Particular integral is given by

$$z = \left(D - 2\,D'\right)\left(D + D'\right)16xe^{2y}$$

We first obtain $(D + D')^{-1}(16xe^{2y})$.
Denote $u = (D + D')^{-1}(16xe^{2y})$, or $(D + D')\,u = 16xe^{2y}$

Figure G000149. Particular integral of type 3 and 4 continued · source slide page 8.

EV0000619

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Problem (contd.)

The auxiliary equations are $\dfrac{dx}{1} = \dfrac{dy}{1} = \dfrac{du}{16xe^{2y}}$.

The first two terms give $y = x + c$

Using the first and third terms, we get

$$\frac{dx}{1} = \frac{du}{16x(e^{2c}e^{2x})}$$

$$\text{Hence} \quad u = 16\,e^{2c} \int x\,e^{2x}\,dx$$

Now, $\int x\,e^{2x}\,dx = \frac{x}{2}\,e^{2x} - \frac{1}{4}\,e^{2x} = \left(\frac{x}{2} - \frac{1}{4}\right)e^{2x}$

Hence, $u = 8\left(x - \frac{1}{2}\right)e^{2(x+c)} = (8x - 4)e^{2y}$

Figure G000150. Particular integral of type 3 and 4 continued · source slide page 9.

EV0000622

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Problem (contd.)

Now, denote $z = (D - D')^{-1}u = (D - D')^{-1}\left((8x - 4)e^{2y}\right)$

or $\quad (D - D')\, z = (8x - 4)e^{2y}$

The auxiliary equations are $\dfrac{dx}{1} = \dfrac{dy}{-1} = \dfrac{dz}{(8x - 4)e^{2y}}$

The first two terms give $x + y = C$

Figure G000151. Particular integral of type 3 and 4 continued · source slide page 10.

EV0000625

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Problem (contd.)

Next, the first and third equations give

$$\frac{dz}{(8x-4)\,e^{2y}} = \frac{dx}{1} \implies dz = (8x-4)\,e^{2C}e^{-2x}\,dx,$$

since $y = C - x$

$$\therefore z = e^{2C}\int (8x-4)e^{-2x}\,dx$$

But,

$$\int (8x-4)e^{-2x}\,dx = \left(-4x\,e^{-2x} - 2\,e^{-2x}\right) + 2\,e^{-2x} = -4x\,e^{-2x}.$$

$$\Rightarrow z = e^{2C}\left[-4x\,e^{-2x}\right] = -4x\,e^{2C-2x} = -4x\,e^{2(C-x)} = -4x\,e^{y},$$

which is the particular integral.

Figure G000152. Particular integral of type 3 and 4 continued · source slide page 11.

EV0000628

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000597 · EV0000598 · EV0000599
2. Source slide page 2 (extracted text): Contents 1 Problems on Case 3 and Case 4 2 / 11 EV0000600 · EV0000601 · EV0000602
3. Source slide page 3 (extracted text): Problem Solve the PDE ∂3z ∂3z ∂3z 4 −3 + = 6 x2y2 ∂x3 ∂x ∂y2 ∂y3 To find CF: The factors are 4D3 −3D(D′)2 + (D′)3 = (D −2D′)2 (D + D′). For D + D′, a = b = 1. For D −2D′, a = 1, b = −2. Hence, the complementary function is z = ϕ1(x −y) + ϕ2(2x + y) + x ϕ3(2x + y) 3 / 11 EV0000603 · EV0000604 · EV0000605
4. Source slide page 4 (extracted text): Problem (contd.) We write 4D3−3D(D′)2+(D′)3 = 4D3h 1−34D−2(D′)2+ 4D−3(D′)3i1 1 3 i−1The PI is given by z = 4D−3h 1− 4D−2(D′)2−14D−3(D′)3 (6x2y2) = 32D−3h 1 + 34D−2(D′)2 −14D−3(D′)3 + · · · i (x2y2) 3 3 i 3 1 1 1 = + 2D−3h x2y2 + 2D−2(x2) = 2D−3h x2y2 8x4i = 80x5y2 + 1120x7 Finally, the general solution is z = C.F + P.I 4 / 11 EV0000606 · EV0000607 · EV0000608

5. Source slide page 5 (extracted text): Problem Solve the PDE: ∂2z ∂2z ∂2z ∂z ∂z 2 + 3 + + + = x −y ∂x2 ∂x ∂y ∂y2 ∂x ∂y To find CF: The factors are 2D2 + 3DD′ + D′2 + D + D′ = (2D2 + 3DD′ + D′2) + (D + D′) = (D + D′)(2D + D′) + (D + D′) = (D + D′) (2D + D′) + 1 = (D + D′)(2D + D′ + 1) For (D + D′), a = b = 1. For 2D + D′ + 1, a = 2, b = 1, c = 1. Hence, the complementary function is −x z = ϕ1(x −y) + e 2 ϕ2(x −2y) 5 / 11 EV0000609 · EV0000610 · EV0000611
6. Source slide page 6 (extracted text): Problem (contd.) We write 2D2 + 3D D′ + (D′)2 + D + D′ h = D 1 + D−1D′ + 3D′ + 2D + D−1(D′)2i . The particular integral is given by i−1 z = D−1h 1 + {D−1D′ + 3D′ + 2D + D−1(D′)2} (x −y) = D−1 1 −{D−1D′ + 3D′ + 2D + D−1(D′)2} + · · · (x −y) = D−1 x −y −{ D−1(−1) + 3(−1) + 2(1)} = D−1[x −y + x + 1] = x2 + (1 −y) x Finally, the general solution is z = C.F + P.I 6 / 11 EV0000612 · EV0000613 · EV0000614
7. Source slide page 7 (extracted text): Problem Solve the PDE D2 −DD′ −2(D′)2 z = 16x e2y To find CF: The factors are D2 −D D′ −2 (D′)2 = D −2 D′ D + D′ For D −2 D′, a = 1, b = −2. For D + D′, a = b = 1. Hence, the complementary function is z = ϕ1(−2x −y) + ϕ2(x −y) = ϕ1(2x + y) + ϕ2(x −y) 7 / 11 EV0000615 · EV0000616 · EV0000617
8. Source slide page 8 (extracted text): Problem (contd.) The Particular integral is given by z = D −2 D′ D + D′ 16xe2y We first obtain (D + D′)−1(16xe2y). Denote u = (D + D′)−1(16xe2y), or (D + D′) u = 16xe2y 8 / 11 EV0000618 · EV0000619 · EV0000620
9. Source slide page 9 (extracted text): Problem (contd.) dx dy du The auxiliary equations are = = . 1 1 16xe2y The first two terms give y = x + c Using the first and third terms, we get dx du = 1 16x(e2ce2x) Z Hence u = 16 e2c x e2x dx Now, R x e2x dx = x e2x − 1 e2x = x −1 e2x 2 4 2 4 Hence, u = 8 x −1 e2(x+c) = (8x −4)e2y 2 9 / 11 EV0000621 · EV0000622 · EV0000623
10. Source slide page 10 (extracted text): Problem (contd.) Now, denote z = (D −D′)−1u = (D −D′)−1 (8x −4)e2y or (D −D′) z = (8x −4)e2y dx dy dz The auxiliary equations are = = 1 −1 (8x −4)e2y The first two terms give x + y = C 10 / 11 EV0000624 · EV0000625 · EV0000626
11. Source slide page 11 (extracted text): Problem (contd.) Next, the first and third equations give dz dx = =⇒dz = (8x −4) e2Ce−2x dx, (8x −4) e2y 1 since y = C −x Z ∴z = e2C (8x −4)e−2x dx But, Z (8x −4)e−2x dx = −4x e−2x −2 e−2x + 2 e−2x = −4x e−2x. ⇒z = e2C −4x e−2x = −4x e2C−2x = −4x e2(C−x) = −4x ey, which is the particular integral. 11 / 11 EV0000627 · EV0000628 · EV0000629

SHORT-NOTE RECAP

Source deck preserved: Particular integral of type 3 and 4 continued. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000597

20 Source slide deck: Particular integral when X is any other form

This entry preserves the complete slide deck titled Particular integral when X is any other form as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_03, to avoid an unsupported match. EV0000630 · EV0000631

Graphs and diagrams

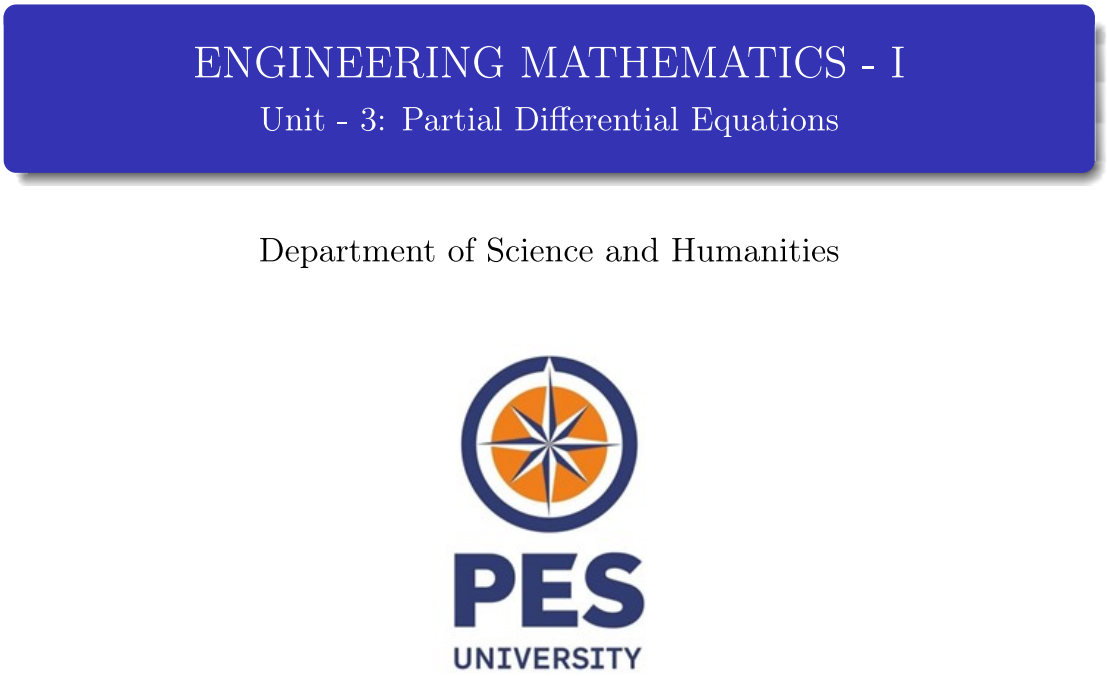


Figure G000153. Particular integral when X is any other form · source slide page 1.

EV0000631

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Contents



1. P.I. when $f(x,y) = x^m y^n$, or a polynomial in x, y
2. P.I. when $f(x,y)$ is not of the form e^{ax+by} or $sin(ax+b)$ or $cos(ax+b)$ or $x^m y^n$ (a polynomial in x, y)

Figure G000154. Particular integral when X is any other form · source slide page 2.

EV0000634

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

P.I. when $f(x, y) = x^m y^n$, or a polynomial in x, y

We write the particular integral as

$$z = [F(D, D')]^{-1} x^m y^n$$

We expand $[F(D, D')]^{-1}$ as an infinite series and operate on $x^m y^n$. If $F(D, D')$ does not contain the constant term, that is, a term independent of D and D', then we expand $[F(D, D')]^{-1}$ in powers of $(D')^{-1}D$ if $m < n$ and in powers of $D^{-1}D'$ if $m > n$

Figure G000155. Particular integral when X is any other form · source slide page 3.

EV0000637

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Problem

Find the particular integral of the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = x^2 + y^2$$

Solution: We have $[D^2 - (D')^2]z = x^2 + y^2$
We write $[D^2 - (D')^2] = D^2\left[1 - (D^{-1})^2(D')^2\right]$

$$\therefore [D^2 - (D')^2]^{-1} = D^{-2}\left[1 - (D^{-1})^2(D')^2\right]^{-1}$$
$$\Rightarrow [D^2 - (D')^2]^{-1} = D^{-2}\left[1 + (D^{-1})^2(D')^2 + (D^{-1})^4(D')^4 + \ldots\right]$$

Figure G000156. Particular integral when X is any other form · source slide page 4.

EV0000640

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

Therefore, the particular integral is given by

$$\begin{aligned} PI &= [D^2 - (D')^2]^{-1}(x^2 + y^2) \\ &= D^{-2}\left[1 + (D^{-1})^2(D')^2 + (D^{-1})^4(D')^4 + \ldots\right](x^2 + y^2) \\ &= D^{-2}\left[(x^2 + y^2) + (D^{-1})^2(D')^2(x^2 + y^2)\right] \\ &= D^{-2}\left[(x^2 + y^2) + (D^{-1})^2(2)\right] \\ &= D^{-2}[x^2 + y^2 + x^2] \\ &= D^{-2}[2x^2 + y^2] \\ &= \frac{x^4}{6} + \frac{x^2y^2}{2}. \quad \textbf{On integrating twice w.r.t. } x \end{aligned}$$

We may also write $[D^2 - (D')^2] = -(D')^2[1 - (D')^{-2}D^2]$.
In this case, we obtain a different form of the particular integral as

$$z = -\frac{y^2}{6}[3x^2 + y^2]$$

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Figure G000157. Particular integral when X is any other form · source slide page 5.

EV0000643

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem

Find the particular integral of the partial differential equation

$$[D^2 + 2DD' + (D')^2]z = 3x + 2y, \quad \text{or} \quad [D + D']^2 z = 3x + 2y$$

We write

$$[D + D']^2 = [D(1 + D'D^{-1})]^2$$

and

$$[D + D']^2 = D^2(1 + D'D^{-1})^2$$

Figure G000158. Particular integral when X is any other form · source slide page 6.

EV0000646

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

The particular integral is given by

$$
\begin{aligned}
z &= D^{-2}(1 + D'D^{-1})^{-2}(3x + 2y) \\
&= D^{-2}[1 - 2D'D^{-1} + 3(D'D^{-1})^2 - \ldots](3x + 2y) \\
&= D^{-2}[3x + 2y - 2D'D^{-1}(3x + 2y)] \\
&= D^{-2}[3x + 2y - 2D'(\frac{3x^2}{2} + 2xy)] \\
&= D^{-2}[3x + 2y - 4x] \\
&= D^{-2}[-x + 2y] \\
&= -\frac{x^3}{6} + x^2y
\end{aligned}
$$

Figure G000159. Particular integral when X is any other form · source slide page 7.

EV0000649

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

We may also write

$$[D + D']^2 = [D'(1 + (D')^{-1}D)]^2$$

The particular integral is given by

$$\begin{aligned} PI &= (D')^{-2}[1 + (D')^{-1}D]^{-2}(3x + 2y) \\ &= (D')^{-2}[1 - 2(D')^{-1}D + 3((D')^{-1}D)^2 - \ldots](3x + 2y) \\ &= (D')^{-2}[3x + 2y - 2(D')^{-1}D(3x + 2y)] \\ &= (D')^{-2}[3x + 2y - 2(D')^{-1}(3)] \\ &= (D')^{-2}[3x + 2y - 6y] \\ &= (D')^{-2}[3x - 4y] \quad \textbf{On integrating twice w.r.t. } y \\ &= \frac{3xy^2}{2} - \frac{2y^3}{3} \end{aligned}$$

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Figure G000160. Particular integral when X is any other form · source slide page 8.

EV0000652

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Let $f(x, y)$ be not of any one of the forms as given in the above cases or the case of failure in Case 2

We use the following procedure:
Assume that $F(D, D')$ is reducible, that is, it can be factorized. Now consider one of the factors as

$$(a_1 D + b_1 D')\, z = a_1 p + b_1 q = f(x, y).$$

Since this is a Lagrange's equation, the auxiliary equations are

$$\frac{dx}{a_1} = \frac{dy}{b_1} = \frac{dz}{f(x, y)}.$$

The first two ratios give the solution

$$b_1\, x - a_1\, y = c, \quad c \text{ arbitrary constant.} \tag{1}$$

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Figure G000161. Particular integral when X is any other form · source slide page 9.

EV0000655

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(Contd.)

Consider now the first and third ratios:

$$\frac{dx}{a_1} = \frac{dz}{f(x,y)} = \frac{dz}{f\big[x,\,(b_1x-c)/a_1\big]},$$

or equivalently

$$a_1\,dz = f\Big[x,\,(b_1x-c)/a_1\Big]\,dx.$$

Integrating, we get

$$a_1\,z = \int f\Big(x,\,\tfrac{b_1x-c}{a_1}\Big)\,dx + c_1 = F(x,c) + c_1,$$

where c_1 is an arbitrary constant. After integration, we replace c by $b_1x - a_1y$ as given in Eq. 1. Since a particular integral is required, we set $c_1 = 0$.

Figure G000162. Particular integral when X is any other form · source slide page 10.

EV0000658

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Figure G000163. Particular integral when X is any other form · source slide page 11.

EV0000661

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000630 · EV0000631 · EV0000632
2. Source slide page 2 (extracted text): Contents 1 P.I. when f(x, y) = xmyn, or a polynomial in x, y 2 P.I. when f(x, y) is not of the form eax+by or sin(ax + b) or cos(ax + b) or xmyn (a polynomial in x, y) 2 / 11 EV0000633 · EV0000634 · EV0000635
3. Source slide page 3 (extracted text): P.I. when f(x, y) = xmyn, or a polynomial in x, y We write the particular integral as z = [F(D, D′)]−1xmyn We expand [F(D, D′)]−1 as an infinite series and operate on xmyn. If F(D, D′) does not contain the constant term, that is, a term independent of D and D′, then we expand [F(D, D′)]−1 in powers of (D′)−1D if m < n and in powers of D−1D′ if m > n 3 / 11 EV0000636 · EV0000637 · EV0000638
4. Source slide page 4 (extracted text): Problem Find the particular integral of the partial differential equation ∂2z −∂2z = x2 + y2 ∂x2 ∂y2 Solution: We have [D2 −(D′)2]z = x2 + y2 We write [D2 −(D′)2] = D2 1 −(D−1)2(D′)2 ∴[D2 −(D′)2]−1 = D−2 1 −(D−1)2(D′)2 −1 ⇒[D2 −(D′)2]−1 = D−2 1 + (D−1)2(D′)2 + (D−1)4(D′)4 + . . . 4 / 11 EV0000639 · EV0000640 · EV0000641

5. Source slide page 5 (extracted text): Problem (contd.) Therefore, the particular integral is given by PI = [D2 −(D′)2]−1(x2 + y2) = D−2 1 + (D−1)2(D′)2 + (D−1)4(D′)4 + . . . (x2 + y2) = D−2 (x2 + y2) + (D−1)2(D′)2(x2 + y2) = D−2 (x2 + y2) + (D−1)2(2) = D−2[x2 + y2 + x2] = D−2[2x2 + y2] x4 x2y2 = + . On integrating twice w.r.t. x 6 2 We may also write [D2 −(D′)2] = −(D′)2[1 −(D′)−2D2]. In this case, we obtain a different form of the particular integral as z = −y2 [3x2 + y2] 6 5 / 11 EV0000642 · EV0000643 · EV0000644
6. Source slide page 6 (extracted text): Problem Find the particular integral of the partial differential equation [D2 + 2DD′ + (D′)2]z = 3x + 2y, or [D + D′]2z = 3x + 2y We write [D + D′]2 = [D(1 + D′D−1)]2 and [D + D′]2 = D2(1 + D′D−1)2 6 / 11 EV0000645 · EV0000646 · EV0000647
7. Source slide page 7 (extracted text): Problem (contd.) The particular integral is given by z = D−2(1 + D′D−1)−2(3x + 2y) = D−2[1 −2D′D−1 + 3(D′D−1)2 −. . .](3x + 2y) = D−2[3x + 2y −2D′D−1(3x + 2y)] = D−2[3x + 2y −2D′(3x2 + 2xy)] 2 = D−2[3x + 2y −4x] = D−2[−x + 2y] = −x3 + x2y 6 7 / 11 EV0000648 · EV0000649 · EV0000650
8. Source slide page 8 (extracted text): Problem (contd.) We may also write [D + D′]2 = [D′(1 + (D′) −1D)]2 The particular integral is given by PI = (D′)−2[1 + (D′)−1D]−2(3x + 2y) = (D′)−2[1 −2(D′)−1D + 3((D′)−1D)2 −. . .](3x + 2y) = (D′)−2[3x + 2y −2(D′)−1D(3x + 2y)] = (D′)−2[3x + 2y −2(D′)−1(3)] = (D′)−2[3x + 2y −6y] = (D′)−2[3x −4y] On integrating twice w.r.t. y 3xy2 −2y3 = 2 3 8 / 11 EV0000651 · EV0000652 · EV0000653
9. Source slide page 9 (extracted text): Let f(x, y) be not of any one of the forms as given in the above cases or the case of failure in Case 2 We use the following procedure: Assume that F(D, D′) is reducible, that is, it can be factorized. Now consider one of the factors as (a1D + b1D′) z = a1p + b1q = f(x, y). Since this is a Lagrange's equation, the auxiliary equations are dx dy dz = = a1 b1 f(x, y). The first two ratios give the solution b1 x −a1 y = c, c arbitrary constant. (1) 9 / 11 EV0000654 · EV0000655 · EV0000656
10. Source slide page 10 (extracted text): (Contd.) Consider now the first and third ratios: dx dz dz = = , a1 f x, y f x, (b1x −c)/a1 or equivalently h i a1 dz = f x, (b1x −c)/a1 dx. Integrating, we get Z b1x−c a1 z = f x, a1 dx + c1 = F(x, c) + c1, where c1 is an arbitrary constant. After integration, we replace c by b1x −a1y as given in Eq. 1. Since a particular integral is required, we set c1 = 0. 10 / 11 EV0000657 · EV0000658 · EV0000659
11. Source slide page 11 (extracted text): (Contd.) Therefore, 1 z = F x, b1x −a1y a1 We repeat the procedure for each factor to obtain the required particular integral. 11 / 11 EV0000660 · EV0000661 · EV0000662

SHORT-NOTE RECAP

Source deck preserved: Particular integral when X is any other form. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000630

21 Source slide deck: Particular integral when X is any other form continued

This entry preserves the complete slide deck titled Particular integral when X is any other form continued as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_03, to avoid an unsupported match.

EV0000663 · EV0000664

Graphs and diagrams

ENGINEERING MATHEMATICS - I

Unit - 3: Partial Differential Equations

Department of Science and Humanities

PES UNIVERSITY

Figure G000164. Particular integral when X is any other form continued · source slide page 1.

EV0000664

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Contents

1 Problems on special case

Figure G000165. Particular integral when X is any other form continued · source slide page 2.

EV0000667

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problems

Find the solution of the differential equation $[2D^2 + 5DD' + 3(D')^2]z = ye^x$.

Solution We write

$$[2D^2 + 5DD' + 3(D')^2]z = (2D + 3D')(D + D')z = ye^x.$$

The complementary function as

$$z = \phi_1(3x - 2y) + \phi_2(x - y).$$

The particular integral is given by

$$z = (2D + 3D')^{-1}(D + D')^{-1}(ye^x).$$

We first obtain $(D + D')^{-1}(ye^x)$ as in case 4. For the sake of completeness, we repeat the procedure used in this case. Denote

$$u = (D + D')^{-1}(ye^x) \quad \text{or} \quad (D + D')u = ye^x.$$

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Figure G000166. Particular integral when X is any other form continued · source slide page 3.

EV0000670

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Problem (contd.)

The auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{ye^x}.$$

The first two terms give $y = x + c$. Using the first and third terms, we get

$$\frac{dx}{1} = \frac{du}{(x+c)e^x}$$

and

$$u = \int (x+c)e^x dx = (x+c-1)e^x = (y-1)e^x.$$

Now, denote

$$z = (2D+3D')^{-1}u = (2D+3D')^{-1}(y-1)e^x$$

or

$$(2D+3D')z = (y-1)e^x.$$

Figure G000167. Particular integral when X is any other form continued · source slide page 4.

EV0000673

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Problem (contd.)

The auxiliary equations are

$$\frac{dx}{2} = \frac{dy}{3} = \frac{dz}{(y-1)e^x}.$$

The first two terms give $2y = 3x + c_1$. The first and third terms give

$$\frac{dx}{2} = \frac{dz}{[(3x + c_1)/2 - 1]e^x}$$

and

$$z = \frac{1}{4}\int (3x + c_1 - 2)e^x dx = \frac{1}{4}(3x + c_1 - 5)e^x = \frac{1}{4}(2y - 5)e^x$$

which is the required particular integral. The general solution of the differential equation is

$$z = \phi_1(3x - 2y) + \phi_2(x - y) + \frac{1}{4}(2y - 5)e^x.$$

Figure G000168. Particular integral when X is any other form continued · source slide page 5.

EV0000676

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Problem

Find the solution of the differential equation

$$\left[D^2 + D\,D' - 2(D')^2\right] z \;=\; 8\,\ln\left(x + 5y\right)$$

Solution:

We write $\left[D^2 + D\,D' - 2(D')^2\right] z = (D + 2D')(D - D')\,z = 8\,\ln\left(x + 5y\right)$.

The complementary function as

$$z = \phi_1(2x - y) \;+\; \phi_2(x + y)$$

Figure G000169. Particular integral when X is any other form continued · source slide page 6.

EV0000679

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

The particular integral is given by

$$z = (D + 2D')^{-1}\,(D - D')^{-1}\Big(8\,\ln(x + 5y)\Big)$$

Denote

$$u \;=\; (D - D')^{-1}\big(8\,\ln(x + 5y)\big), \quad \text{or} \quad (D - D')\,u = 8\,\ln(x + 5y).$$

The auxiliary equations are

$$\frac{dx}{1} \;=\; \frac{dy}{-1} \;=\; \frac{du}{8\,\ln\big(x + 5y\big)}$$

Figure G000170. Particular integral when X is any other form continued · source slide page 7.

EV0000682

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

The first and second terms give $x + y = c$.

The first and third terms give $\dfrac{dx}{1} = \dfrac{du}{8\,\ln(5c - 4x)}$

Thus,

$$u = 8\int \ln(5c - 4x)\,dx = 8\left[x\,\ln(5c - 4x) + \int \frac{4x}{5c - 4x}\,dx\right]$$

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Figure G000171. Particular integral when X is any other form continued · source slide page 8.

EV0000685

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

$$\Rightarrow u = 8\int \ln(5c-4x)\,dx = 8\left[x\,\ln(5c-4x) - \int \frac{4x}{4x-5c}\,dx\right]$$

$$\Rightarrow u = 8\int \ln(5c-4x)\,dx = 8\left[x\,\ln(5c-4x) - \int \left[1+\frac{5c}{4x-5c}\,dx\right]\right]$$

$$\Rightarrow u = 8\left[x\,\ln(5c-4x) - x - \int \frac{5c}{4x-5c}\,dx\right]$$

$$\Rightarrow u = 8\left[x\,\ln(5c-4x) - x + \int \frac{5c}{5c-4x}\,dx\right]$$

Figure G000172. Particular integral when X is any other form continued · source slide page 9.

EV0000688

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

$$\Rightarrow u = 8\left[x\,\ln(5c-4x)\;-\;x\;-\frac{5c}{4}\,\ln(5c-4x)\right]$$

$$\Rightarrow u = 8\left[\left(x-\tfrac{5c}{4}\right)\ln(5c-4x)\;-\;x\right]$$

$$\Rightarrow u = -2(x+5y)\,\ln(x+5y)\;-\;8x$$

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Figure G000173. Particular integral when X is any other form continued · source slide page 10.

EV0000691

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Problem (contd.)

Now, denote $z = (D + 2D')^{-1}u$

or $(D + 2D')\, z = u = -\big[\, 8x + 2(x + 5y)\, \ln(x + 5y)\big]$

The auxiliary equations are $\dfrac{dx}{1} = \dfrac{dy}{2} = \dfrac{dz}{-\big[\, 8x + 2(x + 5y)\ln(x + 5y)\big]}$

The first two terms give $2x - y = c_1$

Figure G000174. Particular integral when X is any other form continued · source slide page 11.

EV0000694

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

The I and III terms give $\dfrac{dx}{1} = \dfrac{dz}{-\left[8x + 2(11x - 5c_1)\,\ln(11x - 5c_1)\right]}$

$$z = -\int \left[8x + 2(11x - 5c_1)\,\ln(11x - 5c_1)\right] dx$$

Integrate w.r.t. x by taking $11x - 5c_1 = t$. Then

$$z = -\left[4x^2 + \frac{2}{11}\left(\tfrac{1}{2}\,(11x - 5c_1)^2\,\ln(11x - 5c_1) - \tfrac{1}{4}\,(11x - 5c_1)^2\right)\right]$$

$$\Rightarrow z = -\left[4x^2 + \frac{1}{11}\,(11x - 5c_1)^2\,\ln(11x - 5c_1) - \tfrac{1}{22}\,(11x - 5c_1)^2\right]$$

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Figure G000175. Particular integral when X is any other form continued · source slide page 12.

EV0000697

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem (contd.)

$$\Rightarrow z = -\left[4x^2 + \frac{1}{22}(11x - 5c_1)^2\{2\ln(11x - 5c_1) - 1\}\right]$$

Since $2x - y = c_1$, $5c_1 = 10x - 5y$. Then $11x - 5c_1$ becomes $x + 5y$. Hence,

$$z = -\left[4x^2 + \frac{1}{22}(x + 5y)^2\{2\ln(x + 5y) - 1\}\right]$$

This is the required particular integral.

The general solution of the differential equation is

$$z = \phi_1(2x - y) + \phi_2(x + y) - \left[4x^2 + \frac{1}{22}(x + 5y)^2\big(2\ln(x + 5y) - 1\big)\right]$$

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Figure G000176. Particular integral when X is any other form continued · source slide page 13.

EV0000700

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS - I Unit - 3: Partial Differential Equations Department of Science and Humanities EV0000663 · EV0000664 · EV0000665
2. Source slide page 2 (extracted text): Contents 1 Problems on special case 2 / 13 EV0000666 · EV0000667 · EV0000668
3. Source slide page 3 (extracted text): Problems Find the solution of the differential equation [2D2 + 5DD′ + 3(D′)2]z = yex. Solution We write [2D2 + 5DD′ + 3(D′)2]z = (2D + 3D′)(D + D′)z = yex. The complementary function as z = ϕ1(3x −2y) + ϕ2(x −y). The particular integral is given by z = (2D + 3D′)−1(D + D′)−1(yex). We first obtain (D + D′)−1(yex) as in case 4. For the sake of completeness, we repeat the procedure used in this case. Denote u = (D + D′)−1(yex) or (D + D′)u = yex. 3 / 13 EV0000669 · EV0000670 · EV0000671
4. Source slide page 4 (extracted text): Problem (contd.) The auxiliary equations are dx dy du = = . 1 1 yex The first two terms give y = x + c. Using the first and third terms, we get dx du = 1 (x + c)ex and Z u = (x + c)exdx = (x + c −1)ex = (y −1)ex. Now, denote z = (2D + 3D′)−1u = (2D + 3D′)−1(y −1)ex or (2D + 3D′)z = (y −1)ex. 4 / 13 EV0000672 · EV0000673 · EV0000674

5. Source slide page 5 (extracted text): Problem (contd.) The auxiliary equations are dx dy dz = = . 2 3 (y −1)ex The first two terms give 2y = 3x + c1. The first and third terms give dx dz = 2 ⌊(3x + c1)/2 −1⌋ex and 1 Z 1 1 z = (3x + c1 −2)exdx = + c1 −5)ex = −5)ex 4 4(3x 4(2y which is the required particular integral. The general solution of the differential equation is 1 z = ϕ1(3x −2y) + ϕ2(x −y) + 4(2y −5)ex. 5 / 13 EV0000675 · EV0000676 · EV0000677
6. Source slide page 6 (extracted text): Problem Find the solution of the differential equation D2 + D D′ −2(D′)2 z = 8 ln x + 5y Solution: We write D2 + D D′ −2(D′)2 z = (D + 2D′)(D −D′) z = 8 ln x + 5y . The complementary function as z = ϕ1(2x −y) + ϕ2(x + y) 6 / 13 EV0000678 · EV0000679 · EV0000680
7. Source slide page 7 (extracted text): Problem (contd.) The particular integral is given by z = (D + 2D′)−1 (D −D′)−1 8 ln(x + 5y) Denote u = (D −D′)−1 8 ln(x + 5y) , or (D −D′) u = 8 ln(x + 5y). The auxiliary equations are dx dy du = = 1 −1 8 ln x + 5y 7 / 13 EV0000681 · EV0000682 · EV0000683
8. Source slide page 8 (extracted text): Problem (contd.) The first and second terms give x + y = c. dx du The first and third terms give = 1 8 ln 5c −4x Thus, " # Z Z 4x u = 8 ln 5c −4x dx = 8 x ln 5c −4x + dx 5c −4x 8 / 13 EV0000684 · EV0000685 · EV0000686
9. Source slide page 9 (extracted text): Problem (contd.) " # Z Z 4x ⇒u = 8 ln 5c −4x dx = 8 x ln 5c −4x − dx 4x −5c " # Z Z 5c ⇒u = 8 ln 5c −4x dx = 8 x ln 5c −4x − 1 + dx 4x −5c " # Z 5c ⇒u = 8 x ln 5c −4x −x − dx 4x −5c " # Z 5c ⇒u = 8 x ln 5c −4x −x + dx 5c −4x 9 / 13 EV0000687 · EV0000688 · EV0000689
10. Source slide page 10 (extracted text): Problem (contd.) " # ⇒u = 8 x ln 5c −4x −x −5c ln(5c −4x) 4 h ⇒u = 8 x −5c ln 5c −4x −xi 4 ⇒u = −2(x + 5y) ln(x + 5y) −8x 10 / 13 EV0000690 · EV0000691 · EV0000692
11. Source slide page 11 (extracted text): Problem (contd.) Now, denote z = (D + 2D′)−1u or (D + 2D′) z = u = − 8x + 2(x + 5y) ln(x + 5y) dx dy dz The auxiliary equations are = = 1 2 − 8x + 2(x + 5y) ln(x + 5y) The first two terms give 2x −y = c1 11 / 13 EV0000693 · EV0000694 · EV0000695
12. Source slide page 12 (extracted text): Problem (contd.) dx dz The I and III terms give = 1 − 8x + 2(11x −5c1) ln(11x −5c1) Z z = − 8x + 2(11x −5c1) ln(11x −5c1) dx Integrate w.r.t. x by taking 11x −5c1 = t. Then 2 1 1 i h z = − 4x2 + 2 (11x −5c1)2 ln(11x −5c1) − 4 (11x −5c1)2 11 1 1 h ⇒z = − 4x2 + (11x −5c1)2 ln(11x −5c1) − 22 (11x −5c1)2i 11 12 / 13 EV0000696 · EV0000697 · EV0000698
13. Source slide page 13 (extracted text): Problem (contd.) 1 h i ⇒z = − 4x2 + (11x −5c1)2 {2 ln(11x −5c1) −1} 22 Since 2x −y = c1, 5c1 = 10x −5y. Then 11x −5c1 becomes x + 5y. Hence, 1 h i z = − 4x2 + (x + 5y)2 {2 ln(x + 5y) −1} 22 This is the required particular integral. The general solution of the differential equation is 1 h i z = ϕ1(2x−y) + ϕ2(x+y) − 4x2 + 22(x+ 5y)2 2 ln(x+5y) −1 13 / 13 EV0000699 · EV0000700 · EV0000701

SHORT-NOTE RECAP

Source deck preserved: Particular integral when X is any other form continued. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000663

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Formation of PDE transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Introduction transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on formation of PDE by eliminating arbitrary functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Problems on formation of PDE by eliminating arbitrary functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__9ce94de93a3b3017.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Session-3 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/002_Session-3__9680cfc6ed04da14.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagranges Linear equation_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagranges Linear equation_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/004_Lagranges_Linear_equation_continued__7d0c7f14ab47b67a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.
Method of separation of variables transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/005_Method_of_separation_of_variables__40649d5bab207a80.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Modeling of One dimensional Heat equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9974; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Solution of one dimensional heat equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Higher order PDE transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Higher_order_PDE__d750329ea5ff8f74.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of type 1 and 2 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Particular integral of type 1 and 2 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Particular_integral_of_type_1_and_2__265cba67cfcb2e18.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on finding the solution when RHS=0 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9965; only the JSON segmentation supplies the displayed timestamps.
Problems on finding the solution when RHS=0 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/007_Problems_on_finding_the_solution_when_RHS_0__0bbcd9c646bfb2ec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral -Type 3 and Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral -Type 3 and Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/010_Particular_integral_-Type_3_and_Type_4__8bacb290bb4f1c00.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Formation of PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Formation of PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Formation of PDE (converted PDF; original hash 6b7579603cfb0efd48bd28bfbf4ef0e5efc3f852b845e4c5d58726437cc5d251)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Introduction to Partial differentiation (converted PDF; original hash 496c28b2e6072e7dd4a2fa45951cc9287e90f3476ebfad3c06ff26caad113eba)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on formation of PDE by eliminating arbitrary functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on formation of PDE by eliminating arbitrary functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on formation of PDE by eliminating arbitrary functions (converted PDF; original hash 6088fd4554e5221d7231bb022216ee5422c61c5a8b21fea2359fb7442ae4649d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Lagrange s Linear Equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Lagrange s Linear Equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Lagrange s Linear Equation (converted PDF; original hash 0f0d47015a775c00d5a2ef1c1ba0ad54770829f306f718a0ec8b0ae8aa28f548)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems of Lagranges equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems of Lagranges equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems of Lagranges equation (converted PDF; original hash 45fa62ce35f0b306eab5a3f9dadf64d5e48dbbca35e38a017405d89268100242)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Lagrange s Linear Equation Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Lagrange s Linear Equation Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Lagrange s Linear Equation Continued (converted PDF; original hash 1629fb7dfc2d5895a175d48266cf8dcad23def6e219976b01adf7120e45ef814)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Method of separation of variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Method of separation of variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Method of separation of variables (converted PDF; original hash 78c0cbeb9322daba54ecea6092cb185fa60b5fa52d896ac35042c6c3e27988b7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Method of separation of variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Method of separation of variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Method of separation of variables continued (converted PDF; original hash 22c930a9df5e782fd0e32a2cc9faf75db1ff5cb668310cbd24d0412b87da02da)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Modeling of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Modeling of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Modeling of one dimensional heat equation (converted PDF; original hash bf2d6d6926b3ddb1f00f641cefc01736225035f2f8799c95ed0da06bac5074a6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Solution of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Solution of one dimensional heat equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Solution of one dimensional heat equation (converted PDF; original hash ae4bcbac05a2c15a9a250781a63db204cafb89f00bdb705917d9af11397ae05b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Higher order PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Higher order PDE	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Higher order PDE (converted PDF; original hash 2b96649655e0e57d5e1312ee2ae72dcc1948ceca02bd1d6edf53d0f772ee6a5d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 1 and 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral of Type 1 and 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 1 and 2 (converted PDF; original hash 3d506acd6683b2c621e0d15c9ca59f1065600dc4cc87198b47d515e6e31afee0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: solutions of PDE when RHS 0	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: solutions of PDE when RHS 0	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for solutions of PDE when RHS 0 (converted PDF; original hash ce9d6509f0e7ebe5773e99c5c9758150bcc91a4cc9d770d5d396e9dd59b58680)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of Type 1 and 2 Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral of Type 1 and 2 Continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of Type 1 and 2 Continued (converted PDF; original hash 0675c88ff179ac22172f95bc9add1e205ebd6cc4ef187b2ed1404995555316e5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of type 3 and 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral of type 3 and 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral of type 3 and 4 (converted PDF; original hash a16d85b951f83cc4d5d75cfe114ee11333442d8e6eb035d99e177fbafcd65086)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral of type 3 and 4 continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral of type 3 and 4 continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
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Slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Particular integral when X is any other form	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral when X is any other form (converted PDF; original hash a16d85b951f83cc4d5d75cfe114ee11333442d8e6eb035d99e177fbafcd65086)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Particular integral when X is any other form continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Particular integral when X is any other form continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Particular integral when X is any other form continued (converted PDF; original hash 6ff6d8dbbbe10faa89598152440e66fd69fddf04c79cd701298c561fa443f4c4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV00000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.

Evidence	Source unit	Status	Decision
EV000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000015	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000016	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000022	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000023	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000030	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000031	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000032	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000033	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000034	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000035	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000036	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000037	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000038	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000039	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000040	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000041	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000042	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000043	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000044	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0047. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000049	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000050	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000051	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000057	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000058	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000065	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000066	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000067	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000068	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000069	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000070	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000071	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000072	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000079	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000080	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000081	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000086	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000088	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0071. Sidecar variants compare at similarity 0.9989; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000091	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000092	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9974; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000093	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 0.9974; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 00	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare at similarity 0.9974; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 01	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 02	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 03	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 04	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 05	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 06	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000107	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000112	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000113	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 14	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 15	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 16	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0065. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 17	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 18	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 19	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 20	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000121	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000126	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000127	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 28	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 29	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 30	SRC00038/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 31	SRC00039/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 32	SRC00040/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0073. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 33	SRC00041/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 34	SRC00038/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0089. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000135	SRC00039/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0089. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00040/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0089. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00041/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0089. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000140	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000141	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000142	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000146	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000147	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000148	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001149	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001150	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001151	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001152	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9965; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001153	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001154	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001155	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000156	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 0.9965; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00050/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00051/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00052/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00053/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00050/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000163	SRC00051/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00052/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00053/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00050/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00051/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000168	SRC00052/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000169	SRC00053/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001770	SRC00050/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001771	SRC00051/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001772	SRC00052/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0066. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001773	SRC00053/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001774	SRC00054/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001775	SRC00056/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001776	SRC00055/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000177	SRC00054/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000178	SRC00056/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000179	SRC00055/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000180	SRC00054/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000181	SRC00056/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000182	SRC00055/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000183	SRC00054/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001184	SRC00056/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001185	SRC00055/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00001186	SRC00054/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001187	SRC00056/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001188	SRC00055/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00001189	SRC00054/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001190	SRC00056/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000191	SRC00055/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000192	SRC00054/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000193	SRC00056/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000194	SRC00055/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000195	SRC00054/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000196	SRC00056/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000197	SRC00055/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00001 98	SRC00054/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 99	SRC00056/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 00	SRC00055/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 01	SRC00054/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 02	SRC00056/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 03	SRC00055/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 04	SRC00054/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000205	SRC00056/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Formation_of_PDE__b03c4069a1ba54fb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000206	SRC00055/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000207	SRC00057/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000208	SRC00059/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000209	SRC00058/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000210	SRC00057/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000211	SRC00059/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000212	SRC00058/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000213	SRC00057/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000214	SRC00059/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000215	SRC00058/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000216	SRC00057/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000217	SRC00059/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000218	SRC00058/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002219	SRC00057/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002220	SRC00059/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002221	SRC00058/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002222	SRC00057/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002223	SRC00059/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002224	SRC00058/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 25	SRC00057/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 26	SRC00059/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 27	SRC00058/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 28	SRC00057/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 29	SRC00059/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 30	SRC00058/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000231	SRC00057/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000232	SRC00059/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000233	SRC00058/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000234	SRC00057/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000235	SRC00059/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Introduction_to_Partial_differentiation__e5e8e9c449cbe3df.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000236	SRC00058/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000237	SRC00060/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000238	SRC00062/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000239	SRC00061/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000240	SRC00060/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000241	SRC00062/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000242	SRC00061/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000243	SRC00060/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000244	SRC00062/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000245	SRC00061/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000246	SRC00060/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000247	SRC00062/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000248	SRC00061/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000249	SRC00060/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000250	SRC00062/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000251	SRC00061/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000252	SRC00060/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000253	SRC00062/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000254	SRC00061/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 55	SRC00060/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 56	SRC00062/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 57	SRC00061/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 58	SRC00060/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 59	SRC00062/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/001_Problems_on_formation_of_PDE_by_eliminating_arbitrary_functions__c988f2952d1bcf9a.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 60	SRC00061/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 61	SRC00063/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 62	SRC00065/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 63	SRC00064/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 64	SRC00063/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 65	SRC00065/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 66	SRC00064/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 67	SRC00063/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 68	SRC00065/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 69	SRC00064/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 70	SRC00063/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 71	SRC00065/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 72	SRC00064/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 73	SRC00063/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 74	SRC00065/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 75	SRC00064/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 76	SRC00063/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 77	SRC00065/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 78	SRC00064/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 79	SRC00063/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 80	SRC00065/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 81	SRC00064/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 82	SRC00063/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 83	SRC00065/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 84	SRC00064/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002285	SRC00063/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002286	SRC00065/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/002_Lagrange_s_Linear_Equation__933d8890824a8e3c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002287	SRC00064/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00002288	SRC00066/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002289	SRC00068/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002290	SRC00067/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002291	SRC00066/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002292	SRC00068/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002293	SRC00067/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00002294	SRC00066/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002295	SRC00068/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002296	SRC00067/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 97	SRC00066/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 98	SRC00068/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 99	SRC00067/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 00	SRC00066/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 01	SRC00068/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 02	SRC00067/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 03	SRC00066/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 04	SRC00068/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 05	SRC00067/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 06	SRC00066/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 07	SRC00068/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 08	SRC00067/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 09	SRC00066/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 10	SRC00068/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/003_Problems_of_Lagranges_equation__c98cf70dfb469813.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 11	SRC00067/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 12	SRC00069/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 13	SRC00071/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 14	SRC00070/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003315	SRC00069/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003316	SRC00071/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003317	SRC00070/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003318	SRC00069/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003319	SRC00071/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003320	SRC00070/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003321	SRC00069/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003322	SRC00071/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003323	SRC00070/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003324	SRC00069/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003325	SRC00071/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003326	SRC00070/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003327	SRC00069/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003328	SRC00071/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003329	SRC00070/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003330	SRC00069/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003331	SRC00071/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003332	SRC00070/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000333	SRC00069/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000334	SRC00071/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000335	SRC00070/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000336	SRC00069/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000337	SRC00071/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000338	SRC00070/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000339	SRC00069/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000340	SRC00071/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/004_Lagrange_s_Linear_Equation_Continued__4d8f53e68af1b6b9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000341	SRC00070/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000342	SRC00072/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000343	SRC00074/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000344	SRC00073/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 45	SRC00072/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 46	SRC00074/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 47	SRC00073/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 48	SRC00072/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 49	SRC00074/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 50	SRC00073/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 51	SRC00072/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 52	SRC00074/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 53	SRC00073/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 54	SRC00072/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 55	SRC00074/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 56	SRC00073/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000357	SRC00072/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000358	SRC00074/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000359	SRC00073/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000360	SRC00072/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000361	SRC00074/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables__f60e784635a308f5.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000362	SRC00073/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000363	SRC00075/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000364	SRC00077/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000365	SRC00076/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000366	SRC00075/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000367	SRC00077/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000368	SRC00076/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003369	SRC00075/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003370	SRC00077/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003371	SRC00076/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003372	SRC00075/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003373	SRC00077/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003374	SRC00076/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003775	SRC00075/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003776	SRC00077/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003777	SRC00076/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003778	SRC00075/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003779	SRC00077/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003780	SRC00076/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000381	SRC00075/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000382	SRC00077/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/005_Method_of_separation_of_variables_continued__d6eae2962166f91d.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000383	SRC00076/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000384	SRC00078/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000385	SRC00080/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000386	SRC00079/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000387	SRC00078/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000388	SRC00080/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000389	SRC00079/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000390	SRC00078/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000391	SRC00080/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000392	SRC00079/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000393	SRC00078/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000394	SRC00080/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000395	SRC00079/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000396	SRC00078/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000397	SRC00080/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000398	SRC00079/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 99	SRC00078/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 00	SRC00080/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 01	SRC00079/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 02	SRC00078/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 03	SRC00080/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Modeling_of_one_dimensional_heat_equation__1d8081c04c0b7180.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 04	SRC00079/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000405	SRC00081/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000406	SRC00083/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000407	SRC00082/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000408	SRC00081/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000409	SRC00083/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000410	SRC00082/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000411	SRC00081/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000412	SRC00083/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000413	SRC00082/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000414	SRC00081/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000415	SRC00083/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000416	SRC00082/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000417	SRC00081/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000418	SRC00083/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00082/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000420	SRC00081/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000421	SRC00083/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/006_Solution_of_one_dimensional_heat_equation__e48fe3b0f3f38d0f.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000422	SRC00082/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000423	SRC00084/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000424	SRC00086/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000425	SRC00085/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000426	SRC00084/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000427	SRC00086/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000428	SRC00085/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000429	SRC00084/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004030	SRC00086/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004031	SRC00085/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00004032	SRC00084/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004033	SRC00086/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004034	SRC00085/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00004035	SRC00084/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004036	SRC00086/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000437	SRC00085/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000438	SRC00084/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000439	SRC00086/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00085/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000441	SRC00084/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000442	SRC00086/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00085/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000444	SRC00084/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000445	SRC00086/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000446	SRC00085/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000447	SRC00084/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000448	SRC00086/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00085/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000450	SRC00084/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000451	SRC00086/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000452	SRC00085/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000453	SRC00084/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000454	SRC00086/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000455	SRC00085/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000456	SRC00084/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000457	SRC00086/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000458	SRC00085/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000459	SRC00084/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000460	SRC00086/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000461	SRC00085/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000462	SRC00084/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000463	SRC00086/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000464	SRC00085/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000465	SRC00084/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000466	SRC00086/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000467	SRC00085/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000468	SRC00084/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000469	SRC00086/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Higher_order_PDE__c1bdbadec9aabc42.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000470	SRC00085/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000471	SRC00087/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000472	SRC00089/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000473	SRC00088/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000474	SRC00087/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000475	SRC00089/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000476	SRC00088/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000477	SRC00087/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004 78	SRC00089/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 79	SRC00088/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 80	SRC00087/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 81	SRC00089/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 82	SRC00088/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 83	SRC00087/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004 84	SRC00089/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 85	SRC00088/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 86	SRC00087/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 87	SRC00089/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 88	SRC00088/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 89	SRC00087/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004090	SRC00089/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004091	SRC00088/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00004092	SRC00087/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004093	SRC00089/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004094	SRC00088/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00004095	SRC00087/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004 96	SRC00089/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 97	SRC00088/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 98	SRC00087/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 99	SRC00089/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 00	SRC00088/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 01	SRC00087/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000502	SRC00089/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_Particular_integral_of_Type_1_and_2__b3554ce747c5c569.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000503	SRC00088/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000504	SRC00090/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000505	SRC00092/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000506	SRC00091/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000507	SRC00090/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 08	SRC00092/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 09	SRC00091/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 10	SRC00090/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 11	SRC00092/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 12	SRC00091/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 13	SRC00090/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000514	SRC00092/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000515	SRC00091/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000516	SRC00090/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000517	SRC00092/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000518	SRC00091/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000519	SRC00090/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000520	SRC00092/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000521	SRC00091/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000522	SRC00090/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000523	SRC00092/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/007_solutions_of_PDE_when_RHS_0__0b6ef7e22c1b84bb.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000524	SRC00091/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000525	SRC00093/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 26	SRC00095/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 27	SRC00094/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 28	SRC00093/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 29	SRC00095/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 30	SRC00094/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 31	SRC00093/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000532	SRC00095/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000533	SRC00094/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000534	SRC00093/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000535	SRC00095/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000536	SRC00094/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000537	SRC00093/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005538	SRC00095/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005539	SRC00094/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00005540	SRC00093/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005541	SRC00095/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005542	SRC00094/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00005543	SRC00093/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005544	SRC00095/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005545	SRC00094/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00005546	SRC00093/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005547	SRC00095/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005548	SRC00094/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00005549	SRC00093/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 50	SRC00095/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 51	SRC00094/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 52	SRC00093/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 53	SRC00095/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 54	SRC00094/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 55	SRC00093/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000556	SRC00095/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000557	SRC00094/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000558	SRC00093/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000559	SRC00095/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000560	SRC00094/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000561	SRC00093/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000562	SRC00095/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/009_Particular_integral_of_Type_1_and_2_Continued__494127478d9b0a44.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000563	SRC00094/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000564	SRC00096/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000565	SRC00098/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000566	SRC00097/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000567	SRC00096/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 68	SRC00098/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 69	SRC00097/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 70	SRC00096/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 71	SRC00098/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 72	SRC00097/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 73	SRC00096/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000574	SRC00098/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000575	SRC00097/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000576	SRC00096/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000577	SRC00098/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000578	SRC00097/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000579	SRC00096/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 80	SRC00098/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 81	SRC00097/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 82	SRC00096/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 83	SRC00098/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 84	SRC00097/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 85	SRC00096/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 86	SRC00098/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 87	SRC00097/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 88	SRC00096/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 89	SRC00098/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00005 90	SRC00097/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 91	SRC00096/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000592	SRC00098/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000593	SRC00097/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000594	SRC00096/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000595	SRC00098/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4__d8db4b61abaf4d95.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000596	SRC00097/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000597	SRC00099/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000598	SRC00101/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000599	SRC00100/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000600	SRC00099/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000601	SRC00101/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000602	SRC00100/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000603	SRC00099/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000604	SRC00101/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000605	SRC00100/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000606	SRC00099/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000607	SRC00101/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000608	SRC00100/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000609	SRC00099/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00006010	SRC00101/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006011	SRC00100/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00006012	SRC00099/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006013	SRC00101/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006014	SRC00100/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00006015	SRC00099/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000616	SRC00101/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000617	SRC00100/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000618	SRC00099/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000619	SRC00101/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000620	SRC00100/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000621	SRC00099/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000622	SRC00101/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000623	SRC00100/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000624	SRC00099/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000625	SRC00101/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000626	SRC00100/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000627	SRC00099/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00006028	SRC00101/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/010_Particular_integral_of_type_3_and_4_continued__23bf4ab0a81ab3c7.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006029	SRC00100/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00006030	SRC00102/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006031	SRC00104/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006032	SRC00103/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00006033	SRC00102/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000634	SRC00104/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000635	SRC00103/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000636	SRC00102/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000637	SRC00104/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000638	SRC00103/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000639	SRC00102/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000640	SRC00104/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000641	SRC00103/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000642	SRC00102/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000643	SRC00104/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000644	SRC00103/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000645	SRC00102/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000646	SRC00104/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000647	SRC00103/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000648	SRC00102/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000649	SRC00104/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000650	SRC00103/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000651	SRC00102/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000652	SRC00104/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000653	SRC00103/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000654	SRC00102/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000655	SRC00104/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000656	SRC00103/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000657	SRC00102/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000658	SRC00104/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000659	SRC00103/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000660	SRC00102/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000661	SRC00104/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form__206fd1d9950921e9.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000662	SRC00103/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000663	SRC00105/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000664	SRC00107/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000665	SRC00106/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000666	SRC00105/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000667	SRC00107/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000668	SRC00106/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000669	SRC00105/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000670	SRC00107/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000671	SRC00106/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000672	SRC00105/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000673	SRC00107/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000674	SRC00106/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000675	SRC00105/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00006776	SRC00107/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006777	SRC00106/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00006778	SRC00105/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006779	SRC00107/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006780	SRC00106/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00006781	SRC00105/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00006 82	SRC00107/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006 83	SRC00106/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00006 84	SRC00105/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 85	SRC00107/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00006 86	SRC00106/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00006 87	SRC00105/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000688	SRC00107/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000689	SRC00106/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000690	SRC00105/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000691	SRC00107/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000692	SRC00106/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000693	SRC00105/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000694	SRC00107/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000695	SRC00106/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000696	SRC00105/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000697	SRC00107/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000698	SRC00106/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000699	SRC00105/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000700	SRC00107/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/Slides/016_Particular_integral_when_X_is_any_other_form_continued__ea709a2f03dc5f0c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000701	SRC00106/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

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